

WHAT DOES PRESERVING A DEFENSE PRESERVE? CAUSAL IDENTIFICATION FOR FEDERATED-LEARNING DEFENSE COMPOSITION

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ABSTRACT

Federated learning composes defenses that were never designed together, so upstream processing can destroy what the downstream defense reads. The natural repair is to require the upstream stage to **preserve** it; for the broad class of *positive per-client rescaling* compositions, where the invariance question is exactly algebraic, that is not enough. The obstruction is structural: a criterion gated on the outcome it predicts conditions on a descendant of both the mechanism and the defense’s standalone strength, so **no screen of this form identifies the mechanism’s contribution, at any sample size**. Ignoring this does not cost power, it costs the sign: on the identical cell the outcome-gated ladder moves -0.272 where the instrument moves $+0.098$. Preservation is a hierarchy, and we break its last two links by intervening rather than comparing: Oracle intervention pinning every adversary’s coefficient closes the attenuation channel, and disturbing the statistic Krum reads then flips its decision in up to 80% of rounds while the *support* of the admitted adversarial mass never changes in any measured round of any arm, and suppression stays inside our pre-specified ± 0.15 equivalence margin. We prove sufficient conditions for the weighting $\rightarrow$ aggregator class, which certify **mechanism preservation, not security**, and bound no ASR. A second, *Coverage* limit follows: suppression emergent in a composition and absent from both constituents cannot satisfy an inherited-suppression gate, capping recall structurally. We replicate the negative in a second invariance class and on a second dataset and architecture. The implication is methodological: evaluate composed mechanisms by intervening on adversarial contribution, not by inferring it from downstream statistics.

1 INTRODUCTION

Federated learning (McMahan et al., 2017) is vulnerable to data poisoning (Chen et al., 2017; Bagdasaryan et al., 2020), and an arms race has produced defenses (Fung et al., 2020; Pillutla et al., 2022; Yin et al., 2018; Blanchard et al., 2017) with accuracy/robustness trade-offs and none universally dominant (Appendix G). *Can we compose them for stronger robustness?* Composition does not even commute: FoolsGold $\rightarrow$ CoordMedian suppresses a pixel backdoor to ASR 0.076 while the reverse order reaches 0.732: a rank aggregator emits one vector rather than per-client updates, so it cannot be an *upstream* stage and the pair collapses to FoolsGold alone, bit-identically. Naive layering fails this way whenever upstream processing destroys what the downstream defense reads, and it bites hardest in decentralized pipelines, foundation models included.

Why the question is sharper without a trusted coordinator. Such stacks compose independently developed mechanisms among heterogeneous, partially trusted, often pseudonymous participants, so preservation is the obvious discipline to demand of them. Concretely: a contributor clips its own update, a scheduler reweights contributors by reputation, and whoever merges runs a robust aggregator: three parties, three mechanisms, nobody owning the stack. The only check available at each seam is that the next stage still reads what it expects, and a preserved statistic invites the reading that the stack is therefore still robust. **That inference is what this paper refutes. The identification problem we isolate is defined independently of model scale**, so we study it in

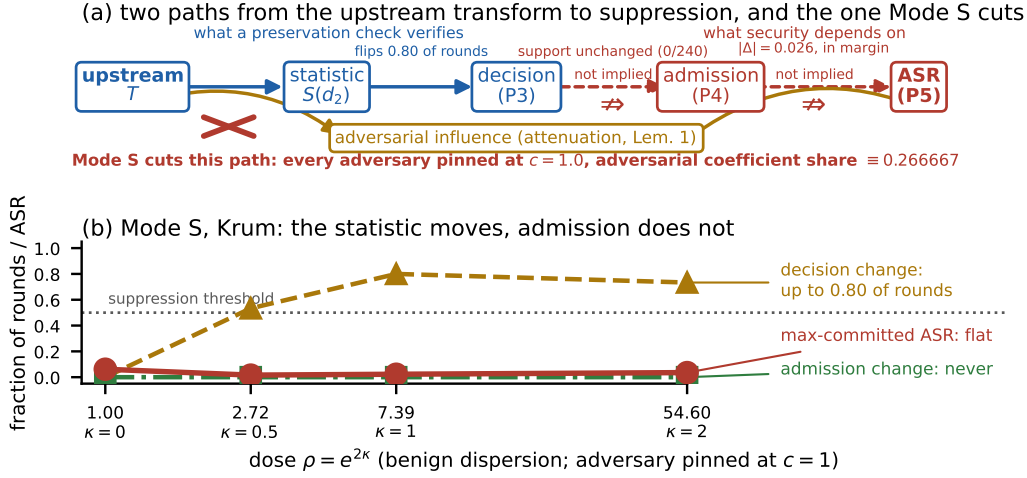


Figure 1: **The paper in one figure: the statistic moves, admission does not, suppression does not follow.** (a) Two paths run from an upstream rescaling T to suppression: the *chain* a preservation criterion reasons along, and a *confounded* path on which T merely attenuates whichever client carries the poison (Lem. 2) and reaches ASR **without passing through d_2 's statistic at all**. **Mode S cuts the second by construction**, pinning every adversary at $c=1$ (the one channel the dose closes, and not the only thing it moves), which is what makes the chain's two broken arrows measurable. The admission count pools the four aggregators of Table 1, with adversarial mass present at baseline in 100 of those 240 rounds; Krum alone admits no adversary at any rung, so its own zero is a floor. **But the non-event is not only a Krum floor effect**: reputation and `coord.median` each admit adversarial mass at baseline in 48 of their 60 measured rounds, and the support of that mass still never moves (App. E.2). (b) Mode S, Krum under model-scaling (§5): as ρ grows to 54.6 the *decision* flips in up to 80% of rounds while the *support* of the admitted adversarial mass does not, and max-committed ASR stays inside the pre-specified ± 0.15 equivalence margin ($|\Delta| = 0.026$). Bars are Student- t 95% intervals ($n=5$), recomputed per seed.

a controlled federated setting rather than claiming transfer to foundation-model training; statistic-preservation checks are therefore not attack-suppression evidence in larger decentralized stacks, and assuming otherwise in a trustless pipeline is worse than uninformative.

Contributions. We propose no new defense. **Our contribution is a causal-identification result about how defense compositions are evaluated**, the design it forces, and one experiment that separates the channels it names. *Attack-suppression preservation under a fixed committed attack* is the object throughout: suppression of that attack surviving composition, instantiated as ASR, never the statistic or the decision that follows. We claim nothing about confidentiality, availability, adaptive adversaries, or robustness beyond that suppression; Fig. 1 is the paper in one picture.

(1) **The identification result (§3).** A screen that keeps only pairs whose constituents already suppress the attack, then asks whether the composition preserves it, gates on a descendant of the quantity it is trying to estimate. **The consequence is non-identification, not bias**: one cell of the relevant 2×2 is empty for *every* defense menu, so the estimand has no support and **no adjustment set, no reweighting and no sample size recovers it**. That is what separates this from ordinary collider bias, which is a bias, and correctable in principle given the right adjustment.

(2) **A second, independent limit on the same design (Prop. 2).** Even where such a screen is admissible it cannot *find* what it is for: suppression that is emergent, present in the composition and absent from both constituents, fails the gate by definition, capping recall at $1 - |\mathcal{E}|/|\mathcal{S}|$ on every menu and every attack. Our own lowest-ASR composition is the witness. This is a coverage statement about a class of screens and does not follow from the collider argument.

(3) **The boundary is constructive (§4).** Each clause of the evaluation protocol is forced by a term in the proposition, so the result arrives with the design that respects it rather than only a prohibition.

(4) **The demonstration is a sign inversion, not attenuation (§5).** On an identical cell the outcome-gated design estimates -0.272 where the intervention estimates $+0.098$. Collider bias is usually taught as attenuation; here the confounded design does not lose power, it **reverses the direction** of the measured effect, inside a live security-evaluation methodology.

Also. The preservation hierarchy (Def. 1); mechanism preservation as invariance under positive per-client rescalings (Prop. 10), with margins on the weighting→aggregator class (Thm. 8) that **certify mechanism preservation and bound no ASR**; counterexamples across four aggregators of three structural kinds; the negative replicated in a second invariance class and on a second dataset and architecture; and, reported as a refutation, the collapse of the cross-arm admission ordering we had ourselves predicted. What each claim does *not* license is inventoried in Table A2, its evidence tier in Table A1, and the limits of the whole in (L1)–(L6). *Roadmap:* §2 names the preservation levels, §3 proves the boundary, §4 states the protocol it forces, §5 intervenes, and §6 replicates both broken links and asks what a screen then buys.

2 MECHANISM PRESERVATION: INVARIANCE CLASSES

What does it mean for a composition to preserve a defense? Every per-client defense we consider acts as a positive scalar rescaling $T(u_i) = c_i u_i$, which gives the invariance question a sharp *algebraic* answer (Prop. 10), but writing preservation as an invariance of the statistic S leaves a gap our own experiments exposed. Three different things can be preserved: what the downstream defense *reads* (P1–P2), what it *decides* and what *survives* it (P3–P4), and whether the attack is still *suppressed* (P5). They come apart, so we name the levels and ask of each whether it determines the one below:

Level	What is preserved	Determines the level below?
(P1)	the statistic’s value	yes, by definition
(P2)	the ordering it induces	yes, by definition
(P3)	the decision d_2 makes	no , by counterexample (§5)
(P4)	the adversarial contribution admitted	no , by counterexample (§6.2)
(P5)	suppression of the committed attack	—

Both breaks are measured, at fixed d_2 and attack: Krum’s decision flips in up to 80% of rounds with the adversarial mass it admits unchanged (§5), and `cos_krum`’s decision and admitted mass are bit-identical while ASR falls 0.173 (§6.2); Fig. A5 draws the chain and both breaks.

The algebra fixes which level a given composition sits at: cosine similarity is exactly invariant, so (P2) holds for every $\cdot \rightarrow$ FoolsGold composition; coordinate ordering survives only under a bounded-ratio condition; consensus distance and residual magnitude are not invariant at all, so $\cdot \rightarrow$ Reputation compositions fail (P2); norm clipping does not blind FoolsGold, though it does blind Reputation.

Definition 1 (Preservation levels). *For an upstream transform T and a downstream defense d_2 reading statistic S :*

- (P1) **Value invariance:** $S(T(U)) = S(U)$.
- (P2) **Ordering invariance:** $S(T(U)) = \varphi(S(U))$ for some φ preserving the ordering S induces on clients.
- (P3) **Decision invariance:** the client or coordinate d_2 actually selects is unchanged.
- (P4) **Adversarial contribution (admission) invariance:** the adversarial input d_2 admits into its aggregate is unchanged. Admission is the pre-registered term throughout for how much adversarial input survives d_2 .
- (P5) **Attack-suppression preservation:** ASR stays below threshold, for the fixed committed attack the arm is run against.

We target (P3) $\nrightarrow$ (P4) in §5 and (P4) $\nrightarrow$ (P5) in §6.2. The mechanism conditions the levels induce are in App. B and C.

(P4) is a level, not one measurable quantity, and we do not claim it is one. Each rule admits adversarial input through whatever its own machinery exposes: Krum’s and `cos_krum`’s selected-client indicator, `coord_median`’s selected-coordinate share, reputation’s adversarial weight share.

So our (P4) result is four within-family statements, not one cross-defense quantity: in each family, that family’s own admission instrument moves (or provably cannot move) without suppression following. The instruments are not commensurable across families and we never difference or rank across them; that is also why App. E.2’s influence column is aggregator-specific and why Table 1 does not order the arms. A single cross-defense admission quantity would be a stronger claim and we do not have one.

What the theory delivers: supporting machinery, not the claim. For a per-client *weighting* into a *rank-based* or *geometric* aggregator, Thm. 8 (App. C.2) gives sufficient conditions for d_2 ’s ordering, levels (P1)–(P3), to survive: for **rank aggregators**, a count majority plus a *signed* coordinate-wise ordering with $\rho < r^*$ against the separation margin, **magnitude alone not sufficing** without the common-sign assumption; for **geometric median / RFA**, a separation condition $\Delta'_{\text{sep}} > 2(R'_B + \delta')$. **Norm-equalizing** d_1 (NormClip with τ below every client norm) is not a bounded reweighting at all: its induced ratio is unbounded, any σ_2 keyed on update norm or consensus distance goes constant across clients, and **(P2) fails**. The theorem **certifies mechanism preservation and bounds no ASR**, constraining the *aggregate* adversarial coefficient only through Cor. 2: it is not a sufficient condition for suppression and is not the paper’s claim. Two facts about it matter downstream and are why ρ alone mispredicts: the conditions are safe at *both* extremes of the weight ratio, and **the margin is what heterogeneity closes**, RFA’s collapsing from 3.24 to 1.01 at Dirichlet $\alpha=0.1$, the limit at which the bound goes vacuous (L5).

3 THE TESTABILITY BOUNDARY

Why can we not simply compare many defense pairs? The natural test keeps pairs whose constituents already suppress the attack and asks whether the composition preserves it. That is exactly where identification is lost, and it is this paper’s central claim, so we state it rather than pointing at it.

Proposition 1 (Identification: constituent-effect contrasts cannot identify composition effects under inherited-effect gating). *Let a screen certify $(d_1 \rightarrow d_2)$ against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$ holds, determined by the two defenses’ definitions alone. Suppose the screen is applied only where the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P ’s contribution to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set occupies at most two, and **the cell needed to separate P from d_2 ’s standalone effectiveness is empty by the gate, for every $\mathcal{D}$** . No contrast varying d_2 identifies P ’s contribution. Identification requires holding d_2, a and therefore E fixed while intervening on the upstream transform: **the admissible designs are within-defense**.*

The proof is three lines and is in App. C.1: the load-bearing condition excludes d_1 from attaining the minimum in (i), so certification forces $M(d_2, a) < \theta$ and both cells with $M(d_2, a) \geq \theta$ go unpopulated whatever $\mathcal{D}$ is. Four cells follow, and that design never fills two of them:

Mechanism condition	d_2 suppresses a alone	it does not
P holds (preserved)	observed: every certified pair	empty by the gate
$\neg P$ (destroyed)	observed: predicted-FAIL pairs	empty by the gate

Reading down a column is the contrast that would identify P ’s contribution, and **the only column in which d_2 is not already doing the work is empty**, for every menu a screen of this form could range over: excluded by design, not undersampled: more data cannot fill a cell the design never permits. The gate cannot be dropped either, because without it the screen’s inherited-suppression premise is vacuous: it would certify pairs neither of whose constituents suppresses the attack. So wherever our conditions certify a pair, a low composition ASR is attributable to d_2 ’s standalone strength rather than to preservation. Equivalently, this is selection on a collider (Elwert & Winship, 2014; Pearl, 2009): the gate is a descendant of both P and d_2 ’s standalone effectiveness, so conditioning on it makes them co-vary. The hazard is standard; the gate reading as hygiene rather than as the step that removes identification is not, and we made that mistake ourselves (§6.1).

Read forwards the boundary is *constructive*, and §4 is what it leaves. A second boundary bounds not what such a screen can be *validated* on but what it can *find*.

Proposition 2 (Coverage: inherited-suppression screens cannot certify emergent synergy). *Call $(d_1 \rightarrow d_2)$ ’s suppression of a **emergent** if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent composition satisfies the eligibility gate of Prop. 1. Hence, writing $\mathcal{S}$ for the compositions with $M < \theta$ and $\mathcal{E} \subseteq \mathcal{S}$ for the emergent ones, any such screen’s recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|\mathcal{S}|$, on every menu and every attack.*

Our own lowest-ASR composition is the witness, which makes this a scope result rather than a defect found by tuning: FG→RFA reaches max-committed ASR 0.093 ($n=30$, CI [0.062, 0.123]) where FoolsGold alone reaches 0.732 and RFA alone 0.845, so it is emergent, it is a false negative of our own criterion, and the bound binds at $|\mathcal{E}|/|\mathcal{S}| = 1/5$: exactly the 40% recall we report. The result is specific neither to our own screen nor to FL, since any screen requiring a constituent to suppress the attack independently inherits it. Screening for *inherited* robustness and searching for synergistic compositions are different problems, and only the first is ours (App. D.3).

4 A CAUSAL EVALUATION PROTOCOL

If the comparison is inadmissible, what replaces it? Not a better criterion but a different design, and Prop. 1 determines it: **do not select across defenses; hold the downstream defense and the attack fixed, intervene on the upstream transformation, and measure the causal channels separately**. Each clause is forced by a term in the proposition rather than chosen for convenience.

- (1) **Hold d_2 and the attack fixed.** *Forced by the gate $E(d_1, d_2, a)$: vary d_2 or a and the two sides of a contrast are conditioned on different events, so they are not comparable whatever the sample size.*
- (2) **Intervene on the upstream transform; do not select a different d_2 .** *Forced by the empty cell: variation in d_2 is exactly what the gate turns into missing support, so the intervention has to live on d_1 .*
- (3) **Pin the adversarial coefficient channel.** *Forced by what the proposition does *not* constrain: neither the gate nor the mechanism condition prevents T from reaching the outcome without passing through d_2 ’s statistic at all, by simply attenuating whichever client carries the poison (Lem. 2). That open path is the confound that inverted our own sign.*
- (4) **Measure the channels, not only the outcome.** *Forced by what P asserts: it is a claim about a mechanism, so a contrast reporting only ASR cannot say which channel carried the effect, and every channel the intervention opens has to be reported, including the ones it opened by accident.*

Operationally, for any composition claim. (i) Name the downstream mechanism and its invariance class. (ii) State *which* preservation level of Def. 1 is being claimed; a claim that does not name its level is not checkable. (iii) Check the upstream transform lies in the class the argument actually covers: one that alters participation, carries cross-round state, or mixes clients across each other is covered by no invariance argument we know of, ours included. (iv) Hold d_2 , the attack and therefore the gate’s input fixed, and **do not select candidates on the outcome**. (v) Intervene on the transform and measure the channels. (vi) Report the level at which the chain broke. Steps (iv)–(v) are what Prop. 1 licenses, and following them turned our own -0.272 into $+0.098$ on an identical cell (§5).

The protocol’s boundary, measured rather than asserted. Every clause above holds the attack fixed. So what the protocol establishes is **attack-suppression preservation under a fixed committed attack**, and nothing wider: an adversary that reoptimizes against the composed pipeline is outside it, which is why we do not call the object “security”. That boundary is measured. Under *one* whitebox composition-aware *adaptive* stress test, which solves the joint FoolsGold/RFA constraint knowing both stages, FG→RFA rises to ASR 0.167 ± 0.069 , $3.3\times$ its seed-matched committed 0.051: **inherited robustness is not adaptive robustness**. That is one adaptive attacker against one composition and we generalize nothing from it; a protocol stated against an adaptive adversary is the sharpest thing this framework lacks (App. L).

Table 1: **The four channels, all four aggregators, under Mode S**; endpoint change $\kappa=0 \rightarrow \kappa=2$. $\Delta \text{agg.}$ is the relative displacement of the aggregate the defense emits, $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$, so it is what enters training and **not** the value of a scoring statistic. $\Delta \text{adm.}$ is the support of the admitted adversarial mass and $\Delta\Lambda_a$ is the separate influence question, the weight the adversary’s direction carries; the two are never interchangeable. The decision/admission dissociation is not one aggregator’s property: it holds for a selector, a cosine-invariant selector, a coordinate-wise order statistic and a weighted averager. The table does **not** order the arms, $\Delta\Lambda_a$ and ΔASR disagreeing in rank, which is this section’s refutation stated in one row-pair, and it is *not* an out-of-sample test. Row 5 replicates row 1 on a second dataset and architecture with its channels measured *before* its ASR existed; row 6 is the pre-registered score-only control, row 1 with the magnitude channel closed, so its decision, admission and influence columns are row 1’s own. Every cell is emitted from frozen artifacts by a single generator (App. E.2).

Aggregator	Kind	$\Delta \text{agg.}$	$\Delta \text{dec.}$	$\Delta \text{adm.}$	$\Delta\Lambda_a$	ΔASR
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	-0.026
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178
Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	-0.425
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098
Krum (FEMNIST)	selector (pairwise distance)	0.842	0.643	0.000	0.000	-0.017
Krum, score-only	selector, magnitude channel closed	0.986	0.733	0.000	0.000	-0.023

5 STATISTIC PRESERVATION IS NOT ATTACK-SUPPRESSION PRESERVATION

Does disturbing what a defense reads actually cost suppression? **The claim: it need not, and the design that says otherwise gets the sign backwards.** Our pre-registered pooled hypothesis (H1), that ASR rises with disturbance of the statistic d_2 reads, is rejected, the effect *negative*: that ladder moved adversarial influence and the statistic together (§6.1). We adjudicate the implication that losing (P3) means losing (P5) by oracle intervention on two arms in two invariance classes, not by a general law.

The instrument, as a counterfactual. The clean question asks what happens when the downstream statistic changes but the adversary cannot become weaker. **Mode S** enforces exactly that: it pins every adversary’s coefficient at $c=1.0$ and spreads only the benign clients over $\rho = e^{2\kappa}$, holding the adversarial coefficient share constant at 0.266667 to within float32 read-back noise, closing Figure 1(a)’s attenuation path, so a rise in ASR here cannot be attenuation. Mode S and Mode A (§6) are causal instruments built from adversary identity: they measure mechanism-level effects rather than emulate deployable defenses (Table A2, App. E.2).

Two readings of preservation, frozen against each other. *The statistic reading:* what matters is disturbance of d_2 ’s statistic, read off its *decisions*. *The admission reading:* what matters is the *adversarial input* d_2 admits. Mode S disturbs Krum’s selection heavily while leaving the mass it admits untouched, so that arm adjudicates (Table A10, the pre-registration’s H-statistic and H-admission): as ρ climbs to 54.6 we find the decision changing in up to 0.80 of rounds while the adversarial mass admitted changes in none of them, and ASR showing no evidence of the predicted increase, remaining inside the pre-specified equivalence margin ($p_{\uparrow}=0.70$, $|\Delta|=0.026 < 0.15$). **The statistic reading is refuted:** statistic and decision disturbance need not change suppression once the adversarial coefficient share is held fixed. *On this arm the zero is a floor:* Krum admits no adversary at any rung, so **the unchanged admission is not by itself decisive here** and the dissociation is read across all four aggregators, each under its own committed attack (Table 1). The floor does not, however, bound the channel decomposition of the supplement (§ S8), **whose outcome is ΔASR and not $\Delta \text{adm.}$** : that experiment decomposes the suppression effect across the channels the dose opens, and needs Krum to admit nobody no more than it needs Krum to admit somebody.

$\Delta \text{adm.}$ is the *support* of the admitted adversarial mass: it never moves in any of the 60 measured rounds of any arm, including the 48 per arm where reputation and `coord_median` carry adversarial mass at baseline. $\Delta\Lambda_a$ is the separate question of how much weight the adversary’s direction carries.

The table contains its own positive control, and it is what makes the negative readable. A design that only ever reports “we moved something and nothing happened” cannot distinguish a real dissociation from an instrument too blunt to move anything that matters. Here the two arms whose *influence* channel does move are the two whose suppression moves with it: reputation, $\Delta\Lambda_a=0.020 \Rightarrow \Delta\text{ASR}=+0.178$, and `coord.median`, $\Delta\Lambda_a=0.033 \Rightarrow +0.098$. Every arm with $\Delta\Lambda_a=0.000$ has $|\Delta\text{ASR}| \leq 0.026$ and sits inside the equivalence margin. So **the instrument does register suppression changes when adversarial influence changes** ($\Delta(P4) \neq 0 \Rightarrow \Delta\text{ASR} \neq 0$), and registers none when it does not ($\Delta(P4) = 0 \Rightarrow \Delta\text{ASR} \approx 0$) *even as the decision changes in up to 0.733 of rounds*. The dissociation is between the decision and suppression, not between our measurements and reality. What the table does **not** support is an *ordering*: $+0.178$ against 0.020 and $+0.098$ against 0.033 rank $\Delta\Lambda_a$ and ΔASR in opposite directions, which is §6.2’s refutation of our own predicted ordering and is why we claim the implication and not the magnitude. `cos.krum`’s -0.425 is the full-ladder endpoint, below our accuracy floor; the witness we rest on is its first, gate-passing rung (0.173 , §6.2); App. E.2 gives the emitter and the per-artifact provenance, and Table 1 here adds the FEMNIST replication and the score-only control.

What the instrument does not hold fixed, and the pre-registered control that closes it. Mode S pins the *adversarial* coefficients and nothing else: the dose also moves Krum’s pairwise distances, *which* benign client it selects, and the **magnitude and geometry of the update actually aggregated**. That third channel is measured: App. E.2’s Δ agg. column, the aggregate the defense *emits*, reads **0.892** here (supplement § S8). **Score-only Krum** closes it: Krum scores the transformed stack and so makes the same selection, but the *untransformed* selected update is aggregated: the decision channel preserved, the magnitude and geometry change removed. Rules were frozen at 35788d9 before any score-only ASR existed, $\Delta > +0.15$ meaning the published result was partly a magnitude artifact. We find $\Delta = -0.023$ ($0.062 \rightarrow 0.039$) against -0.026 uncontrolled on the identical cell, at the same Δ dec. of 0.733 , the refuting threshold 0.212 nowhere approached and every rung clearing the accuracy floor, so the central claim is instrumented against two independent confounds, attenuation and magnitude (App. E.2).

A pre-registered replication that refutes the positive half, and the sign reversal. `coord.median` under the pixel backdoor is *rank-discordant*: of the interpretable arms it has the smallest decision change (0.482) and the largest admission-level change ($\Delta\Lambda_a=0.033$), so the two readings predict opposite rank positions for its rise, and thresholds were frozen against the two published rises (confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$). We find $\Delta = +0.098$, which **refutes the admission ordering**: the arm with the largest admission-level change rose *less* than one with a smaller change. The decision reading does not predict it either ($R^2=0.239$ pooled). What replicates is the negative on a coordinate-wise order statistic: a decision change of 0.482 costs no suppression by the frozen margin, at accuracy 0.698 . Identity ASR here is 0.443 , so equivalence is to the *identity rung*; the rise is positive in $5/5$ seeds (post-hoc): small and consistent, not zero.

The confounded design gets the causal sign wrong. On this identical cell (matched aggregate displacement, matched decision change, matched clean accuracy, Table A3), closing one channel reverses the sign of the measured effect:

Design	The adversarial coefficient	Δ ASR
Outcome-gated ladder (§6.1)	free to attenuate	-0.272 ($p_{\downarrow}=0.0015$)
Mode S intervention (§5)	pinned at $c=1$	$+0.098$

The confounded design does not lose power, it gets the sign wrong. Closing the attenuation channel does not sharpen a weak effect, it **inverts it**: a criterion calibrated against the confounded contrast would have been calibrated against the confound; the boundary of §3 is not a technicality about power. **Downstream-statistic preservation is not, in general, sufficient to establish attack-suppression preservation** (statistic-level non-identifiability, Prop. 11): no predicate of the downstream statistic alone determines the admitted adversarial mass, and none of the admitted mass alone determines suppression: the title claim, by counterexample *within this class* (Table A2).

6 WHAT THE BOUNDARY LICENSES, AND WHAT WE FOUND

Does the pattern hold beyond the adjudicating arm, and what does a screen then buy? We validate on CIFAR-10 with a 4-layer CNN, $N=10$ clients, $f=0.2$ adversarial, two committed attacks (model-scaling, pixel backdoor), at $n=5$ seeds per cell ($n=8$ for `cos_krum`, $n=3$ on FEMNIST); ASR means *max-committed ASR*; the rest of the protocol is fixed in the supplement (§ S5). The core evidence is a seed-matched intervention on one cell, not a mean comparison across cells: what §3 excludes is a cell, not a sample size.

6.1 THE CONFOUNDED LADDER THAT MOTIVATED THE INSTRUMENT, AND THE SIGN IT GOT WRONG

Our earlier suites tried to validate ordering preservation by *choosing a different downstream defense*, the move Prop. 1 rules out. Varying only the *dose* at fixed d_2 and attack ($\kappa=0$ bit-identical to d_2 standalone) refutes that suite’s pooled prediction that disturbance predicts each arm’s rise: $\rho_s = +0.351$, one-sided $p=0.091$ against a frozen $p < 0.05$, and in three of four arms ASR *falls* as the dose rises (80 runs, one arm confirming decisively; Table A9). **The fault is ours and is now measured:** fixing $\text{mean}(c)=1$ leaves relative weights free, so a large spread mostly *dilutes* whichever client carries the poison: attenuation, not statistic disturbance (supplement § S6, § S9).

6.2 WHAT THE INSTRUMENT DOES TO THAT LADDER, ACROSS FOUR AGGREGATORS AND A SECOND DATASET

Most of the one confirmation was attenuation: with the adversary’s share pinned, reputation’s rise at $\rho=54.6$ shrinks from $+0.762$ to $+0.178$, and what remains tracks a small *admission* change, not the large decision change (App. E.2). The class-(a) control refutes us the other way: `cos_krum`’s selection is bit-identical at every rung yet ASR is 0.584 at the identity (it never suppresses) and *falls* 0.173 at the first rung alone, where clean accuracy moves only $0.599 \rightarrow 0.567$ and our accuracy gate passes (one-sided $p=0.044$, 5/8 seeds); across the full ladder it falls 0.425 at rungs below the accuracy floor, so **the gate-passing rung is the witness we rest on**. **Mode A** (payload share at fixed dispersion) separates Thm. 8’s two regimes: confirmed for `coord_median`, refuted for reputation. A *pre-registered* replication on EMNIST-byclass/simple_cnn moves the decision on 0.643–0.714 of rounds with the adversary never admitted, suppression inside the pre-specified ± 0.15 equivalence margin: $\Delta = -0.017$ beside CIFAR-10’s -0.026 , at accuracy $0.767 \rightarrow 0.835$: **replicated** on a second dataset and architecture, never “generalizes” (App. E.2, D).

And what a screen then buys: less than one number suggests. The conditions were developed on the same menu they are scored on, so the only sub-tier that risks falsification is a *prospective* one: three defenses the paper had never used (FLTrust, Krum, Multi-Krum), with a predicted label and mechanism recorded for all 20 pairs and git-committed (262cf35) before any run existed and never revised, carrying the sanity gate our out-of-sample battery lacked (a predicted-FAIL control counting only if the attack clears ASR 0.5 against FedAvg, which both committed attacks do: 0.667 scaling, 0.768 pixel). It scores 13/20, and **we do not offer that as validation of the screen**. FLTrust suppresses both committed attacks alone (max-committed ASR 0.048), the first defense in our menu that does, so all 13 pairs containing it land within 0.030 of FLTrust alone in either slot: seven fall outside the load-bearing scope condition, **which we added retrospectively rather than pre-registering it**, and six more are near-identities, because FLTrust renormalizes every update to $\|g_{\text{srv}}\|$ and so cancels a positive rescaling exactly, which Prop. 10(a) permits no other outcome for. Only the seven FLTrust-free pairs could have surprised us (5/7), and none of those is predicted-LOW, so they bear on specificity alone. Two further pre-registered attempts to validate the screen were defeated by the confound itself, which makes them evidence for Prop. 1 rather than for the screen. **This suite is the paper’s only prospective evidence and its only fully out-of-sample one**, which is why we report both its score and the design flaw that limits it in the same breath rather than in separate places (§D.1 for the suite and the two misses, Table A4, App. D, D, and the supplement § S1 arm by arm).

What this does not establish. The caveats are collected here rather than attached to each claim, so that no claim above is hedged twice and none is hedged nowhere.

- (L1) **The witnesses are narrow even though the identification result is not.** Prop. 1 is a statement about a design and involves no dataset, architecture or scale. Every witness for it here is CIFAR-10 with a 4-layer CNN, $N=10$, $K=5$, $f=0.2$, two committed attacks (model-scaling, pixel backdoor), replicated on EMNIST-byclass/simple_cnn. We ran neither DBA nor Neurotoxin through the intervention, and the score-only control that closes the magnitude channel is a single arm: one aggregator, one attack.
- (L2) **The attack is fixed, by construction.** What we establish is attack-suppression preservation under a fixed committed attack; the protocol has no clause for an adversary that reoptimizes against the pipeline, and the one adaptive test we ran breaks the guarantee it is pointed at (§4).
- (L3) **(P4) is per-family, and on Krum it is a floor.** The four admission instruments are not comensurable, so the (P4) result is four within-family statements. On both Krum arms the adversary is admitted in no round at any rung, so their zeros corroborate the dissociation and cannot establish it alone; reputation and coord_median, which do admit adversarial mass in 48 of 60 baseline rounds each, carry that weight.
- (L4) **Scale is a prediction, not a measurement.** The *pre-registered* PASS predictions hold at $N=100/\text{ResNet18}$ (FG→RFA 0.069) but far from converged accuracy, so they bound nothing about converged training, and **the negative was never measured at that scale.**
- (L5) **The theorem’s conditions are hard to use.** $\rho < r^*$ and $\Delta'_{\text{sep}} > 2(R'_B + \delta')$ are not readable off a (d_1, d_2) definition pair; they need a pilot measurement. Worse, the margin is what heterogeneity closes ($3.24 \rightarrow 1.01$ at Dirichlet $\alpha=0.1$), so the bound goes vacuous in the regime where a practitioner would most want it.
- (L6) **The screen is not validated and we no longer present it as the contribution.** Pooled, disturbance explains little ($R^2=0.239$ over 26 cells); the cross-arm admission *ordering* we predicted is refuted; recall is capped structurally at $1 - |\mathcal{E}|/|\mathcal{S}|$ and observed at 40%; and the one prospective number is dominated by near-identities. That negative space is what the identification result predicts, which is why we report it rather than a screen.

7 CONCLUSION

The finding. Downstream-statistic preservation is not, in general, attack-suppression preservation. An intervention changes a defense’s statistic and decision substantially without changing the adversarial mass it admits or the suppression that follows; conversely a bit-identical statistic and decision coexist with a change in suppression, each direction with a measured witness.

Why. An outcome-gated cross-defense comparison conditions on the same outcome it seeks to explain, so it cannot separate mechanism preservation from standalone effectiveness: a structural identification boundary (Prop. 1) no amount of additional data within that design removes. Within-defense intervention fixes d_2 and the attack instead: on the identical cell **the confounded design does not lose power, it gets the sign wrong.**

The implication. Evaluate composed defenses by measuring the adversarial contribution that survives the downstream mechanism, not by inferring suppression from the statistic it consumes. That object is (P4), which Mode S reads only by knowing adversary identity, so our experiments identify the causal channel rather than screen for it (App. I; supplement § S10). That leaves the question: **can (P4) be estimated without knowing which clients are malicious?**

ETHICS STATEMENT

This paper studies poisoning attacks in order to evaluate defenses against them, so its subject matter is dual-use. Three things bound the risk. Every attack we run is already published (model-scaling (Bagdasaryan et al., 2020), pixel-pattern backdoors (Chen et al., 2017), DBA (Xie et al., 2020)) and we introduce no new attack or new attack capability. All experiments are simulations on public benchmark datasets (CIFAR-10, EMNIST-byclass, UCI Spambase) with no human subjects, no personal data and no deployed system involved. Our headline result is defensive and cautionary rather than offensive: it tells practitioners that a preserved downstream statistic is *not* evidence that a composed defense stack still suppresses an attack, which removes a false assurance rather than supplying an attacker with one. The instruments we build (Mode S, Mode A, score-only Krum)

require knowing which clients are malicious and are therefore measurement devices, not deployable defenses; we state this at each use so that no reader mistakes them for a recommendation.

REPRODUCIBILITY STATEMENT

Every claim in the paper is tagged *general*, *formal* or *empirical* at the point it is made, and Table A2 states the scope conditions on each. For the formal claims, Theorem 8 and Propositions 1, 10 and 2 are stated in the body with their full statements and proofs in the appendix. For the empirical claims: each pre-registered arm has a decision rule git-committed before the corresponding runs existed, with the commit hash recorded in the text and a runner that refuses to start unless its own recorded commit matches; every table cell is emitted by a generator that reads only frozen artifacts rather than being transcribed; sample sizes, seed ranges and per-seed values are given for every arm, with per-seed detail in the appendix and the supplement; and the figures are recomputed per seed by the same scripts. Where a number was revised we report the revision and its cause rather than the corrected value alone, and where a derived condition turned out to be empty on our data we report it as empty rather than omitting it (§C.2). **No code or data package accompanies this submission**, and throughout this paper “the supplement” means the companion PDF only. The analysis code, the pre-registration documents and the frozen result artifacts underlying every table and figure will be released on publication; what the paper itself provides is the pre-registration commit hashes, the frozen decision rules they fix, and per-seed values for every arm, which is what is needed to check the reported numbers against a reimplementation.

AI USE STATEMENT

In this work, we used generative AI tools as a writing and analysis assistant: for prose editing and revision of the manuscript, for L^AT_EX restructuring, and for writing verification and analysis scripts that check the manuscript against the frozen result artifacts. We did not use generative AI tools to generate research ideas, to formulate the hypotheses, to design the experiments or pre-registrations, or to produce any experimental result, and no claim, number, figure or table in this paper was authored by an AI tool. We have reviewed all AI-assisted work: AI-assisted prose was checked against the underlying artifacts sentence by sentence, and AI-written analysis scripts were run against frozen inputs whose outputs are reproducible independently of the tool. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

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A THE THREE TIERS OF EVIDENCE

Three tiers of evidence, of very unequal weight, never pooled. Every empirical claim about the preservation hierarchy or the criterion is tagged with one of these three words at the point it is made, so a reader can look up what licenses it instead of reconstructing it from hedges.

Table A1: The three tiers of evidence, of very unequal weight and never pooled. Every empirical claim in the paper carries one of these three words at the point it is made; the tiers are never combined into a single verdict.

tier	design	what it licenses
Causal	within-defense intervention at fixed d_2 , attack and C1 input: Mode S, Mode A (§E.2)	statements about the <i>mechanism</i> , scoped to these cells
Replication	FEMNIST, second dataset <i>and</i> architecture ($\Delta = -0.017$), and coord_median, a second invariance class (+0.098)	“replicated on a second dataset and architecture”, never “generalizes”
Screening	the outcome-gated 42-pair suite	evaluation prioritization only ; by Proposition 6 it cannot identify the mechanism’s contribution at <i>any</i> sample size

B PRELIMINARIES AND THE PRESERVATION LEVELS

Federated learning. N clients each hold a local dataset $\mathcal{D}_i \sim P_i$; the server learns $w^* = \arg \min_w \sum_i \frac{|\mathcal{D}_i|}{|\mathcal{D}|} F_i(w)$. In FedAvg (McMahan et al., 2017), each round the server broadcasts w^t , selected clients run local SGD, and the server aggregates $w^{t+1} = w^t + \frac{1}{|S_t|} \sum_{i \in S_t} \Delta_i^t$.

Poisoning and robust aggregation. An adversary controlling a fraction f of clients can flip labels (Chen et al., 2017), inject backdoor triggers (Bagdasaryan et al., 2020; Xie et al., 2020), or scale malicious updates to dominate aggregation (Bagdasaryan et al., 2020). The server may replace FedAvg with a Byzantine-robust rule: Krum (Blanchard et al., 2017), Trimmed Mean, Coordinate-wise Median, Norm Clipping (Sun et al., 2019), or RFA (geometric median) (Pillutla et al., 2022), each imposing an accuracy/robustness trade-off.

Game theory. A normal-form game is $G = (\mathcal{N}, \{\mathcal{S}_i\}, \{u_i\})$; an equilibrium is *pure* when every player’s strategy puts all its mass on one action and *mixed* otherwise. **No claim of this paper is game-theoretic:** this is needed only to read the residual scale check of Appendix M.1, and the type-space and Bayes-Nash machinery an earlier version used is not needed here.

C THE SCREEN, THE IDENTIFICATION BOUNDARY, AND THE PROOFS

The paper’s central practical question is *which* defense pairs compose well under deterministic joint application. Specifically, the criterion screens for *mechanism-preserving composition*: when one constituent defense is already individually effective (C1), can the other defense be added without breaking that suppression? This is the operationally relevant question (the criterion is not a general characterization of all composable pairs, but a sufficient condition for *mechanism preservation* among pairs where at least one component has known effectiveness), never a sufficient condition for low ASR. We answer it first, with a checkable criterion, and treat the alternative, temporal mixing, as outside the scope of this paper. **But the practical question is not what the paper establishes:** what it establishes is that preservation of the statistic the screen checks is not causally tied to the security the screen predicts, so the criterion below should be read as the application §D.3 scopes it to (an ordering over an evaluation budget) and not as a deployment recommendation.

Observe that each robust aggregation defense d operates through an *admission signal* $\sigma(u)$ applied to client update u : it accepts, clips, or down-weights u based on whether $\sigma(u)$ exceeds a threshold. Crucially, the admission signal varies by defense family:

- **Weighting defenses** (reputation, FoolsGold): $\sigma(u)$ = distance from consensus (L2) or pairwise similarity.
- **Clipping defenses** (NormClip): $\sigma(u) = \|u\|_2$.
- **Rank-based defenses** (TrimmedMean, CoordMedian): $\sigma_k(u)$ = coordinate-wise rank of u_k .
- **Geometric defenses** (RFA): $\sigma(u)$ = distance from geometric median.

When two defenses are applied sequentially (d_1 transforms the updates, d_2 aggregates the result), d_1 's transformation may *corrupt* d_2 's signal or vice versa.

Criterion 3 (Signal-Preservation Composability). *Let d_1 and d_2 be defense mechanisms. The following sufficient conditions guarantee that composition ($d_1 \rightarrow d_2$) suppresses attack a . They are of two different kinds, and we group them accordingly, because conflating an empirical screen with a mechanistic property is the main source of confusion about what the criterion claims.*

Stage 0: applicability. *Before either kind of condition is informative, the question the criterion asks has to be live.*

(C0) The downstream mechanism is load-bearing. *ASR(d_1, a) ≥ 0.5 for at least one committed attack a . C2 and C3 concern preservation of d_2 's admission signal, so they say nothing when d_1 alone already suppresses everything: the composition inherits that suppression, destroying d_2 's signal costs nothing, and a correct prediction would be correct for a reason the criterion did not supply. The asymmetry is deliberate: a strong d_2 is the intended case, since asking whether upstream processing breaks an effective downstream defense is the framework's whole question, whereas a strong d_1 makes the question moot. C0 was **formulated post-hoc** and its provenance is in Remark 5.*

Mechanism-preservation conditions. *These are what Theorem 8 proves for the weighting→aggregator class, and they are determined by d_1 's and d_2 's definitions alone, no experiment required. What we recommend is an evaluation-prioritization criterion, C1∧C2, and we name it that way rather than a "screen" because that is the whole of what our evidence supports. We disclaim predictive validity explicitly, here at first use: the criterion orders candidate compositions for evaluation, and it does not predict whether a composition is secure. The pooled two-channel model over 26 cells is unconfirmed ($R^2=0.239$), and our own pre-registered replication **refuted** the cross-arm admission ordering the criterion would need in order to predict (§E.2); recall is 40% (Proposition 7), so the negative direction carries no information either. It is never a security screen, and a PASS is an evaluation ordering rather than a deployment decision (Table A2). C3 is retained as a margin diagnostic rather than a gate, because on our data it never binds on a pair C1∧C2 already admits and shows no incremental predictive value (§D.3). The pre-registered labels of §D were frozen with C3 in the gate and are not relabelled; the demotion is a recommendation going forward, supported by a post-hoc factorial ablation, and it changes no reported number: C1∧C2 selects exactly the same two in-distribution pairs.*

(C2) Signal preservation. *d_1 's per-client transformation T_1 does not destroy d_2 's admission signal: $\sigma_2(T_1(u_a)) \approx \sigma_2(u_a)$ for malicious updates u_a .*

(C3) Boundary preservation. *d_1 's transformation does not compress the adversarial–benign separation below d_2 's discrimination threshold. Formally: if d_2 suppresses a by detecting $\tau_{d_2}(u_a) > \theta$ (triggering trimming/rejection), then $\tau_{d_2}(T_1(u_a)) > \theta$ must hold, i.e., d_1 must not map adversarial updates from above to below d_2 's rejection boundary.*

When C1 holds but C2 fails (signal destruction), composition is no better than d_2 alone. When C1 and C2 hold but C3 fails (boundary compression), d_2 retains its admission signal direction but its threshold no longer fires: the attack passes because d_1 's rescaling places adversarial magnitudes within d_2 's acceptance region.

Writing C2 as preservation of d_2 's admission signal elides a distinction our own experiments later force, so we name the levels before relying on the condition. They are the paper's main conceptual instrument, and every claim below is stated at one of them.

Definition 2 (Preservation levels). *For an upstream transform T_1 and a downstream defense d_2 reading admission signal σ_2 :*

- (P1) **Value invariance:** $\sigma_2(T_1(U)) = \sigma_2(U)$.
- (P2) **Ordering invariance:** $\sigma_2(T_1(U)) = \varphi(\sigma_2(U))$ for some φ preserving the ordering σ_2 induces on clients. This is the level C2 is stated at (made explicit in (2)); the distinction from (P1) matters, because exact values are sufficient but never necessary.
- (P3) **Decision invariance:** the client or coordinate d_2 actually selects is unchanged.
- (P4) **Adversarial contribution (admission) invariance:** the adversarial input d_2 admits into its aggregate is unchanged. Admission is the term we use throughout for how much adversarial input survives

d_2 , and it is a **level, not one measurable quantity**: for a selector it is the indicator that an adversarial update is selected at all; for a weighted averager, the adversarial share of the weight mass; for a coordinate-wise order statistic, the adversarial share of selected coordinates, which is why Table A11’s influence column is aggregator-specific by necessity. The committed pre-registrations say “admission”, so that word is retained in every frozen rule name and no scored quantity is relabelled.

(P5) **Attack-suppression preservation**: ASR stays below threshold.

(P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) **hold, and hold by definition**: equal values induce the same ordering, and a rule reading only that ordering cannot decide differently. **These three are logical implications under the definitions above, not empirical claims about how defenses behave**, and nothing is asserted by proving them: their content is that they stop at (P3). **Below (P3) the chain breaks, and it breaks in both directions**. Both breaks are measured. Holding decision and admission invariance does not deliver suppression: `cos_krum`’s selection is bit-identical at every dose and its admitted adversarial mass is 0.000 throughout, yet its ASR falls 0.173 by the first rung alone through the magnitude channel, at $p=2.72$, where clean accuracy moves only $0.599 \rightarrow 0.567$ and our accuracy gate passes (paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds). Across the whole ladder it falls 0.425, but that endpoint sits below our own accuracy floor, so **the gate-passing rung is the witness we rest on** (§E.2). Losing decision invariance does not cost suppression: Krum’s selection flips in up to 80% of rounds under an intervention that pins the adversarial mass, and its ASR stays inside the pre-specified equivalence margin ($|\Delta|=0.026 < 0.15$) (§E.2). A criterion stated at (P1)–(P2) therefore constrains security in neither direction. **(P4) is where the security content sits, but we do not claim it is sufficient for (P5), and it is a level rather than one universal scalar security quantity**. It is instantiated per aggregator by the measure that aggregator’s own rule exposes, Krum’s and `cos_krum`’s selected-client indicator, `coord_median`’s share of selected coordinates, reputation’s adversarial weight share (`experiments/build_channel_table.py`), so **(P4) names what to measure, not a single number comparable across defenses**. It is the first level that refers to adversarial influence on the aggregate rather than to the downstream defense’s internal decision, and `cos_krum` is our own witness that (P4) $\nRightarrow$ (P5): admission identically 0.000 at every rung while suppression moves. **So the hierarchy answers “which level does a security claim have to be stated at?” and not “which level predicts ASR?”**: no level in it is offered as a predictor.

Remark 4 (Which level C2 asserts, and which one is load-bearing). At $\rho = 54.6$ the dose ladder of §E changes Krum’s decision in 86.7% of rounds while changing its admission of adversarial input in 13.3%, and Krum’s suppression does not deteriorate at all. A selection that flips from one benign client to another disturbs the statistic maximally and the mechanism not at all. **Terminology, once and for all: “mechanism” always means the downstream defense’s discriminative operation (what separates adversarial from benign updates) and never its security behaviour, since that implication is precisely what this paper refutes**. We reserve “mechanism preservation” for the formal property Theorem 8 establishes, saying statistic or decision preservation where only the weaker property holds. C2 above is the (P2) condition (which is what makes it checkable from definitions alone) but the load-bearing reading is (P4). **This is the paper’s central tension: our theorem operates at (P1)–(P3), while the experiments indicate security behaviour is decided at (P4)**; §E.2 puts the two against each other in an experiment where they predict opposite outcomes, and Figure A5 is that argument in one picture.

Remark 5 (C0’s provenance: an applicability condition formulated post-hoc). C0 sits in Criterion 3 as a stage-0 applicability condition because Lemma 1 is stated over $C0 \wedge C1$, but it **does not have the same evidential status as C1–C3**, and we record that here rather than let its promotion obscure it. It is **vacuous on the 7-defense menu of §D**, where the strongest single defense reaches only 0.519, and we formulated it only after the prospective suite of §D.1 introduced a defense for which it binds. It is a lesson from prospective evaluation, applied there retrospectively; it is not part of the criterion validated on the 42 pairs, and none of the accuracy figures in §D depend on it.

A scope condition on the criterion itself, not a limitation of our runs. C1 and every prediction built on it are defined against a *committed* adversary, one whose attack is fixed before the composition is chosen. An adversary who observes the composition can re-target whichever channel the screen certifies, so **what the screen bounds is inherited robustness under a fixed attack, never robustness against an adaptive one**. Our single whitebox composition-aware stress test is consistent with that reading ($0.051 \rightarrow 0.167$ seed-matched, $3.3\times$ the committed-attack ASR; §F); **its role is not to measure adaptive robustness but to show that an inherited-robustness guarantee is not adaptive robustness**, and a screening condition stated against an adaptive adversary is left open.

C.1 THE TESTABILITY BOUNDARY

C0 and C1 together constrain what any validation of this criterion can look like. We did not anticipate this; it turns out to explain the central weakness of our own prospective suite (§D.1, L7), and (once stated in general form) to *derive* the class of within-defense designs in which C2’s contribution is identifiable (§E). Its content is narrower than “a general identification boundary” and worth stating precisely: **under inherited-effect gating**, cross-defense contrasts cannot identify the mechanism’s contribution, because the gate conditions on a descen-

Table A2: **Scope of claims: three kinds, and the one place each scope condition is stated. Claims do not transfer upward:** the formal guarantee does not certify the general framing, and the empirical figures certify neither level above them. This table is the conservation record for every caveat in the paper; §H restates the same conditions individually as (L1)–(L6).

kind	content	status, and what it does <i>not</i> license
General	An evaluation gated on the outcome it predicts cannot identify the mechanism’s contribution, and admissible designs are within-defense; an inherited-suppression gate cannot certify emergent synergy (Prop. 6, Prop. 7).	Proved, and not specific to FL, to backdoors, or to our conditions. Licenses no claim that any composition <i>is</i> secure, and none that mechanism preservation implies suppression. Recall is bounded, not achieved: 40% here, from $ \mathcal{E} / \mathcal{S} =1/5$ (L5, L7).
Formal	Mechanism preservation as <i>invariance</i> under positive per-client rescalings $T(u_i) = c_i u_i$, $c_i > 0$, into specific rank and geometric aggregators, with explicit margins (Thm. 8, Lem. 2, Cor. 2, Prop. 10, Prop. 9).	Proved on that class only: a <i>laboratory</i> for composition, not a model of FL defenses in general. A d_1 that alters participation, carries cross-round state, mixes clients, or aggregates first admits no such T and is covered by nothing here . Bounds no ASR. C0 was formulated post-hoc and applied retrospectively to arms frozen before it (Rem. 5, L7). Regime A’s condition is not operationally checkable, S not being a stable per-attack constant (L5). Cor. 2’s bound goes vacuous as $m \rightarrow 1$, which heterogeneity realizes.
Empirical	§E.2’s interventions and §D–§D.1’s 42-pair menu: CIFAR-10, CifarCNN/ResNet18, $N \in \{10, 100\}$, $K \in \{5, 10\}$, $f \in [0.05, 0.6]$, Dirichlet $\alpha \in \{0.1, 0.5\}$, star topology , three attack families; one arm on FEMNIST/simple_cnn.	Measured at $n=5$ seeds per cell ($n=8$ for cos_krum, $n=3$ on FEMNIST, $n=30$ for the FG→RFA characterization), all fixed before the runs, so intervals are wide and no null is read as flatness without a pre-registered equivalence margin. ASR is thresholded at 0.5, the level at which the trigger fires on the majority of triggered inputs; a cell below the 0.35 clean-accuracy floor is <i>uninterpretable</i> , not suppressed. Dataset transfer is tested on the flagship arm alone: “replicated on a second dataset and architecture”, never “generalizes” (L1). “Refuted” always means a pre-registered <i>directional</i> hypothesis was rejected against a rule frozen for it, never that a mathematical claim was disproved. The $\gamma \approx 0.66$ boundary is architecture-conditional (L3, L6); the strongest development-set figures are <i>fit</i> , not prediction (L2).
Both instruments are oracles	Mode S and Mode A build c_i from adversary identity, as does Δ_c (§E.2, supplement §S9).	They identify a causal channel and are not screens: no deployed defense knows which clients are adversarial, and estimating (P4) without that knowledge is the open problem of §I. Admission is <i>a direction to measure</i> , not a validated predictor: the pooled two-channel model is unconfirmed over 26 cells ($R^2=0.239$) and its cross-arm ordering was refuted by our own replication (L6). C1 and everything built on it assume a committed adversary, so what is bounded is <i>inherited</i> robustness under a fixed attack, never adaptive robustness .

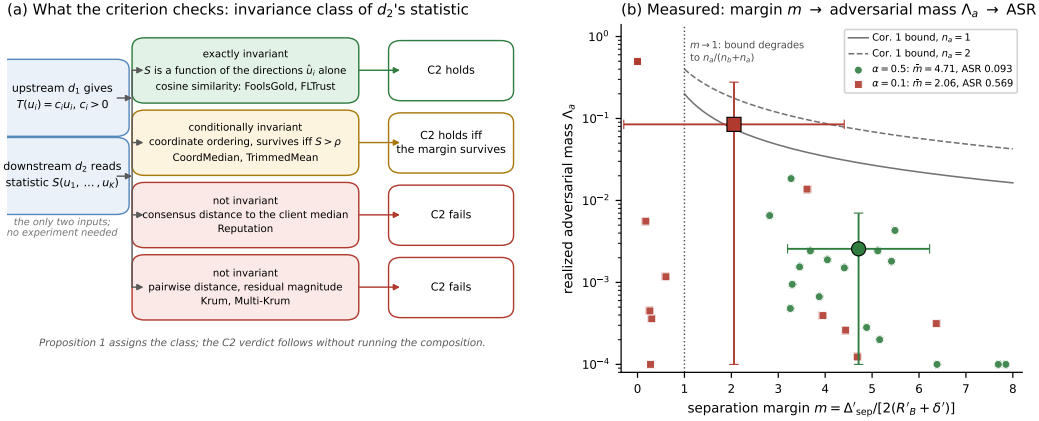


Figure A1: **The criterion is a mechanism test, and the mechanism is measurable.** (a) What C2 checks: the upstream transform T and the statistic S that d_2 reads determine an *invariance class* (Proposition 10), and the class determines the verdict: no composition needs to be run. This is why norm clipping does not blind FoolsGold but does blind Reputation. (b) The measured chain for $\text{FG} \rightarrow \text{RFA}$, per round, on the transformed points RFA actually receives. Each marker is one round with $w_a > 0$ (rounds where FoolsGold zeroes every adversary are Lemma 2 and carry no payload); large markers are mean $\pm$ s.d. Corollary 2's bound holds in every round, and the $\alpha=0.1$ collapse is visible as a collapse in the theorem's own quantity: m falls from 4.71 ± 1.51 ($n=17$ rounds) to 2.06 ± 2.35 ($n=12$), with several rounds at $m < 1$ where the bound degrades to $n_a / (n_b + n_a)$, and realized Λ_a rises from 0.003 to 0.085 (max 0.50) as composition ASR goes $0.093 \rightarrow 0.569$. Values below 10^{-4} are drawn at 10^{-4} .

dant of both the mechanism and the outcome. What we offer as the transferable contribution is not that negative but **the admissible design it derives**, set out below and run in §E.2.

Lemma 1 (Testability boundary). *Let $(d_1 \rightarrow d_2)$ satisfy C0 and C1, and let a^* be a committed attack witnessing C0, so $\text{ASR}(d_1, a^*) \geq 0.5$. Then $\text{ASR}(d_2, a^*) < 0.5$: the downstream defense suppresses a^* on its own.*

Proof. C1 at a^* gives $\min_{d \in \{d_1, d_2\}} \text{ASR}(d, a^*) < 0.5$. By C0 the minimum is not attained at d_1 , hence it is attained at d_2 . $\square$

Nothing in the argument is specific to federated learning, to backdoors, or to our defense menu. Lemma 1 is **the enabling observation for the FL instance**; the substantive claim is the general one, and we state it as such.

Proposition 6 (Why constituent-effect contrasts cannot identify composition effects under inherited-effect gating). *Let a screen certify $(d_1 \rightarrow d_2)$ against attack a only if both (i) an **eligibility gate** $E(d_1, d_2, a) := \min_{d \in \{d_1, d_2\}} M(d, a) < \theta$ holds, where M and θ are the same outcome metric and threshold the screen predicts for the composition, and (ii) a **mechanism condition** $P(d_1, d_2)$ holds, determined by the two defenses' definitions alone. Suppose the screen is applied only where the downstream mechanism is load-bearing, $M(d_1, a) \geq \theta$. Consider estimating P 's contribution to $M(d_1 \rightarrow d_2, a)$ from contrasts across a defense menu $\mathcal{D}$. Then every certified pair satisfies $M(d_2, a) < \theta$, so of the four cells of $\{P, \neg P\} \times \{M(d_2, a) < \theta, M(d_2, a) \geq \theta\}$ the certified set occupies at most two, and **the cell needed to separate P from d_2 's standalone effectiveness is empty by the gate, for every $\mathcal{D}$. No contrast varying d_2 identifies P 's contribution. Identification requires holding d_2, a and therefore E fixed while intervening on the upstream transform: the admissible designs are within-defense.***

Proof. The load-bearing condition excludes d_1 from attaining the minimum in (i), so certification forces $M(d_2, a) < \theta$; both cells with $M(d_2, a) \geq \theta$ are therefore unpopulated whatever $\mathcal{D}$ is, and P is never observed against a downstream defense that fails on its own. Any estimator of P 's contribution from such contrasts is unidentified. $\square$

A second boundary: what such a screen cannot find. The gate that limits what a screen can be *validated* on also limits what it can *cover*, which converts the most obvious objection to our criterion, that its recall is poor, into a scope result.

Table A3: **The confounded ladder against the Mode-S instrument, on the one arm that has both.** Same downstream defense, same attack, same seeds, same nominal dose, and the two designs share their $\kappa=0$ cell *numerically* (mean and worst per-seed gap $0.00\text{e}+00$, asserted at 10^{-12}). At the top rung they are matched on aggregate displacement, decision change and mean clean accuracy, so **the ladder’s fall is not an accuracy collapse**; they differ chiefly in the **adversarial coefficient channel**, and the ASR effect **changes sign**. **The one non-match, disclosed:** aggregate adversarial influence Λ_a moves in *both* designs, 0.081 against 0.033 ($2.4\times$), so neither column is a clean statistic-only intervention, which is why the contrast is “differ chiefly in the coefficient channel”, never “only in it”. *Vocabulary:* this row is the $\Delta\Lambda_a$ column of the channel table (the adversarial share of selected coordinates, the same field in the same artifact) and **not** its Δ adm. column, which counts rounds where the *support* of the adversarial mass changes and reads 0.000 on every row: what moves here is how much weight the adversary’s direction carries, not whether it enters at all. This is a two-column contrast on one arm: it locates where the two designs differ, and attaches no p -value to the difference between the columns. Every cell recomputed from per-seed and per-round rows by `build_comparability_table.py`; nothing here is retyped.

	Confounded ladder dose_kappa	Instrument (Mode S) doseS_kappa
downstream d_2	coord_median	coord_median
attack	committed pixel	committed pixel
seeds	42–46	42–46
nominal dose	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6	κ 0.0 $\rightarrow$ 2.0, ρ 1 $\rightarrow$ 54.6
benign coefficients	mean 1 over <i>all</i> clients	mean 1 over <i>benign</i> clients
adversarial coefficients	drawn from the same spread	pinned $c = 1.0$ exactly
adv. coefficient share	0.267 $\rightarrow$ 0.187	0.266667, constant
Δ_c	+0.000 $\rightarrow$ -0.336	0 by construction (read back 9e-08)
aggregate displacement	0.000 $\rightarrow$ 0.722	0.000 $\rightarrow$ 0.670
decision change	0.000 $\rightarrow$ 0.487	0.000 $\rightarrow$ 0.482
adversarial influence $\Delta\Lambda_a$	0.000 $\rightarrow$ 0.081	0.000 $\rightarrow$ 0.033
mean clean accuracy	0.767 $\rightarrow$ 0.690	0.767 $\rightarrow$ 0.698
ASR	0.443 $\rightarrow$ 0.170	0.443 $\rightarrow$ 0.541
ASR effect	-0.272 ($p_{\downarrow} = 0.0015$)	+0.098 ($p_{\uparrow} = 0.069$)

Proposition 7 (Inherited-suppression screens cannot certify emergent synergy). *Call $(d_1 \rightarrow d_2)$ ’s suppression of a **emergent** if $M(d_1 \rightarrow d_2, a) < \theta$ while $\min_{d \in \{d_1, d_2\}} M(d, a) \geq \theta$. No emergent composition satisfies the eligibility gate of Proposition 6. Hence, writing S for the compositions with $M < \theta$ and $\mathcal{E} \subseteq S$ for the emergent ones, any such screen’s recall obeys $\text{recall} \leq 1 - |\mathcal{E}|/|S|$ on every menu and every attack.*

Proof. Emergence is the negation of the gate, so each member of $\mathcal{E}$ is rejected while belonging to S : a false negative. The bound is the resulting recall. $\square$

Mechanism–effectiveness confounding: Proposition 6 in our instance. The boundary has an uncomfortable consequence for validation, and it is worth spelling out concretely. In *every* in-scope pair the criterion certifies, d_2 already suppresses the attack that d_1 fails. A certified composition’s low ASR can therefore always be attributed to d_2 ’s standalone strength (C1) rather than to d_1 having preserved d_2 ’s mechanism (C2, C3): the two are confounded **by the conditions themselves**, not by an unlucky choice of defense menu. We name this **mechanism–effectiveness confounding** and offer it as a methodological observation about validating composition criteria of this shape, applicable beyond our own experiments.

In particular it rules out the experiment we previously named as decisive: a d_2 whose admission signal is invariant under positive rescaling but which is *weak in isolation* cannot satisfy C1 in scope, so no such certified pair exists to test. The achievable substitute is **per-attack complementarity**: a d_2 that suppresses the attack d_1 fails while remaining weak on a *different* committed attack, so its max-committed ASR stays high and it cannot suppress the composition single-handedly. Reputation is such a defense (0.017 on model-scaling, 0.842 on the pixel backdoor), and §D.2 builds the test on that observation.

Argument sketch. (C1 sufficient) If at least one of d_1, d_2 individually suppresses a , composition inherits that suppression provided $C2 \wedge C3$ preserve the signal chain. C1 is sufficient, not necessary: two individually-weak defenses may suppress synergistically through complementary mechanisms (e.g., partial downweighting + geometric rejection), but predicting such synergy requires the quantitative margins below. The criterion is therefore *conservative*; it may reject working compositions but will not approve broken ones.

(C2 role) Suppose d_2 individually suppresses a via signal $\sigma_2(u_a) > \theta$. If d_1 's transformation T_1 destroys this signal ($\sigma_2(T_1(u_a)) \not\approx \sigma_2(u_a)$), d_2 cannot detect the attack. Example: NormClip as d_1 equalizes all norms to $\leq \tau$, so reputation's distance signal $\sigma_2(u) = \|u - \hat{\mu}\|$ collapses; adversarial and benign updates become equidistant from consensus. This is measured, not hypothetical: NormClip $\rightarrow$ reputation leaves model-scaling ASR at 0.807 ± 0.048 and the pixel backdoor at 0.842 ± 0.047 over 5 seeds, against reputation alone at 0.017 (per-seed values in §L.1). Quantitatively: C2 holds when relative signal distortion $\Delta_{\text{signal}} := |\sigma_2(T_1(u_a)) - \sigma_2(u_a)| / \sigma_2(u_a) < \delta_2$ for a defense-specific tolerance δ_2 .

(C3 role) Even when d_1 preserves the *direction* of d_2 's signal (C2 holds), it may compress magnitudes below d_2 's threshold. Example: FoolsGold as d_1 downweights adversarial clients by $\alpha \ll 1$. TrimmedMean's coordinate-wise rank signal still distinguishes adversarial coordinates (C2 holds), but after rescaling by α , per-coordinate values $\alpha \cdot u_{a,k}$ fall within the trimming range: the coordinate-wise extreme that would trigger trimming is no longer extreme. Quantitatively: C3 holds when the margin ratio $\Delta_{\text{margin}} := (\tau_{d_2}(T_1(u_a)) - \theta) / (\tau_{d_2}(u_a) - \theta) > 0$, i.e., adversarial scores remain above d_2 's rejection boundary after d_1 's transform.

(Sufficiency) If C1 holds, C2 holds ($\Delta_{\text{signal}} < \delta_2$), and C3 holds ($\Delta_{\text{margin}} > 0$), then d_2 detects and suppresses u_a even after d_1 's transformation. For the weighting $\rightarrow$ aggregator class, Theorem 8 instantiates these margins explicitly: $\rho < r^*$ for the rank aggregators, and the transformed-point separation condition $\Delta'_{\text{sep}} > 2(R'_B + \delta')$ for the geometric one, which Corollary 1 restates in a form that can be checked without running the composition. $\square$

C.2 A RESTRICTED THEOREM: C2 $\wedge$ C3 FOR WEIGHTING $\rightarrow$ RANK AND WEIGHTING $\rightarrow$ GEOMETRIC COMPOSITIONS

What follows is a boundary theorem, and deliberately not a security theorem. It bounds what an *invariance* argument about composition can establish: preservation of the downstream defense's discriminative mechanism, levels (P1)–(P3) of Definition 2, and it says nothing whatever about suppression, which lives at (P4)–(P5). We state that in the same breath as the theorem because the whole point of the paper is the gap between the two: the theorem is the best case for statistic-level reasoning, proved with explicit margins, and §E.2 then shows that even exact invariance neither implies nor preserves suppression. Read it as a *formal* claim in the sense of Table A2: proved, on one class, bounding no ASR.

The criterion above is an operational heuristic. For the class of compositions that matter most in practice, a per-client *weighting* defense d_1 (reputation, FoolsGold) followed by a *rank-based* or *geometric* aggregator d_2 (TrimmedMean, CoordMedian, RFA), which contains the flagship FoolsGold $\rightarrow$ RFA, C2 and C3 can be *proven* rather than asserted, with an explicit tolerance on how aggressively d_1 may reweight. The key structural fact is that d_2 's admission score is scale-equivariant, so it commutes with d_1 's reweighting up to a bounded ratio; the composition fails only when d_1 compresses that ratio past a margin set by d_2 's separation.

Theorem 8 (Bounded reweighting preserves the downstream discriminative mechanism). *Consider a round with n_b benign and n_a adversarial updates under honest majority $n_b > n_a$. Let d_1 be a per-client reweighting $T_1(u_i) = w_i u_i$ with $w_i \geq 0$, and define the ratio over strictly positive weights, $\rho := \max_i w_i / \min_{i: w_i > 0} w_i$. Suppose d_2 suppresses attack a through a client admission score $\sigma_2(\cdot; U)$ that is scale-equivariant, $\sigma_2(c u_i; c U) = c^k \sigma_2(u_i; U)$ for all $c > 0$ and some degree $k \geq 0$. Assume $w_a > 0$ for adversarial clients; the $w_a = 0$ case is Lemma 2. The theorem establishes preservation of d_2 's suppression mechanism; it does not bound ASR. Then:*

1. **(Rank aggregators; C2 $\wedge$ C3 hold up to a margin.)** If $d_2 \in \{\text{TrimmedMean}, \text{CoordMedian}\}$, so σ_2 is a coordinate-wise rank, then d_2 's per-coordinate suppression of a is preserved when two conditions hold. (i) **Count majority:** $n_a < K/2$ for CoordMedian, and $n_a \leq \lfloor \beta K \rfloor$ for TrimmedMean with β -trimming. (ii) **Signed ordering:** on each suppression-relevant coordinate k , the adversarial transformed value lies strictly beyond every benign one, $w_a u_{a,k} > \max_b w_b u_{b,k}$ (upper tail) or $w_a u_{a,k} < \min_b w_b u_{b,k}$ (lower tail). Sufficient magnitude condition for (ii): if $\text{sign}(u_{a,k})$ is common to all adversaries and $|u_{a,k}| > \rho \cdot \max_b |u_{b,k}|$, then (ii) holds; writing $r^* := \min_k \min_b |u_{a,k}| / |u_{b,k}|$ for the coordinate-wise separation margin, this is $\rho < r^*$. Magnitude alone is not sufficient without the sign assumption: an adversarial coordinate can be large in absolute value yet sit between the benign values. At $\rho = 1$ ranks are preserved exactly, recovering Proposition 9.
2. **(Geometric median / RFA; C2 $\wedge$ C3 hold up to a margin.)** The composition hands RFA the transformed points $u'_i := w_i u_i$ and RFA runs its standard unweighted geometric median on them; it is not a weighted geometric median, and w_i enters only by moving the points. All quantities must therefore be defined on the transformed set. Let μ'_B be the geometric median of the benign u'_b alone, μ' that of all of U' , $R'_B := \max_b \|u'_b - \mu'_B\|$ the transformed benign radius, and $\delta' := \|\mu' - \mu'_B\|$ the displacement caused by including the adversarial points. Suppression is preserved when

$$\Delta'_{\text{sep}} := \min_{a,b} \|u'_a - u'_b\| > 2(R'_B + \delta'),$$

under which every adversarial residual $\|u'_a - \mu'\|$ strictly exceeds every benign residual $\|u'_b - \mu'\|$, so RFA's inverse-residual weight on a stays below the benign floor.

3. (**Norm-equalizing d_1 ; C2 fails: the degenerate case.**) If instead d_1 maps every client norm to a common value (NormClip with τ below all client norms), then any σ_2 keyed on update norm or consensus distance becomes constant across clients, so σ_2 no longer separates a and C2 fails. This is not a bounded reweighting: the induced per-coordinate weight ratio is unbounded, so no ρ -margin applies.

Parts (1) and (2) are sufficient conditions for d_2 to preserve the downstream suppression ordering, not to exclude adversarial contributions outright. Neither is a sufficient condition for low ASR, and part (2) orders residuals per client only; it constrains the aggregate adversarial coefficient solely through Corollary 2.

Proof. (1) TrimmedMean/CoordMedian act on coordinate k through the rank of $w_i u_{i,k}$ among $\{w_j u_{j,k}\}_j$; a is suppressed on k because its transformed coordinate is extreme (trimmed, or not selected by the median). Given (ii), at most n_a of the extreme order statistics are adversarial, so with (i) the selected value is benign. For the magnitude condition, $w_a > 0$ gives $\rho \geq \max_b w_b/w_a$, so $|u_{a,k}| > \rho \max_b |u_{b,k}|$ implies $|w_a u_{a,k}| > w_a \rho \max_b |u_{b,k}| \geq (\max_b w_b) \max_b |u_{b,k}| \geq \max_b |w_b u_{b,k}|$; with a sign common to all adversaries this places $w_a u_{a,k}$ strictly beyond every benign value on that side. Scale-equivariance ($k=0$ for pure rank) makes the uniform case $\rho=1$ exact. (2) All quantities are on the transformed points $u'_i = w_i u_i$, which is what RFA receives. Every benign residual obeys $\|u'_b - \mu'\| \leq R'_B + \delta'$ by the triangle inequality. For adversarial u'_a , $\|u'_a - \mu'_B\| \geq \Delta'_{\text{sep}} - R'_B$, hence $\|u'_a - \mu'\| \geq \Delta'_{\text{sep}} - R'_B - \delta'$. Demanding that this exceed the benign bound gives $\Delta'_{\text{sep}} - R'_B - \delta' > R'_B + \delta'$, i.e. the stated condition. No weighted-majority assumption is needed, and none would be appropriate: RFA weights every input point equally and the w_i act only by relocating the points. Our implementation runs Weiszfeld for 50 iterations at tolerance 10^{-6} , so the argument transfers to it. (3) Clipping to a common τ sends $\|u_i\| \mapsto \tau$ for all i , so $\sigma_2(u_i) = \|u_i\|$ (NormClip-as- d_2) or the consensus distance $\|u_i - \hat{\mu}\|$ collapses to a client-independent constant; separation vanishes, which is precisely C2-failure. $\square$

The condition as stated cannot be checked before running the composition, and that is fixable. Part (2) is stated on the transformed points, and $\delta' = \|\mu' - \mu'_B\|$ is by definition the displacement that including the adversarial points causes: evaluating it requires executing the very composition the criterion is meant to screen. In a screening criterion that is a real defect. It is also removable, because the displacement is controlled by the benign spread and the client counts alone.

Corollary 1 (Checkable separation margin). *In the setting of Theorem 8(2), write $S'_B := \sum_b \|u'_b - \mu'_B\|$ and let $\bar{R}'_B := S'_B/n_b$ be the mean transformed benign residual. Under honest majority $n_b > n_a$,*

$$\delta' \leq \frac{2S'_B}{n_b - n_a} = C_A \bar{R}'_B, \quad C_A := \frac{2n_b}{n_b - n_a},$$

and therefore the separation condition of Theorem 8(2) holds whenever

$$\Delta'_{\text{sep}} > 2R'_B + 2C_A \bar{R}'_B. \quad (1)$$

No quantity in (1) is computed from the composed aggregate.

Proof. μ' minimizes $F(x) := \sum_i \|x - u'_i\|$, so $F(\mu') \leq F(\mu'_B)$. Write $t := \delta'$. By the triangle inequality each benign term obeys $\|\mu' - u'_b\| \geq t - \|u'_b - \mu'_B\|$ and each adversarial term obeys $\|\mu' - u'_a\| \geq \|\mu'_B - u'_a\| - t$, so $F(\mu') \geq n_b t - S'_B + \sum_a \|\mu'_B - u'_a\| - n_a t$. Since $F(\mu'_B) = S'_B + \sum_a \|\mu'_B - u'_a\|$, the inequality $F(\mu') \leq F(\mu'_B)$ gives $(n_b - n_a)t \leq 2S'_B$, which is the displacement bound. Substituting it into $\Delta'_{\text{sep}} > 2(R'_B + \delta')$ yields (1). $\square$

Honest majority is exactly what the geometric case needs, and it enters on the displacement, not on the weight ratio. C_A diverges as $n_a \rightarrow n_b$, so Corollary 1 is the precise point at which $n_b > n_a$ becomes indispensable, while the theorem itself needs no majority assumption on the *weights*. An earlier version of this paper summarized part (2) as a weight-ratio bound $\rho < n_b/n_a$; that summary was wrong, and the correction is not cosmetic, because the majority condition belongs on δ' instead. **Measured at Dirichlet $\alpha=0.5$:** the displacement bound holds in all 17 rounds with $w_a > 0$, and is conservative there by a factor of at least 10.5 (mean 215), because the realized displacement is set by a force balance among the residuals rather than by the worst case a bound must cover. Condition (1) is satisfied in 16 of those 17 rounds: 15/15 at $n_a=1$, where $C_A = 2.67$ and the requirement at the measured mean $\bar{R}'_B/R'_B = 0.429$ is $\Delta'_{\text{sep}} > 4.29 R'_B$ against a measured $\Delta'_{\text{sep}}/R'_B$ never below 5.88; and 1/2 at $n_a=2$, where $C_A = 6.00$ tightens the requirement past what one of the two rounds supplies. **The heterogeneity limit recorded for Corollary 2 applies here too:** at $\alpha=0.1$ condition (1) holds in only 5 of 12 positive-weight rounds. The checkable form certifies less exactly where the condition it replaces also stops binding, and it does not rescue the guarantee there.

A further reduction to ρ alone is vacuous on our runs, and we report it as vacuous. The natural next ask is the analogue of part (1)’s $\rho < r^*$: a threshold on ρ computable from the *pre-composition* round geometry, so that no pilot is needed at all. Such a threshold exists formally. Bounding $\Delta'_{\text{sep}} \geq w_{\min}[\Delta_{\text{sep}} - (\rho - 1)N_A]$ and $R'_B \leq 2w_{\min}[\rho R_B + (\rho - 1)\|\mu_B\|]$, where Δ_{sep} , R_B and μ_B are the untransformed analogues and $N_A := \max_a \|u_a\|$, makes $w_{\min}$ cancel and turns (1) into a linear inequality in ρ with a critical value ρ^* depending only on $(\Delta_{\text{sep}}, R_B, \|\mu_B\|, N_A, n_a, n_b)$. **It is empty on our data.** It requires $\Delta_{\text{sep}} > 4(1+C_A)R_B$, and the measured $\Delta_{\text{sep}}/[4(1+C_A)R_B]$ never reaches 1. Its maximum is 1.015 at $\alpha=0.5$, attained in an $n_a=1$ round where $C_A = 2.67$ makes the requirement loosest, with mean 0.754 over those 15 rounds and a maximum of only 0.389 over the two $n_a=2$ rounds; at $\alpha=0.1$ the maximum is 0.853. Consequently $\rho^* \leq 1$ in 16 of 17 rounds at $\alpha=0.5$ and in all 12 at $\alpha=0.1$, against a measured ρ of at least 1.034 and 1.015 in the two settings respectively: *no* value of ρ satisfies the reduced condition, which is a stronger negative than a merely conservative threshold. The looseness is in the composition of the two bounds rather than in either one separately, and closing it would need a curvature assumption on the benign configuration that we are not prepared to assume. **We therefore claim no pilot-free screen for the geometric class**, and the calibration protocol below prices it accordingly.

What part (2) does not assume. Recorded explicitly, because the step invites readings it does not support: there is no honest-majority assumption on the *weights*, no requirement that adversarial updates lie outside the benign convex hull, and the count majority $n_b > n_a$ is used in part (2) only through Corollary 1, nowhere in the theorem’s own proof.

Theorem 8(2) orders residuals *per client*: each adversarial point receives less inverse-residual weight than each benign point. That alone does not bound the adversary’s *total* share of the aggregate: several adversaries could each be down-weighted yet collectively hold substantial mass. The separation margin controls this, but only quantitatively:

Corollary 2 (Bounded aggregate adversarial influence). *Let $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ be the realized separation margin, so Theorem 8(2) reads $m > 1$. Every benign residual is at most $D_b := R'_B + \delta'$ and every adversarial residual at least $D_a := \Delta'_{\text{sep}} - R'_B - \delta' = (2m - 1)D_b$. Bounding $\sum_j 1/d_j$ below by the benign terms alone and each adversarial term above by $1/D_a$,*

$$\Lambda_a = \sum_a \lambda_a \leq \frac{n_a/D_a}{n_b/D_b + n_a/D_a} = \frac{n_a}{n_b(2m - 1) + n_a},$$

so μ' is a convex combination whose adversarial share is at most Λ_a : writing $\mu' = (1 - \Lambda_a)\bar{u}_B + \Lambda_a\bar{u}_A$ for its benign and adversarial parts, the displacement from the benign-only combination $\bar{u}_B$ is at most $\Lambda_a\|\bar{u}_A - \bar{u}_B\|$.

The bound is only as good as the margin, and **the separation condition alone is not enough**: as $m \rightarrow 1$ it degrades to $n_a/(n_b + n_a)$, permitting the adversary 40% of the mass at $n_a=2, n_b=3$. A margin comfortably above 1 does the work. We measure $m \geq 2.82$ in every round with $w_a > 0$ (mean 4.71 over those rounds), capping the adversarial share at 5.1% for $n_a=1$ and 12.6% for $n_a=2$. The realized figure is far tighter than the bound: measuring Λ_a directly, it never exceeds 1.8% in any round, and the bound is satisfied in all 17 measured rounds with $w_a > 0$. With Lemma 2 this completes the picture, and the mechanism survives for *opposite* reasons: at $w_a=0$ the adversaries hold most of the Weiszfeld mass ($\Lambda_a = 0.71$ measured) but carry no payload; at $w_a > 0$ they carry payload but hold at most 12.6%.

This is the one place in the paper where the theory predicts a number and the measurement lands inside it, so we state the pair plainly: **bound 12.6%, realized $\leq 1.8\%$** , in every measured round with $w_a > 0$. **And it is also the one place where heterogeneity makes the guarantee measurably empty.** The bound is a function of the realized margin m alone, and at Dirichlet $\alpha=0.1$ RFA’s **all-round mean margin collapses from $m=3.24$ to $m=1.01$** , exactly the limit at which Corollary 2 degrades to $n_a/(n_b+n_a)$ and certifies nothing, with two of three certified compositions crossing the ASR threshold there (§D.4(E)). So the theorem’s explanatory value is genuine and measured in *both* directions: it predicts a number that holds where heterogeneity is moderate, and its own quantity records where it stops holding. **That is not a robustness check; it is one of the paper’s three central empirical findings** (§E.2, §I).

Λ_a is level (P4) in measurable form, but (P4) is a level, not a single metric. Before the instantiation, one clarification the rest of the paper depends on: **(P4) is a conceptual level of the hierarchy and is instantiated defense-specifically**, because different aggregator families have different notions of what it means for adversarial input to enter the aggregate at all. A *selector* admits or does not, so its (P4) is an indicator in $\{0, 1\}$; a *weighted averager* has an adversarial coefficient share in $[0, 1]$; a *coordinate-wise order statistic* has a fraction of selected coordinates that are adversarial. There is no single scalar all three compute, and this is why the influence column of Table A11 is **necessarily aggregator-specific** rather than a common measurement across rows: a fact about the level, not a shortcoming of the table. With that said: Definition 2’s levels are not equally meaningful. (P1)–(P3) are properties of a statistic and of a decision rule reading it; **(P4) is the first level whose content is the adversary’s influence on the aggregate**, and for a geometric aggregator Λ_a is exactly that level made measurable: Corollary 2 bounds it, Lemma 2 drives it to a benign-only convex combination,

and §E.2’s instruments intervene on it directly. So the negative results do not leave the theory empty-handed: **the level the evidence points to is one the theory can bound, just not from a (d_1, d_2) definition pair alone**, which is what a screen would need. That gap is the open problem we state in §I. **Three quantities are distinguished throughout and never used interchangeably**: *admitted adversarial mass*, whether an adversarial update enters d_2 ’s aggregate at all, binary for a selector and a coordinate fraction for an order statistic; the *adversarial coefficient share*, the fraction of the upstream coefficient vector’s total mass held by adversaries, which is what the instruments of §E.2 pin; and *aggregate adversarial influence* Λ_a , the weight the adversary’s direction actually carries in the aggregate. Unqualified, “admission” means the first. (Symbols are collected in Appendix K.)

Lemma 2 (Zero-weight annihilation). *Zero-weight clients are not removed from the composition: d_2 receives them as zero vectors and still sees all K points. Our implementation runs the ε -smoothed Weiszfeld iteration ($\varepsilon = 10^{-8}$), whose fixed point is $\mu' = \sum_i \lambda_i u'_i$ with $\lambda_i \propto 1/\max(\|u'_i - \mu'\|, \varepsilon)$: a convex combination for every configuration, including the singular case $\mu' \in U'$ that a zero-weight client at the origin can create, so no non-degeneracy assumption is required. If $w_a = 0$ for every adversarial client, then $u'_a = 0$ and, writing $\Lambda_a := \sum_a \lambda_a$, the adversarial terms vanish identically: $\mu' = \sum_{b \text{ benign}} \lambda_b u'_b = (1 - \Lambda_a) \bar{u}$ with $\bar{u}$ a convex combination of benign transformed updates. The aggregate is a contraction by $1 - \Lambda_a \in (0, 1)$ of a benign-only weighted average: the attack direction never enters μ' . The origin points’ only effect is that contraction: a reduction in effective step size, i.e. a utility cost, not a security failure. The same holds for CoordMedian, where a zeroed coordinate carries no payload but shrinks the aggregate when selected.*

Why the positive-weight restriction matters. Defining ρ over all weights would make it *infinite in every round*: FoolsGold’s normalization subtracts the minimum weight, so 2 of 5 clients receive exactly zero (measured, $N=10, K=5$). A bound of the form $\rho < 4$ would then never be satisfiable, and Theorem 8(2) would be vacuous on the very composition it is stated over. Restricting ρ to strictly positive weights and handling $w_a=0$ separately via Lemma 2 is what makes the analysis apply to the algorithm we actually run.

Operational calibration protocol. To evaluate the criterion for $(d_1 \rightarrow d_2)$: **C1** is available from standard single-defense baselines (zero cost). **C2/C3 for the theorem-backed class** (FG or reputation as d_1 ; RFA, CM, TM as d_2): for the *rank* aggregators, Theorem 8(1)’s bound $\rho < r^*$ needs only ρ and the coordinate-wise separation, both read off the C1 baseline runs, so this case really is free. **For the geometric case an earlier version of this paper claimed the same, and that was false**: Corollary 1 removes the one quantity that would have required running the composition, but the benign spread that replaces it is measured on the transformed points, so RFA carries the same short-pilot cost as the other families and we no longer exempt it. **C2/C3 for other families**: one short pilot (5–10 rounds, $\sim 10\%$ of full cost) measuring $\sigma_2(T_1(u_a))$ vs. $\sigma_2(u_a)$ suffices. The full 50-round experiment runs only for criterion-PASS candidates. $\delta_2 = 0.5$ separates all PASS from C2-FAIL cases on the development set; insensitive to choice: any $\delta_2 \in [0.3, 0.7]$ yields identical classifications on all 42 pairs.

Proposition 9 (Rank-invariance of coordinate-wise aggregators). *Let d_2 be a coordinate-wise rank aggregator (TrimmedMean or CoordMedian) with admission statistic $\sigma_k(u) = \text{rank}_k(u)$ among the received updates $\{u_1, \dots, u_m\}$. Let $T_1(u_i) = \alpha \cdot u_i$ be a uniform positive rescaling ($\alpha > 0$, same for all clients). Then:*

1. $\sigma_k(T_1(u_i)) = \sigma_k(u_i)$ for all i, k , i.e., C2 and C3 hold trivially.
2. Under heterogeneous rescaling $T_1(u_i) = \alpha_i \cdot u_i$ with $\alpha_i \neq \alpha_j$, rank preservation fails when rescaling reorders coordinates across clients: $\exists k \text{ s.t. } \text{rank}_k(\alpha_i u_{i,k}) \neq \text{rank}_k(u_{i,k})$.

Proof. (1) Rank is invariant to positive scalar multiplication: $u_{i,k} > u_{j,k} \iff \alpha u_{i,k} > \alpha u_{j,k}$ for $\alpha > 0$. (2) Counterexample: let $u_{1,k} = 3, u_{2,k} = 2$, with $\alpha_1 = 0.1, \alpha_2 = 1$. Then $\alpha_1 u_{1,k} = 0.3 < \alpha_2 u_{2,k} = 2$, reversing the rank. $\square$

Corollary 3. *CoordMedian (pure rank selection) is robust to C3 boundary compression under any uniform rescaling but vulnerable under heterogeneous client-specific rescaling (e.g., FoolsGold or reputation weights). This predicts that compositions $X \rightarrow \text{CoordMedian}$ will show intermediate ASR when d_1 applies heterogeneous weights: consistent with observed $\text{NC} \rightarrow \text{cm}$ (0.433) and $\text{rfa} \rightarrow \text{cm}$ (0.485), both near the 0.5 threshold.*

Proposition 9 concerns rank under *uniform* rescaling. The sharper question is which admission signals survive *heterogeneous* per-client rescaling, since that is what every per-client defense in our menu applies: NormClip ($c_i = \min(1, \tau/\|u_i\|)$), reputation ($c_i \propto e^{-d_i/s}$) and RFA-reweighting ($c_i \propto 1/\max(\|u_i - \mu'\|, \varepsilon)$) are all of the form $T(u_i) = c_i u_i$ with $c_i > 0$ strictly, each positive by construction. FoolsGold is of the same form with $c_i \geq 0$ but attains $c_i = 0$ in every round (its shift-and-clamp gives the least-trusted client weight exactly zero, 2 of 5 clients here), so it falls outside the strict hypothesis below and is governed by Lemma 2 instead: the same distinction we draw for FLTrust’s ReLU zeros in §D.1. The strictly-positive case has a sharp answer, and it partitions the menu:

Proposition 10 (Invariance classes under positive per-client rescaling). *Let $T(u_i) = c_i u_i$ with $c_i > 0$, strictly: the form taken by NormClip ($c_i = \min(1, \tau/\|u_i\|)$), Reputation ($c_i \propto e^{-d_i/s}$) and RFA-reweighting ($c_i \propto$*

1/ $\max(\|u_i - \mu'\|, \epsilon)$, each positive by construction. **FoolsGold is deliberately not in that list:** its shift-and-clamp gives the least-trusted client weight exactly 0 in every round (2 of 5 clients here), so a zero coefficient is not a positive rescaling and FoolsGold-as- d_1 falls outside the hypothesis, governed by Lemma 2 rather than by (a)–(c). Then:

- (a) **Cosine similarity is exactly invariant:** $\cos(c_i u_i, c_j u_j) = \cos(u_i, u_j)$. FoolsGold’s weight vector is a function of the pairwise-cosine matrix alone, so $w(T(U)) = w(U)$ identically, and **C2 holds for every $\cdot \rightarrow$ FoolsGold composition.**
- (b) **Coordinate ordering is not invariant** in general; it survives under the bounded-ratio condition of Theorem 8(1).
- (c) **Consensus distance and residual magnitude are not invariant.** Heterogeneous c_i can invert their ordering, so $\cdot \rightarrow$ reputation compositions fail C2, and $\cdot \rightarrow$ RFA needs the separation condition of Theorem 8(2).

Proof. (a) $c_i u_i / \|c_i u_i\| = u_i / \|u_i\|$ for $c_i > 0$, so every pairwise cosine, each client’s normalized maximum similarity, and hence every FoolsGold weight is unchanged. (c) Counterexample for consensus distance: with $u_1=(1, 0)$, $u_2=(0, 1)$, $u_3=(3, 3)$ the coordinate-wise median is $(1, 1)$ and u_3 is farthest; under $c=(1, 1, 0.1)$ the median becomes $(0.3, 0.3)$ and u_3 becomes the *closest* point, reversing the ordering reputation depends on. $\square$

This corrects a natural but wrong intuition. Norm clipping does *not* blind FoolsGold: it cannot, by (a), though it does blind reputation, whose signal is consensus distance rather than similarity. We verified (a) in the shipped implementation: FoolsGold’s weight vector changes by at most 3×10^{-8} (float32 rounding) under NormClip, across 1000 synthetic trials and live FL rounds in which 1–2 of 5 clients were genuinely clipped (raw norms up to 24.5 against $\tau=5$). Note this also means Theorem 8(3)’s norm-equalizing degenerate case requires τ below *all* client norms; at our $\tau=5$ only 1–2 of 5 clients clip per round, so norms are not equalized and that case does not describe our runs.

C.3 THE FRAMEWORK IN GENERAL FORM

Positive rescaling is the instantiation that matters for current defenses, but the structure is more general, and stating it separates what is essential to the framework from what is an artifact of today’s defense menu. Write d_2 ’s admission mechanism as a statistic $S(U)$ of the round’s updates together with a decision rule that consumes only the *ordering* S induces over clients, which are extreme, which are far from consensus, which are down-weighted. Let T be d_1 ’s per-client transform. Then $(d_1 \rightarrow d_2)$ is **mechanism-preserving** exactly when

$$S(T(U)) = \varphi(S(U)) \quad \text{for some } \varphi \text{ preserving the ordering } d_2 \text{ acts on,} \quad (2)$$

which is C2; C3 additionally asks that φ not compress the quantitative gap below d_2 ’s operating threshold. Proposition 10 is the case $T(u_i) = c_i u_i$: for $S =$ cosine similarity, $\varphi = \text{id}$, so (2) holds unconditionally; for $S =$ coordinate value, φ is a per-client positive scaling, order-preserving only under the bounded-ratio condition $\rho < \tau^*$ of Theorem 8(1); for $S =$ consensus distance or residual magnitude, no φ exists, because the reference point (median, geometric median) moves with T , so $S(T(U))$ is not a function of $S(U)$ at all. Three consequences worth stating. First, this is why the conditions are checkable without running an experiment: one reads S off d_2 ’s definition and T off d_1 ’s, and the question becomes algebraic. Second, it identifies what a genuinely new instantiation would require: an S and a non-rescaling T , e.g. a d_1 that projects updates onto a subspace or applies a coordinate-wise mask, for which (2) would need to be re-derived rather than inherited. Third, it marks the framework’s real boundary: a d_1 whose transform is not per-client (one that mixes updates across clients, or carries cross-round state) admits no T of this form, and nothing here applies to it.

Predicted composability for our defense menu. Criterion 3 makes precise predictions for each two-defense pair from our 7-defense menu. We validate against targeted pairs spanning all prediction categories. Key predictions:

D WHAT THE SCREEN BUYS, AND WHAT IT DOES NOT

Criterion 3 predicts for each two-defense pair whether composition achieves low max-committed ASR. All experiments in this section and §F use one deployment regime throughout (CIFAR-10, $N=10$ clients, $K=5$ per round, adversarial fraction $f=0.2$, 50 rounds, 4-layer CNN), with committed attacks model_scaling and backdoor_pixel. **Convention:** unless stated otherwise, *ASR* means *max-committed ASR* (the maximum over the two committed attacks of the per-attack mean across seeds) thresholded at 0.5: the level at which the trigger fires on the majority of triggered inputs, so that “suppresses” means the backdoor is not the model’s dominant response; adaptive-attack, order-sensitivity and per-seed figures are labelled explicitly where they differ. A second gate is needed because a destroyed model has a low ASR for the wrong reason: every cell in §E–§E.2 must also hold

mean clean accuracy ≥ 0.35 (our **accuracy floor**), below which we call the cell *uninterpretable* rather than suppressed. **Sample sizes are small and were fixed before the runs:** $n=5$ seeds per cell in the interventional suites ($n=8$ for `cos_krum`, $n=3$ for the second-dataset replication), $n=30$ for the FoolsGold $\rightarrow$ RFA characterization, so intervals are wide, and no null result is read as flatness unless it clears a pre-registered equivalence margin. **“Refuted” means one thing throughout:** a pre-registered *directional* hypothesis was rejected against the rule frozen for it, never that a mathematical claim was disproved. Adversaries are the fixed client identities $\{0, \dots, fN - 1\}$ and each round samples K participants uniformly without replacement from all N , with no stratification, so the per-round adversary count is *hypergeometric* rather than fixed: for $N=10$, $f=0.2$, $K=5$, $P(n_a=0) = P(n_a=2) = 56/252 = 0.222$ and $P(n_a=1) = 140/252 = 0.556$ (measured over 100 rounds: $0.19/0.54/0.27$). Two consequences the data bear out: CoordMedian’s count-majority condition $n_a < K/2 = 2.5$ holds for *every* attainable $n_a \in \{0, 1, 2\}$, i.e. in 100% of rounds, while TrimmedMean’s budget $\lfloor \beta K \rfloor = \lfloor 0.2 \cdot 5 \rfloor = 1$ is exceeded whenever $n_a=2$: 22.2% of rounds, ≈ 11 of 50. That is a second, count-based reason FoolsGold $\rightarrow$ TrimmedMean fails (0.508) while FoolsGold $\rightarrow$ CoordMedian succeeds (0.107) under an identical upstream transform; we retain the pre-registered C3-FAIL label, the budget violation being an additional mechanism rather than a relabelling. We validate in two waves. The **development set** (18 targeted pairs, Table A4) spans all criterion prediction categories and was used to refine the criterion (adding C3). The **held-out set** (24 pairs, Table A5) was *pre-registered*: all 24 predictions were committed to version control (commit `7e4d9b1`) *before* any held-out experiment was run, eliminating post-hoc classification concerns. The criterion predicts LOW (<0.5) or HIGH (≥ 0.5) max-committed ASR. The verdict categories are:

Table A4: Composability criterion validation ($n=3$ seeds; max across committed_scaling and committed_pixel; $\pm$ is per-seed std where reported). Threshold: LOW < 0.5 , HIGH ≥ 0.5 . $^\dagger n=30$ from the main composition experiment.

Pair	Failure mode	Predicted	Max ASR	Correct?
<i>PASS: C1$\wedge$C2$\wedge$C3 all hold (predicted LOW)</i>				
foolsgold $\rightarrow$ rfa	—	LOW	0.093 ± 0.082	✓
foolsgold $\rightarrow$ coord_median	—	LOW	0.107 ± 0.034	✓
reputation $\rightarrow$ coord_median	—	LOW	0.423 ± 0.155	✓
<i>C1-FAIL: neither defense individually suppresses all attacks (predicted HIGH)</i>				
reputation $\rightarrow$ trimmed_mean †	C1 (pixel)	HIGH	0.679	✓
reputation $\rightarrow$ rfa	C1 (pixel)	HIGH	0.825	✓
reputation $\rightarrow$ foolsgold	C1 (pixel)	HIGH	0.807	✓
reputation $\rightarrow$ norm_clip	C1 (pixel)	HIGH	0.813	✓
norm_clip $\rightarrow$ foolsgold ‡	C1	HIGH	0.887	✓
<i>C2-FAIL: d_1 destroys d_2’s signal direction (predicted HIGH)</i>				
norm_clip $\rightarrow$ reputation	C2	HIGH	0.819	✓
rfa $\rightarrow$ reputation	C2	HIGH	0.882	✓
<i>C3-FAIL: d_1 compresses boundary below d_2’s threshold (predicted HIGH)</i>				
norm_clip $\rightarrow$ trimmed_mean	C3	HIGH	0.780	✓
norm_clip $\rightarrow$ rfa	C3	HIGH	0.853	✓
rfa $\rightarrow$ trimmed_mean	C3	HIGH	0.678	✓
rfa $\rightarrow$ coord_median	C3	HIGH	0.485	×
foolsgold $\rightarrow$ trimmed_mean	C3	HIGH	0.508	✓
norm_clip $\rightarrow$ coord_median	C3	HIGH	0.433	×
<i>DEGEN: d_1 has no per-client transform (predicted HIGH)</i>				
fedavg $\rightarrow$ norm_clip	DEGEN	HIGH	0.935	✓
fedavg $\rightarrow$ trimmed_mean	DEGEN	HIGH	0.846	✓
Overall: 16/18 = 89%. HIGH direction: 13/15 correct (87%). Under the tightened C1 the certified PASS set is 2/2 correct; fg $\rightarrow$ rfa is reclassified as a synergy false negative.				
‡ Reclassified from C2-FAIL: by Proposition 10(a) clipping preserves FoolsGold’s similarities exactly, so C2 holds and the pair fails C1. Prediction and outcome unchanged.				

PASS precision: 100%, all compositions the criterion predicts to work *do* work. Under the tightened per-attack C1 (§C) the certified in-distribution set is 2 pairs, fg $\rightarrow$ cm (0.107) and rep $\rightarrow$ cm (0.423), both correct; fg $\rightarrow$ rfa (0.093) is no longer certified and is reported below as a false negative of our own criterion. This is the practitioner-relevant result: the criterion’s positive predictions are reliable, at the cost of 40% recall. Overall development-set accuracy: **16/18 = 89%**. Two misses, both in the HIGH direction:

One certified pair is not resolved below threshold. Rep $\rightarrow$ CM’s interval $[0.231, 0.616]$ crosses 0.5, and one of its five seeds reaches 0.665. Its LOW label therefore rests on a mean whose uncertainty admits the HIGH

side, and we report it as correct on the mean while flagging it as the one certified pair whose margin is not statistically established. The certified in-distribution set is 2/2 correct either way, but only $\text{fg} \rightarrow \text{cm}$ is certified and resolved below threshold at the seed budget we can afford. Clean accuracy on the three cells is 0.50, 0.68, 0.76, all well clear of the collapse regime.

Two boundary conditions, stated alongside the primary result. Both are developed in full later, but neither should be met late. (i) **Extreme heterogeneity breaks two of the three PASS pairs.** At $\alpha=0.1$, $\text{fg} \rightarrow \text{rfa}$ reaches 0.569 and $\text{rep} \rightarrow \text{cm}$ 0.548; only $\text{fg} \rightarrow \text{cm}$ survives, at 0.433. What makes this more than a negative datapoint is that the failure appears in the theorem’s own quantity: RFA’s separation margin $m := \Delta_{\text{sep}}' / [2(R_B' + \delta')]$, measured on the transformed points, collapses from an all-round mean of 3.24 to 1.01 (exactly the $m \rightarrow 1$ boundary at which Corollary 2’s bound degrades to 40% adversarial mass), so a practitioner running the 5–10 round pilot would measure $m \approx 1$ and decline to deploy. We emphasise that this is a post-hoc measurement of a known failure, not a pre-registered prediction of it; full sweep and caveats (bimodality at $n=3$, accuracy collapse to 0.20–0.48) in §D.4(E). (ii) **Mechanism preservation is not adaptive-attack robustness.** A criterion-aware adversary solving FoolsGold’s and RFA’s constraints jointly (decorrelated coordinate blocks to evade the similarity test, ϵ -ball projection to stay inside the geometric median’s radius) raises $\text{fg} \rightarrow \text{rfa}$ from 0.051 (seed-matched) to 0.167 within our evaluated composition-aware attack family (§F). Theorem 8 bounds what the preserved mechanism admits, not what an adversary reshaping its update to satisfy both constraints at once can do, and the $3.3\times$ increase is the size of that gap on our flagship pair.

Held-out validation (pre-registered): 22/24 = 92% (Screening: an evaluation ordering, and by Proposition 6 not an identification of the mechanism’s contribution). Table A5 reports the 24 pre-registered pairs. Combined across both waves: **38/42 = 90.5%**. **Read against its base rate that figure is nearly uninformative, which is why it appears here, where it is computed, and in no summarizing position.** Only 5 of the 42 pairs are low-ASR (the 12% base rate of §D.3), so the constant predictor that calls every pair HIGH already scores $37/42 = 88.1\%$; 90.5% is one pair better than that, and the tally is carried by correct negatives. The informative quantities are precision 2/2 and recall 40% (§D.3). The two held-out misses are both LOW-predicted pairs that were flagged as *speculative* in the pre-registration (neither is a clean $\text{C1} \wedge \text{C2} \wedge \text{C3}$ PASS):

Table A5: Held-out pre-registered validation (24 pairs, $n=5$ seeds; max across committed_scaling and committed_pixel). All predictions committed to version control (commit 7e4d9b1) before experiments. Threshold: $\text{LOW} < 0.5$, $\text{HIGH} \geq 0.5$. Only the two LOW predictions are shown individually; the 22 HIGH predictions (all correct) are summarized.

Pair	Category	Predicted	Max ASR	Correct?
<i>Speculative LOW predictions (not clean $\text{C1} \wedge \text{C2} \wedge \text{C3}$ PASS)</i>				
$\text{fedavg} \rightarrow \text{coord_median}$	DEGEN	LOW	0.519	×
$\text{foolsgold} \rightarrow \text{norm_clip}$	C1 (uncertain)	LOW	0.927	×
<i>HIGH predictions (22 pairs: 6 CM-as-d_1, 6 TM-as-d_1, 4 fedavg-as-d_1, 3 weighting-d_1, 3 RFA-as-d_1; range 0.70–0.94)</i>				22/22 ✓
Held-out: 22/24 = 92%. Combined with development set: 38/42 = 90.5%.				
Constant-HIGH baseline on the same 42 pairs: 37/42 = 88.1%.				

Rep→tm: composition dominates despite C1-FAIL. Reputation→TrimmedMean is correctly classified as C1-FAIL (pixel ASR 0.679 > 0.5), yet it is the paper’s main composition result. The resolution: the criterion predicts *max-committed* ASR, while the deployment-relevant metric is *oracle* ASR (0.316). The oracle cannot exploit the pixel gap because scaling (which reputation *does* suppress) dominates its utility. The criterion thus correctly identifies that $\text{rep} \rightarrow \text{tm}$ does not suppress every committed attack, while the game-theoretic analysis shows it is strategically dominant.

Tightening C1 costs us our own flagship, and reveals a synergy case. Under the per-attack quantifier in C1, **fg→rfa no longer certifies:** neither FoolsGold (0.732) nor RFA (0.845) suppresses the pixel backdoor individually, and FoolsGold’s 0.200 against model-scaling is hollow because it comes with 10% clean accuracy. The composition nonetheless reaches 0.093. This is genuine *synergy* (suppression neither constituent possesses) and Proposition 7 proves that *no* inherited-suppression screen can certify it, so this is a scope boundary rather than a tuning failure. We therefore report our lowest-ASR composition as a false negative of our own criterion. The certified in-distribution set is $\text{fg} \rightarrow \text{cm}$ (0.107) and $\text{rep} \rightarrow \text{cm}$ (0.423), both correct: precision 2/2, recall 2/5 = 40%.

D.1 A PROSPECTIVE TEST ON A NEVER-RUN DEFENSE MENU

Result: 13/20, and the shortfall is in our experimental design. At the same 0.5 threshold: TP = 8, FP = 0, TN = 5, FN = 7 (Table A6). Every predicted-LOW pair came out LOW, 8/8, and that figure is far weaker

Table A6: Prospective suite: 20 pairs, labels frozen at commit 262cf35 before any run. Max-committed ASR over 2 attacks $\times$ 3 seeds; threshold 0.5. Categories are verbatim from the pre-registration, including its own uncertainty hedges.

Pair	Frozen category	Pred.	Max ASR	Acc	Hit
<i>In scope (C0 holds): d_2 load-bearing</i>					
Rep $\rightarrow$ FLTrust	certified	LOW	0.046	0.577	✓
NC $\rightarrow$ FLTrust	certified	LOW	0.046	0.576	✓
RFA $\rightarrow$ FLTrust	certified	LOW	0.047	0.574	✓
Krum $\rightarrow$ FLTrust	DEGEN	LOW	0.048	0.574	✓
MKrum $\rightarrow$ FLTrust	DEGEN	LOW	0.048	0.574	✓
FG $\rightarrow$ FLTrust	certified	LOW	0.064	0.547	✓
FG $\rightarrow$ Krum	C1-fail	HIGH	0.078	0.099	×
Krum $\rightarrow$ CM	DEGEN	HIGH	0.450	0.770	×
Krum $\rightarrow$ RFA	DEGEN	HIGH	0.870	0.780	✓
MKrum $\rightarrow$ RFA	DEGEN	HIGH	0.870	0.780	✓
RFA $\rightarrow$ MKrum	C1-fail	HIGH	0.875	0.782	✓
Rep $\rightarrow$ Krum	C1-fail	HIGH	0.880	0.530	✓
NC $\rightarrow$ MKrum	C1-fail	HIGH	0.943	0.761	✓
<i>Out of scope (C0 violated): FLTrust as d_1 suppresses alone at 0.048</i>					
FLTrust $\rightarrow$ CM	certified-uncertain	LOW	0.041	0.505	✓
FLTrust $\rightarrow$ MKrum	C2-fail	HIGH	0.045	0.576	×
FLTrust $\rightarrow$ TM	C3-fail-uncertain	HIGH	0.046	0.529	×
FLTrust $\rightarrow$ NC	C2-fail	HIGH	0.047	0.576	×
FLTrust $\rightarrow$ RFA	certified	LOW	0.048	0.512	✓
FLTrust $\rightarrow$ Rep	C2-fail	HIGH	0.048	0.518	×
FLTrust $\rightarrow$ Krum	C2-fail	HIGH	0.078	0.361	×
Full suite 13/20 (TP 8, FP 0, TN 5, FN 7); in scope 11/13; non-FLTrust arm 5/7.					

than it looks, because **FLTrust suppresses both committed attacks on its own** (max-committed ASR 0.048 at 0.575 clean accuracy), the first defense in our menu that does. All 13 pairs containing FLTrust land in $[0.041, 0.078]$, within 0.030 of FLTrust alone, in either slot. The two slots fail for different reasons, and the distinction matters:

The two slots fail differently, and the difference is the finding. (i) *FLTrust as d_1 (7 pairs, 2/7)*. d_1 alone already suppresses, so the composition inherits that suppression whatever d_2 does; all five predicted-HIGH misses have real C2/C3 failures, but destroying d_2 's mechanism is costless when d_2 was never load-bearing. This is C0, invisible on the original 42 pairs because no defense there suppresses alone (the minimum single-defense max-committed ASR is CoordMedian's 0.519, and the lowest ASR among all 39 non-PASS pairs is 0.433), so we count these 7 pairs out of scope *retrospectively*: a weaker claim than a pre-registered exclusion. (ii) *FLTrust as d_2 (6 pairs, 6/6)*, but the prediction is nearly an identity. FLTrust gates on the scale-invariant $\cos(u_i, g_{\text{srv}})$ and then renormalizes every update to $\|g_{\text{srv}}\|$, discarding magnitude entirely, so a positive per-client rescaling is *exactly* cancelled: four of the six pairs are bit-identical to FLTrust alone or differ only within seed noise. The single genuine deviation is FG $\rightarrow$ FLTrust (0.064), the one d_1 producing exact *zeros* rather than positive rescalings and hence a Lemma 2 interaction. This arm is a consistency check between theory and implementation, **not** a falsifiable test: every d_1 in our menu is a positive rescaling and Proposition 10(a) permits no other outcome. (iii) *Neither defense is FLTrust (7 pairs, 5/7)*: the only arm that could have surprised us. Five predicted-HIGH pairs are correct (0.870–0.943); both misses are diagnosable and neither is a mechanism error: fg $\rightarrow$ krum measures 0.078 at 9.9% clean accuracy (the collapsed-model artifact, not suppression), and krum $\rightarrow$ cm measures 0.450 against a 0.5 threshold as a degenerate pair reducing to CoordMedian alone, whose own baseline is 0.519. The arm contains *zero* predicted-LOW pairs, so it bears on specificity only. Per-pair numbers for all three arms are in the supplement (§ S1).

What the suite does and does not establish. Counting only in-scope pairs gives 11/13; the arm that carried real risk of falsification gives 5/7 with both misses explained. But we will not present this as the prospective positive result the framework needs, because **it does not contain one**: the 8 predicted-LOW pairs are 6 near-identities plus 2 out-of-scope inheritances. The three matched contrasts holding d_1 fixed while varying d_2 's invariance class (rep $\rightarrow$ fltrust 0.046 vs. rep $\rightarrow$ krum 0.880; nc $\rightarrow$ fltrust 0.046 vs. nc $\rightarrow$ mkrum 0.943; rfa $\rightarrow$ fltrust 0.047 vs. rfa $\rightarrow$ mkrum 0.875) show a 19–20 $\times$ ASR gap turning on exactly the property Proposition 10 identifies, which is the most suggestive evidence here. They are still confounded: in this menu the invariant-signal d_2 is also the only d_2 that suppresses alone, so C1 and C2 co-vary. Lemma 1 explains why no other choice of defense would have helped: in scope, $C0 \wedge C1$ force d_2 to suppress a^* alone, so the confounding is structural. Breaking it requires holding the suppression principle and standalone effectiveness fixed while varying only the invariance class, which is the design of §D.2.

D.2 BREAKING THE CONFOUND: A MATCHED METRIC-SWAP ABLATION

Lemma 1 says mechanism–effectiveness confounding cannot be escaped by picking a different defense off a menu: in scope, C0/C1 force the invariant-signal d_2 to be one that suppresses a^* alone. So we break the confound by construction instead, holding the suppression *principle* fixed and varying only the invariance class of the statistic d_2 reads.

The design’s premise was refuted by its own baselines. Matching failed in both arms, for two independent reasons, and because the C1 inputs are measured before the freeze we learned this before the suite ran: the only reason it can be reported as a finding rather than a post-hoc excuse. `cos_reputation` does not merely weaken relative to its Euclidean original, it collapses, from Reputation’s 0.017 on model-scaling to **0.982**. `cos_krum` reaches 0.078 on model-scaling, which passes C1’s numeric threshold, but at **0.146** clean accuracy, the hollow-suppression artifact already documented for FoolsGold (0.200 at 0.10) and `fg→krum` (0.078 at 0.099), so C1 fails under the accuracy gate we apply everywhere else. The mechanism is visible in the baselines rather than asserted: undefended FedAvg under model-scaling is itself at 0.100 accuracy, so Krum’s 0.577 is *restoring* utility the attack destroyed and `cos_krum`’s 0.146 fails to. **Magnitude information is what lets this defense family handle a magnitude attack:** a direction-only mirror of a Euclidean scorer cannot see a scaling attack at all, so removing the magnitude dependence removes the suppression with it: exactly the coupling the pre-registered contingency anticipated, stated there as the possibility that “invariance and standalone effectiveness may be inherently coupled” for this family. On the magnitude-neutral pixel backdoor the mirror does survive (0.300 at 0.598), but there its control Krum fails C1 (0.583), so those arms are unmatched in the opposite direction and the pixel cells cannot carry the test either.

The consequence is that **the confound re-materialized inside an experiment built specifically to break it**, which we report as the round’s methodological finding rather than as a failed run. Lemma 1 establishes that mechanism–effectiveness confounding is structural in scope; this is what trying to escape it looks like when the mechanism being varied is the mechanism the defense relies on. Two responses were available and we took neither. We did not switch to a pixel-only protocol, which would have made `cos_krum` pass C1 and manufactured two predicted-LOW cells: choosing the attack after seeing the C1 inputs is attack-shopping regardless of the resulting labels. We did not adjust the accuracy gate to admit 0.146, which would have produced a predicted-LOW pair by weakening the very criterion that catches hollow suppression elsewhere. The pre-registration’s second non-negotiable (“no switching of committed attack, threshold, seed count, or accuracy gate”) forecloses both, and either would have bought a positive test at the cost of the property that makes the test worth running.

Outcome: 8/8 correct, and zero positive tests. Because C1 binds for all eight pairs (on model-scaling through the two baseline failures above, on the pixel backdoor by design) every frozen label was HIGH, so the suite bears on specificity only, as a consequence of inputs measured before the freeze rather than a choice made afterwards. We ran all 8 pairs $\times$ 2 attacks $\times$ 3 seeds anyway, because a suite whose labels are all one class is still falsifiable in the direction that matters: a single LOW cell would have been a false positive against a criterion claiming 100% precision. All eight came out HIGH (8/8 correct, TP = 0, FP = 0, TN = 8, FN = 0, max-committed ASR spanning 0.508–0.935) with labels frozen in `pre_registration_metric_swap.md` at commit 356897b, timestamped before `results/metric_swap/` existed and reproduced verbatim in the runner’s PAIRS list so scoring cannot drift. We state the limitation in the same breath as the result: the suite contains **no falsifiable prospective positive test**, and 8/8 is not validation of the framework’s predictive claim: eight correct negatives on pairs our own conditions could only have called HIGH is weak evidence, and it supports a claim about specificity, not prediction.

One prediction in the suite was genuinely two-sided, and it is the round’s only prospective positive result. `cos_krum`’s C1 verdict on model-scaling turns entirely on whether the accuracy gate is applied (0.078 passes the numeric threshold; 0.146 accuracy fails the gate), so rather than choose a reading afterwards we froze *both*, with the resolution rule fixed in advance: accuracy-gated $\Rightarrow$ HIGH, numeric-only $\Rightarrow$ LOW, and a LOW outcome at accuracy ≥ 0.35 would have recorded our two primary labels as misses. Both pairs came out **HIGH** (NormClip $\rightarrow$ `cos_krum` 0.508, RFA $\rightarrow$ `cos_krum` 0.566), refuting the secondary reading and vindicating the accuracy gate against a pre-registered alternative specification of itself on data neither of us had seen. The weaker half is flagged in the supplement (§ S2): the NormClip arm sits barely above threshold and it is the RFA arm that carries the conclusion.

A post-hoc contrast that is not confounded. The cross-pair scoring above is confounded in exactly the way Lemma 1 predicts, but the frozen suite happens to contain a comparison that is not, and it is the strongest C2 evidence in either paper. Compare a composition against *the same* d_2 ’s own standalone ASR on the same attack, under two different upstream transforms. C1 is then identical by construction (same downstream defense, same attack, same measured standalone effectiveness on both sides), so the residual $\Delta = \text{ASR}(d_1 \rightarrow d_2, a) - \text{ASR}(d_2, a)$ cannot be attributed to d_2 being a stronger or weaker defense. What does vary is how much the upstream transform actually disturbs d_2 ’s statistic, and that quantity was measured on the mechanism itself (§D.2, `results/cos_invariance_check.json`) rather than inferred from any ASR. The frozen label scoring is fixed at $n=3$ and must not move, so we topped the six powered cells up

to $n=5$ in a separate directory, with the tested direction fixed in the top-up script before the new seeds ran (`results/metric_swap_topup/`). For Krum under model-scaling (standalone 0.061 at 0.577 accuracy): NormClip changes Krum’s selection in 0/9 rounds and the composition reaches 0.062 ($\Delta = +0.001$, per-seed 0.120/0.018/0.047/0.058/0.068); RFA changes it in 6/9 rounds and the composition reaches 0.779 ($\Delta = +0.718$, per-seed 0.974/0.116/0.982/0.904/0.916): a gap of +0.716 between two compositions sharing a downstream defense, an attack, and a baseline, and at $n=5$ it is statistically resolved (Welch $t = 4.28$, two-sided $p = 0.012$; paired $t = 4.60$, $p = 0.010$; Mann–Whitney one-sided $p = 0.008$), which it was not at $n=3$.

The pattern holds across the suite once cells without power are excluded: Δ is observable only where standalone d_2 genuinely suppresses, since a baseline already above 0.5 shows $\Delta \approx 0$ by ceiling effect whatever the transform did. Restricting to standalone ASR < 0.5 at accuracy ≥ 0.35 leaves six cells, and at $n=5$ measured disturbance still splits them perfectly: the three undisturbed cells (0/9) give $\Delta \in [+0.001, +0.190]$, the three disturbed ones ($> 0/9$) give $\Delta \in [+0.718, +0.878]$: a separation of +0.527 with no overlap. Two of the three undisturbed cells are `cos_krum` on the pixel backdoor, where the mirror suppresses standalone (0.300 at 0.598 accuracy) and carries that suppression through under *both* upstreams. Here the two extra seeds *weaken* the confirming half rather than strengthen it, and we report it as measured: Δ moves from +0.000 at $n=3$ to +0.190 under NormClip and from -0.027 to +0.147 under RFA, leaving both compositions still below threshold (0.490 and 0.447 at 0.61 accuracy) but no longer intact. Exact invariance of the *selection* therefore does not by itself carry suppression through unchanged: the magnitude channel of Lemma 2 is a second route by which Δ can grow, and the ladder’s arm 3 (§E) is built to measure exactly that prospectively rather than describe it post-hoc. The ten excluded cells are listed rather than dropped silently in the supplement (§ S2).

The suite contains a second such contrast, on Reputation under model-scaling (standalone 0.017 at 0.777 accuracy), and the two are informative in opposite ways. NormClip disturbs Reputation’s weight ordering in 3/9 rounds and the composition reaches 0.806; RFA disturbs it in 9/9 and the composition reaches 0.895. The gap is only +0.089 (both arms are far above threshold, because Reputation’s ordering is disturbed under *both* upstreams) but within-arm variance is small, all five seeds move in the predicted direction (per-seed deltas +0.149/ +0.089/ +0.030/ +0.061/ +0.115), and it is resolved at both $n=3$ and $n=5$ (Welch $t = 3.06$, two-sided $p = 0.018$; paired $t = 4.31$, $p = 0.013$). So the two contrasts trade off exactly as the disturbance measurements imply: Krum has the large effect because one of its upstreams leaves it untouched, and Reputation was resolvable even at $n=3$ because neither upstream does.

This is post-hoc, and we do not present it otherwise. It was not pre-registered, it re-uses a suite frozen for a different purpose, the two extra seeds per cell were added *after* seeing the $n=3$ result, and the six-cell split is a description of that suite rather than a test on new data. The top-up cut in both directions, which is why we report all of it: it resolved the Krum gap (Welch $p = 0.16$ at $n=3$, where per-seed paired deltas +0.854/ +0.097/ +0.935 left one seed not separating, to $p = 0.012$ at $n=5$) and at the same time degraded the invariant control’s Δ from ≈ 0 to +0.19. What the analysis does have, which neither the FLTrust contrasts of §D.1 nor the cross-pair scoring has, is C1 held exactly fixed and a disturbance measurement taken on the statistic rather than read back from the outcome. We offer it as the cleanest available piece of C2 evidence and as the shape a properly prospective version of this test would take: fix a downstream defense that suppresses the attack, then vary the upstream along measured disturbance rather than along defense identity. Table A7 sets out, claim by claim, what this design can and cannot conclude.

D.3 AN APPLICATION: SCREENING AS EVALUATION PRIORITIZATION

What follows is a consequence of the identification boundary, not a claim the paper argues for. Proposition 6 says a criterion gated on the outcome it predicts cannot be validated by cross-defense contrast; what such a criterion can still do is *order an evaluation budget*, and that is the only use we claim for it. Proposition 7 bounds what it can find. Read the numbers below as a measurement of a prioritization heuristic’s cost, not as validation of the framework.

Does every condition earn its place? C3 does not: reported as a finding. Scoring all 42 pairs on C1, C2 and C3 *independently* rather than recording only the first failure: against a 12% base rate of low-ASR pairs, C1 alone selects 4 at 50% precision, C2 alone 14 at 36%, and C1∧C2 selects 2 at 100%. **The surviving C2 failures are exactly the non-invariant signal class:** after the Proposition 10 reclassification every one of them is a $\cdot \rightarrow \text{reputation}$ pair ($\text{nc} \rightarrow \text{rep}$ 0.819, $\text{rfa} \rightarrow \text{rep}$ 0.882, $\text{fg} \rightarrow \text{rep}$ 0.798), consensus distance being the signal a positive rescaling does not preserve, while all three non-degenerate $\cdot \rightarrow \text{FoolsGold}$ pairs are C1 failures instead. **C3 has no demonstrated incremental predictive value in this study:** alone it selects 35 pairs at 9% precision, below the base rate, and it never binds on a pair C1∧C2 already admits. **So we demote C3 from a gate to a margin diagnostic and recommend C1∧C2 as the criterion**, retaining C3 on theoretical grounds only: Theorem 8 shows margin compression is a real failure mode, and the $\alpha=0.1$ regime below is precisely a margin collapse. **The demotion changes no number**, C1∧C2 selects the same two pairs at the same precision (recomputed by `analyze_screening_cost.py`), which is why we report it as a finding about the criterion rather than a revision of results. *The pre-registered PASS/FAIL labels and the 16/18 and*

Table A7: **What this design can and cannot conclude.** The middle column is about *identifiability*, not statistical power: a “no” marked *structural* cannot be repaired by more seeds, defenses or attacks (Proposition 6). Every entry is scoped to the measured setting of §D–§E.2.

question	identifiable?	answer, and its evidence
Does disturbing d_2 ’s statistic change its <i>decision</i> ?	measured	Yes: Krum’s selection flips in up to 0.80 of rounds
Does it change the <i>admitted adversarial mass</i> ?	by construction	No: unchanged in every one of the 60 measured rounds
Does it change <i>suppression</i> ?	instrumented	No: $ \Delta =0.026$, inside the frozen margin (Prop. 11(a))
Does <i>adversary-correlated</i> reweighting change suppression?	instrumented	Yes, non-monotonically: single-peaked (Mode A)
Does admitted mass <i>order</i> outcomes across defenses?	pre-registered	No, refuted: +0.098 against a frozen +0.150
Is (P4) <i>sufficient</i> for (P5)?	control arm	No: <code>cos.krum</code> admits 0.000 while ASR falls
Can (P4) be screened <i>without</i> adversary identity?	no, open	Both instruments are oracles: the stated open problem
Does C1 identify C2’s contribution?	no, structural	Empty cell by the gate, for every menu (Prop. 6)
Does any of this certify <i>deployed</i> security?	no, scope	Committed attack only; one adaptive test breaks it

22/24 tallies were frozen with C3 in the gate and are not relabelled; this ablation is post-hoc and labelled as such, and the recommendation is for use going forward.

D.4 OUT-OF-SAMPLE PASS VALIDATION

The development and held-out validation above tests predictions from the *same* attack families (model_scaling, backdoor_pixel) and scale ($N=10$, cifar_cnn) used to design the criterion. To confirm that PASS predictions generalize we pre-registered three out-of-sample tests (commit ca627e5, before any of them ran); all held, the accounting is in the summary below, and per-cell values for every test are in the supplement (§S3). We state plainly what this is and is not: it is evidence that the conditions keep identifying compositions which survive a change of attack family, scale and architecture; it is *not* an out-of-sample classification result, because every prediction in it is positive and the matching FAIL controls proved uninformative. Negative-direction evidence rests entirely on the in-distribution tiers.

(E) Heterogeneity: where the criterion breaks. Earlier heterogeneity sweeps in this work covered *single defenses* only; no composition had been run at any $\alpha \neq 0.5$. We ran all three PASS pairs at $\alpha \in \{0.1, 1.0, 10.0\}$ (3 seeds, both committed attacks, 50 rounds). Reporting max-committed ASR / clean accuracy:

Composition	$\alpha=0.1$	$\alpha=0.5$	$\alpha=1.0$	$\alpha=10.0$
fg→rfa	0.569 / 0.20	0.093 / 0.50	0.051 / 0.57	0.063 / 0.62
fg→cm	0.433 / 0.40	0.107 / 0.68	0.133 / 0.74	0.129 / 0.78
rep→cm	0.548 / 0.48	0.423 / 0.76	0.377 / 0.79	0.326 / 0.81

The criterion transfers cleanly across $\alpha \in \{0.5, 1.0, 10.0\}$, including the near-IID regime where FoolsGold might plausibly founder on legitimately similar benign updates; it does not, and $\alpha=10$ is indistinguishable from $\alpha=0.5$. **Under extreme heterogeneity ($\alpha=0.1$), two of the three PASS compositions cross the threshold** (fg→rfa 0.569, rep→cm 0.548); only fg→cm survives, at 0.433. Two caveats cut in opposite directions: fg→rfa at $\alpha=0.1$ is strongly bimodal across seeds (0.065–0.923), so at $n=3$ the direction is clear but the magnitude is not; and clean accuracy collapses to 0.20–0.48, so this regime stresses utility as much as security. Crucially Theorem 8 supplies a **diagnostic quantity that retrospectively explains this boundary** rather than predicting it: re-measuring on the transformed points at $\alpha=0.1$, RFA’s separation margin $m := \Delta'_{\text{sep}}/[2(R'_B + \delta')]$ collapses from an all-round mean of 3.24 (met in 68% of rounds) to 1.01 (met in 20%). Over the payload-carrying rounds alone (the ones Figure A1 plots) the same collapse reads 4.71 ± 1.51 ($n=17$) to 2.06 ± 2.35 ($n=12$), with the per-round minimum falling $2.82 \rightarrow 0.00$; the all-round means are lower because they average in the Lemma 2 rounds, where FoolsGold zeroes every adversary and m is recorded as 0. Either way the collapse is measured, and the all-round figure is the one that reaches the vacuous limit. This sits exactly at the $m \rightarrow 1$ boundary where Corollary 2’s bound degrades to 40% adversarial mass: extreme heterogeneity inflates the benign radius R'_B until the benign cluster is no longer separated from the adversarial points, which is precisely the precondition the theorem requires. A practitioner running the 5–10 round pilot would mea-

sure $m \approx 1$ and decline to deploy, so the criterion is mechanistically diagnostic at this boundary, though we emphasise this is a post-hoc measurement of a known failure, not a pre-registered prediction of it.

The one axis not validated with composition experiments is *dataset* (CIFAR-100, with 100-class output): the conditions are dataset-agnostic in mechanism, but the downstream aggregator’s decision boundary may shift with the number of classes (L1).

E THE INTERVENTION EVIDENCE: THE CONFOUNDED LADDER, AND THE INSTRUMENT IT MOTIVATED

The controlled upstream transform. We introduce a synthetic d_1 whose only parameter is the *dispersion* of a positive per-client rescaling. For the K participants of a round, at position j in a (seed, round)-keyed pseudorandom permutation,

$$c_j \propto \exp\left(\kappa \cdot \left(\frac{2j}{K-1} - 1\right)\right), \quad \text{normalized so } \frac{1}{K} \sum_j c_j = 1. \quad (3)$$

Table A8: The four arms of the dose–response ladder, and the frozen predictions. The statistic each d_2 reads is, in order, summed pairwise Euclidean distance, distance to the client median, pairwise cosine, and coordinate ordering. *Standalone* is d_2 ’s own ASR (clean accuracy) on that attack, the C1 input, held fixed within every arm. *Measured disturbance* is the fraction of live rounds on which the rung changes that arm’s decision (for `coord_median`, the mean fraction of *coordinates* whose median-attaining client changes) measured before the freeze on the same raw updates, with no ASR computed (`results/dose_disturbance.json`). Shapes and decision rules are frozen at commit e711a95.

arm (d_2 / attack)	class (Prop. 10)	standalone ASR (acc)	measured disturbance $\kappa=0$ 0.5 1.0 2.0				predicted shape in κ
1 krum / scaling	(c) not inv.	0.061 (0.577)	0.00	0.60	0.73	0.87	monotone rise
2 reputation / scaling	(c) not inv.	0.017 (0.777)	0.00	0.87	0.87	0.93	monotone rise
3 cos.krum / pixel	(a) exact	0.300 (0.598)	0.00	0.00	0.00	0.00	FLAT (control)
4 coord.median / pixel	(b) cond. inv.	0.443 (0.767)	0.00	0.25	0.38	0.49	rise, shallower

The abscissa is measured, not nominal. What matters to C2 is not ρ but what the rescaling does to the statistic d_2 actually reads, so before freezing any prediction we pushed live FL updates through the shipped transform and recorded each arm’s decision at every κ on the *same* raw updates, computing no ASR. That establishes three things before any outcome exists (Table A8): *there is a dose*: disturbance is exactly 0 at the identity rung and rises monotonically with κ in every non-invariant arm; *the class-(a) invariance claim holds in the shipped code*: `cos.krum`’s selection is unchanged in **0** of 120 cells at ρ up to 54.6; and *class (b) is graded rather than binary*, at roughly half the class-(c) disturbance (0.487 vs. 0.867/0.933 at $\kappa=2.0$), which is what arm 4’s prediction turns on.

Frozen predictions. The *primary* test is pooled: across all 16 (arm, κ) cells, the rise in ASR over that arm’s own $\kappa=0$ rung is predicted to be positively rank-correlated with the measured disturbance rate (Spearman, one-sided, $\alpha = 0.05$). Its predictor is a *measured property of the statistic*, not the identity of the defense and not its standalone effectiveness, both held fixed within every arm. The *secondary* predictions are the per-arm shapes of Table A8, tested by a one-sided Jonckheere–Terpstra trend test across the four ordered rungs at $n=5$, with arm 3’s “flat” scored as an *equivalence* claim at a 0.15 margin fixed and justified in advance (under a quarter of the +0.629 within-defense gap for Krum (the $n=3$ value on record at the freeze; the later top-up puts it at +0.716, so the margin is if anything conservative) and $0.300 + 0.15 = 0.450 < 0.5$, so “within the margin” and “still suppressing” coincide for this arm) and arm 4’s class-(b) signature being the *shallower slope*.

One prediction we deliberately do not make. Theorem 8(1)’s ordering-survival condition $\rho < S$ would give arm 4 a crossing rung fixed by a measured S , but **no frozen prediction here uses S** : we measured it and it is **not a stable per-attack constant**, ranging 0.00–12.59 on model-scaling (median 0.57) and 0.61–1.63 on the pixel backdoor (median 0.92) across the 12 rounds that sampled an adversary, so the $S = 10$ used illustratively in §C.2 is a stipulation rather than a measurement. This is a limitation of our own theorem that the dose-response measurement surfaced, reported as one (L5) rather than quietly left unchecked.

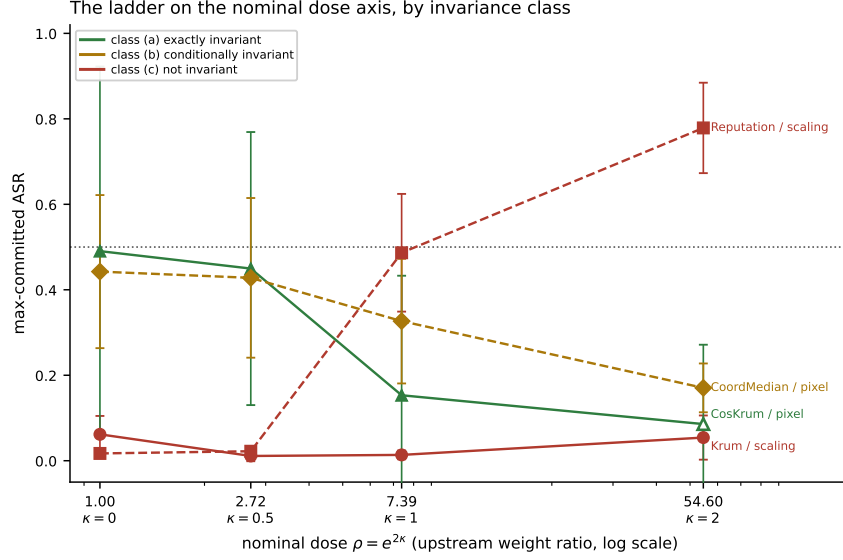


Figure A2: **The ladder and its frozen predictions.** The 16 cells on the nominal dose axis $\rho = e^{2\kappa}$ ($n=5$ per point, Student-t 95% intervals), coloured by Proposition 10 class. Of the two class-(c) arms whose statistics the framework’s fragility claim is stated over, only reputation rises; ASR *falls* with the dose in three of the four arms, against the predicted rises. Hollow markers fail the 0.35 clean-accuracy gate and are uninterpretable. Drawn by `paper/figures/dose_response_figure.py`, which recomputes every value from `results/` and imports the arm construction from the scoring script so the two cannot diverge.

E.1 OUTCOME OF THE CONFOUNDED DESIGN: ONE ARM CONFIRMED, TWO REFUTED, ONE INDETERMINATE, AND THE POOLED TEST FAILS

All 80 runs completed (16.5 h wall clock). Table A9 is the whole ladder; Figure A2 plots it on the nominal dose axis against the frozen shapes and Figure A3 on the independently measured disturbance axis with the real-transform anchors, scored by `analyze_dose_response.py` against the rules frozen at e711a95, every number recomputed per seed from `results/dose_response/`. The **primary** pooled test does not confirm: across the 16 cells, the rise in ASR over each arm’s own $\kappa=0$ rung correlates with the measured disturbance at Spearman $\rho_s = +0.351$, one-sided $p = 0.091$, so the frozen criterion ($\rho_s > 0$ at $p < 0.05$) is not met and **we report the primary prediction as refuted**. Of the four frozen shapes one is confirmed, two are refuted, and one is indeterminate; **in three of the four arms ASR falls as the dose rises**. This is the round’s headline result and it is a negative one. It is also the outcome the pre-registration anticipated and named in advance: “pooled correlation absent while individual arms move: the mechanism is defense-specific and measured disturbance is not a transferable predictor.”

Table A9: The dose–response ladder as measured, $n=5$ per cell, per-attack mean ASR with mean clean accuracy in parentheses. *Predicted* is the shape frozen at commit e711a95 before any run; $p_{\uparrow}$ is the one-sided Jonckheere–Terpstra probability for an *increasing* trend across the four ordered rungs, which is the only direction any frozen rule tests. Verdicts apply the frozen criteria unchanged (Table A8 and the supplement § S5); an arm satisfying neither its confirmation nor its refutation criterion is reported as *indeterminate* rather than scored in our favour. [†] below the 0.35 clean-accuracy gate: a collapsed model, not suppression. Student-t 95% intervals and per-seed values are in the supplement (§ S7).

arm (d_2 / attack)	predicted shape	mean ASR (clean accuracy)				JT verdict $p_{\uparrow}$
		$\kappa=0$	0.5	1.0	2.0	
1 krum / scaling	monotone rise	0.062 (0.57)	0.011 (0.65)	0.014 (0.68)	0.054 (0.62)	0.63 refuted
2 reputation / scaling	monotone rise	0.017 (0.78)	0.022 (0.78)	0.487 (0.77)	0.779 (0.70)	1.7×10^{-6} confirmed
3 cos_krum / pixel	FLAT (control)	0.490 (0.61)	0.450 (0.57)	0.153 (0.47)	0.086 (0.22) [†]	0.999 indeterminate
4 coord_median / pixel	rise, shallower	0.443 (0.77)	0.428 (0.76)	0.327 (0.75)	0.170 (0.69)	0.998 refuted

Arm 2 confirms: the ladder yields one prospective confirmation, while the pooled prediction is rejected. Reputation under model-scaling rises monotonically from 0.017 at the identity rung to 0.487 at $\rho = 7.39$ and 0.779 at $\rho = 54.6$ (Jonckheere–Terpstra $z = +4.64$, one-sided $p = 1.7 \times 10^{-6}$; permutation $p < 5 \times 10^{-5}$),

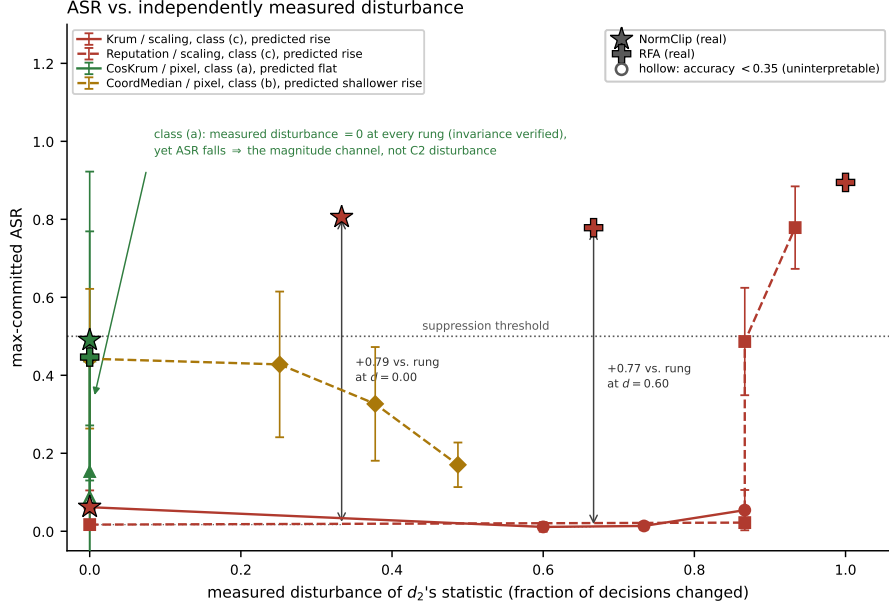


Figure A3: **Why a transform’s flip rate is not a sufficient statistic for what it does.** ASR against the *independently measured* disturbance of each arm’s own statistic ($n=5$, Student-t 95% intervals), the abscissa fixed before the freeze. Stars and crosses are the two *real* upstream transforms measured on these same (d_2 , attack) cells (NormClip, RFA), placed at their own measured disturbance rate: at matched disturbance they sit +0.77 and +0.79 ASR *above* the synthetic curve, while their realized weight ratios (mean 3.61 and 11.93, maximum 4.90 and 17.38) fall *inside* the ladder’s span, which is why we conclude that flip rate does not determine what a transform does to a defense. The class-(a) arm’s four rungs coincide at disturbance exactly 0 (0/120 selections changed, maximum relative score drift 1.1×10^{-4}) and its ASR falls anyway: Lemma 2’s magnitude channel, not C2 disturbance.

while clean accuracy moves only $0.777 \rightarrow 0.701$, so what is lost is suppression, not the model. Same d_2 , same attack, same standalone baseline at every rung: nothing about effectiveness varies, and $55\times$ dispersion in a *positive per-client rescaling* (a transform the criterion treats as benign for invariant statistics) carries a defense from 0.017 across the 0.5 threshold. Two qualifications keep this in proportion. It validates the *destruction* half of C2’s mechanism (a non-invariant statistic, actually disturbed, loses suppression); the *preservation* half (invariance $\Rightarrow$ suppression survives) is arm 3, which came out indeterminate. And even within this arm the measured disturbance is not what orders the curve: it saturates at 0.867 by $\kappa=0.5$, where ASR is still 0.022, and the jump to 0.487 happens at the next rung at the same measured disturbance, so the *dose* moves ASR here while the flip rate does not.

Ecological validity: the real transforms do not lie on the synthetic curve. Pre-registered caveat (iv) required this check, and it fails in an informative way. RFA $\rightarrow$ Krum sits at ASR 0.779 with measured disturbance 0.667, where the synthetic ladder’s nearest rung (0.600) gives 0.011: a residual of +0.767. NormClip $\rightarrow$ Reputation sits at 0.806 with disturbance 0.333, against 0.017 at the identity rung: +0.789. Two anchors do land: RFA $\rightarrow$ Reputation, 0.895 at disturbance 1.000 against 0.779 at 0.933 (+0.116), and both *cos_krum* anchors, which coincide with its curve (+0.000 and -0.044) because that statistic is invariant under all three transforms. This is not a range mismatch: read off the same live rounds, NormClip realizes ρ of mean 3.61 (max 4.90) and RFA mean 11.93 (max 17.38), both inside the ladder’s $\{1.00, 2.72, 7.39, 54.60\}$ span. The conclusion is about our abscissa: **the rate at which a transform flips a defense’s decision is not a sufficient statistic for what that transform does to the defense**: two transforms at matched flip rate differ by up to 0.77 in ASR. **And the axis they are unmatched on is now measured, not conjectured.** Adversary-correlated dispersion $\Delta_c := \mathbb{E}[c_i | i \in A] - \mathbb{E}[c_i | i \notin A]$ (§C) separates exactly these two anchors from the ladder: RFA reaches $\Delta_c = -1.21$ and NormClip -0.72 , against -0.34 at the ladder’s most dispersed rung: $2\text{--}3.6\times$ more extremely targeted at ρ that is 3.7 and $13\times$ smaller, so **matched on the statistic, badly unmatched on the targeting, and ρ does not order Δ_c .** This rescues no predictor: Δ_c ’s magnitude does not order suppression either; RFA is the most extremely targeted of the four real transforms and *rfa* $\rightarrow$ *krum* still fails at 0.779, while FoolsGold’s Δ_c is *positive* (+0.663). These figures are **post-hoc and exploratory**, computed with no ASR and rescoring nothing, and are tabulated in the supplement (§ S9). Arm 4 has no real-transform anchor at all, since *coord_median* was not on the metric-swap menu, so its ecological validity is untested.

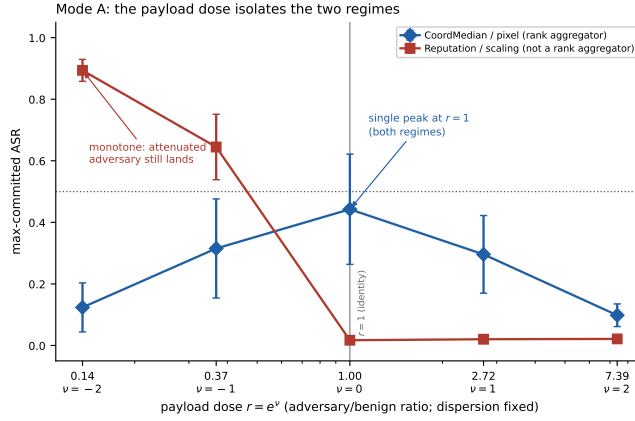


Figure A4: **Mode A, the payload dose: the two mechanism-preserving regimes, and where they hold.** With dispersion fixed and only the adversary/benign weight ratio r swept, `coordmedian` (a rank aggregator, the case Theorem 8’s regimes are stated over) is **single-peaked at $r=1$** as the theorem predicts (over-attenuated adversaries carry no payload, over-amplified ones are discarded) while reputation is monotone, off-spec and refuted for that shape. Mode A is the Δ_c -swept companion to Mode S’s Δ_c -pinned design (Proposition 12); Mode S itself is Figure 1. Bars are Student- t 95% intervals; numbers recomputed per seed by `analyze_targeted_dose.py`.

What we now claim less of. C2’s causal content is established prospectively for one statistic (distance to the client median) and refuted for another (summed pairwise distance) at p up to 54.6; the pooled claim that measured disturbance predicts loss of suppression *across* defenses is refuted, and we no longer make it. Within-defense disturbance remains the only design the testability boundary admits, and §C.1’s general statement stands untouched by this outcome; it is a statement about admissible designs, and this is the admissible design, run and reported as it landed. What changes is the empirical status of C2: one confirmed arm, not a validated screening mechanism.

E.2 THE TARGETED INTERVENTION: ISOLATING C2 FROM THE ATTENUATION CHANNEL

What Mode S does to the ladder’s confirmation. Reputation was the ladder’s one confirmed arm (§E.1); Mode S shows most of that confirmation was attenuation. With the adversary’s share held fixed, reputation’s rise at $\rho=54.6$ shrinks from the ladder’s $+0.762$ to $+0.178$ (one-sided Welch $t = -5.22$, $p=0.0017$ that it is smaller), and what remains tracks a small *admission* change ($0.000 \rightarrow 0.020$) rather than the large decision change ($0.000 \rightarrow 1.000$), so H-admission is confirmed here too (Table A10). The confound is thus *explained*, not explained away: a genuine, much smaller, admission-driven effect survives it. The class-(a) control cuts the other way and refutes us: `cos_krum`’s selection is bit-identical at every rung (decision and admission both 0.000), yet ASR is 0.584 at the identity (it never suppresses) and *falls* 0.425 across the ladder ($p_{\downarrow}=0.0008$; the $\kappa=2$ rung is below the accuracy gate). Exact invariance of the statistic neither implies suppression nor holds it fixed; the magnitude channel moves it regardless. We report this as a refutation of Proposition 10’s practical content, not a confirmation of its algebra.

Mode A: the two regimes, and where they hold. The payload dose gives Theorem 8 a single falsifiable shape: ASR single-peaked at $r=1$, since an over-attenuated adversary carries no payload (Regime B) and an over-amplified one is discarded by a rank aggregator (Regime A). It is **confirmed for `coordmedian`**, the rank aggregator the regimes are stated for: ASR rises $0.124 \rightarrow 0.315 \rightarrow 0.443$ into $r=1$ and falls $0.443 \rightarrow 0.296 \rightarrow 0.098$ past it, both sides significant ($p_{\downarrow}=3.0 \times 10^{-4}$ amplified, 6.4×10^{-4} attenuated) at accuracy ≥ 0.74 throughout (Figure A4). It is **refuted for reputation**, which is monotone: ASR is maximal at the most attenuated rung (0.894 at $r=0.135$) and near zero once $r \geq 1$, because an attenuated adversary reads as *consistent* to a reputation score and stops being down-weighted. Reputation is not a rank aggregator, so the two-regime shape was never licensed for it; the split is the theorem behaving exactly where it applies and not where it does not.

A pre-registered replication on a second downstream defense, and what it refutes. One arm cannot establish which channel matters, and the two Mode-S arms above move together on both candidate predictors: reputation is higher on *both* decision and admission change and also had the larger rise, so nothing in that pair separates them. `coordmedian` under the pixel backdoor is *rank-discordant* and therefore does: of the interpretable Mode-S arms it has the **smallest decision change** (0.482) and the **largest admission change** (0.033), so the decision reading predicts its rise is the *smallest* of the three and the

Table A10: Mode S (statistic-only), $n=5$ per arm ($n=8$ for `cos_krum`), per-attack mean ASR with the two measured channels. The decision channel moves far more than ASR throughout. [†] below the 0.35 accuracy gate. [‡] the pre-registered replication arm, *rank-discordant* by design (smallest decision change, largest admission change of any interpretable arm) and scored against thresholds frozen separately at 06d9bce before `results/dose_replication/` existed: $\Delta=+0.098$ falls below reputation's $+0.178$, so the cross-arm admission *ordering* is refuted while the negative replicates on a different invariance class (§E.2). Verdicts otherwise scored against the readings frozen at 5130cec; per-seed values in the supplement (§ S7).

arm	channel	$\rho=1$	2.72	7.39	54.6	test	verdict
krum / scaling	ASR	0.062	0.016	0.023	0.036	$p_{\uparrow}=0.70$	H-admission confirmed, H-stat. refuted
	decision	0.000	0.533	0.800	0.733		
	admission	0.000	0.000	0.000	0.000		
reputation / scaling	ASR	0.017	0.019	0.025	0.195	rise $+0.178$ vs. ladder $+0.762$ ($p=0.0017$)	H-admission confirmed (mostly atten.)
	decision	0.000	0.933	1.000	1.000		
	admission	0.000	0.002	0.008	0.020		
cos_krum / pixel	ASR	0.584	0.411	0.295	0.159 [†]	$p_{\downarrow}=8 \times 10^{-4}$ $ \Delta =0.425$	refuted (Prop. 10 practical)
	decision	0.000	0.000	0.000	0.000		
	admission	0.000	0.000	0.000	0.000		
coord_median / pixel [‡]	ASR	0.443	0.442	0.490	0.541	rise $+0.098$ < $+0.150$ ($p_{\uparrow}=0.069$)	ordering refuted; negative repl.
	decision	0.000	0.232	0.358	0.482		
	admission	0.000	0.005	0.013	0.033		

admission reading predicts the *largest*. It is also a structurally different defense, a coordinate-wise order statistic (Proposition 10(b)) against Krum's selector (class (c)), and it suppresses the pixel backdoor at usable accuracy, so there is suppression to lose. Thresholds were frozen against the two published rises (confirm if $\Delta > +0.178$, refute if $\Delta < +0.150$, indeterminate between) at commit 06d9bce before `results/dose_replication/` existed, with the identity rung imported bit-identical and 15 new runs spent (`experiments/pre_registration_dose_replication.md`).

The outcome is $\Delta = +0.098$: the admission ordering is refuted. The arm with the largest admission change rose *less* than an arm with a smaller one, so **admission magnitude does not order outcomes across arms**. Neither does the decision reading, which predicted the *smallest* rise of the three when $+0.098$ is the middle one, so this arm refutes a rank prediction from *each* candidate channel. The pooled model agrees: re-fit with these cells it is still not confirmed ($R^2=0.239$; the admission coefficient's one-sided p moves $0.055 \rightarrow 0.041$, but the adversarial-share coefficient is negative, so the pre-specified conjunction fails). **What does replicate is the negative:** a decision change of 0.482 costs no suppression by the frozen margin, at mean accuracy 0.698, so the accuracy gate does not bind and the flagship result now holds on two downstream defenses in *different* invariance classes. Two qualifications, the first pre-specified in the freeze and the second post-hoc: this cell's identity ASR is 0.443, so equivalence is to the *identity rung* and not to a low absolute ASR (an "every rung < 0.5 " criterion has no headroom here and was deliberately not used); and the rise is positive in 5/5 seeds (paired $t=3.39$, one-sided $p=0.014$, *not* pre-registered), so on this class the cost of disturbing the decision is **small and consistent rather than zero**. All numbers are recomputed per seed by `analyze_dose_replication.py`, which imports its rung builder and trend test from the frozen suite's scorer.

The confounded design gets the causal sign wrong. This is the paper's central empirical result. On this identical cell (matched aggregate displacement, decision change and clean accuracy, Table A3), **the confounded design does not lose power, it gets the sign wrong:** the outcome-gated ladder moves -0.272 ($p_{\downarrow}=0.0015$; $0.443 \rightarrow 0.170$) where the instrument moves $+0.098$. Closing the attenuation channel does not sharpen a weak positive effect, it **inverts it**, so a criterion calibrated against the confounded contrast would have been calibrated against the confound, and a cross-defense contrast would have read the fall as evidence about the statistic. That is why the identification boundary of §C.1 is not a technicality about power.

What the two arms jointly establish, stated formally. The arms above and the class-(a) control (`cos_krum`) are between them a *constructive* answer to the question C2 was written to settle. Both halves are witnessed by runs already reported, so the following formalizes measurements rather than adding a claim.

Proposition 11 (Statistic-level non-identifiability). *Within the positive per-client rescaling class there exist transform families T_1, T_2 and downstream defenses d_2 such that (a) T_1 and T_2 induce arbitrarily different statistics and decisions at d_2 while inducing the same admitted adversarial mass and the same suppression; and (b) T_1 and T_2 induce the same admitted adversarial mass (indeed the same statistic and the same decision) while inducing different suppression. Consequently no predicate of the downstream statistic alone determines the admitted adversarial mass, and no predicate of the admitted mass alone determines suppression.*

Proof. Both halves are exhibited by runs reported in this section. (a) Mode S into Krum under model-scaling, $\kappa=0$ against $\kappa=2$: ρ moves from 1 to 54.6 and Krum’s decision changes in up to 0.80 of rounds, while the support of the admitted adversarial mass changes in none of the 60 measured rounds (a floor on this arm: no adversary is admitted at any rung), the adversarial coefficient share is constant at 0.266667, and suppression shows **no evidence of a practically meaningful change under the pre-specified ± 0.15 equivalence margin** ($|\Delta|=0.026$). (b) `cos_krum` under the pixel backdoor: by Proposition 10(a) its statistic is invariant, and its decision and admission are measured bit-identical (0.000) at every rung, yet ASR falls 0.173 by the first rung alone, at $\rho=2.72$, where the accuracy gate still passes ($0.599 \rightarrow 0.567$; paired $t=-1.976$, one-sided $p=0.044$, falling in 5/8 seeds). $\square$

The corollary is the paper’s title claim: no metric defined *solely* on preservation of the downstream statistic can identify security equivalence, because (a) exhibits statistic disturbance without security change and (b) exhibits security change without statistic disturbance. **Scope, stated immediately:** this is a non-existence claim *within the positive-rescaling class on these defenses and attacks*, established by exhibited counterexample: **not** a claim that no metric of any kind can order security, and not a claim about transforms outside the class.

Proposition 12 (ρ and Δ_c are independently controllable). *Within the same class, the statistic-level dispersion $\rho = \max_i c_i / \min_i c_i$ and the adversary-correlated dispersion $\Delta_c = \mathbb{E}[c_i | i \in A] - \mathbb{E}[c_i | i \notin A]$ are not functionally related: there is a family with $\Delta_c \equiv 0$ exactly while ρ ranges over $[1, 54.6]$, and a family with the statistic’s dispersion held fixed while Δ_c ranges over $[-1.138, +2.469]$. Hence neither quantity determines the other.*

Proof. By construction, with the realized values tabulated in the supplement (§ S9) and computed with no ASR involved. Mode S pins every adversarial coefficient at $c=1$ and spreads only the benign clients, giving $\Delta_c \equiv 0$ at every ρ up to 54.6; Mode A holds the benign weights uniform and sweeps only the adversary-to-benign ratio r , moving Δ_c over $[-1.138, +2.469]$ at fixed statistic dispersion. $\square$

This is why the two instruments were the right ones, and it makes the supplement’s §S9 warning precise: **a transform reported by ρ alone leaves Δ_c entirely free**, so ρ cannot be a sufficient statistic for what the transform does to a defense. It does *not* follow that Δ_c is a predictor: §E.1 reports that its magnitude fails to order suppression either.

The dissociation is not Krum-specific: all four aggregators, all four channels. Table A11 collects the endpoint change ($\kappa=0$ against $\kappa=2$) in each measured channel under Mode S for every downstream defense in the paper. **Recomputed, not transcribed:** every cell is emitted by `build_channel_table.py`, which only reads frozen artifacts under `results/`, `admission_measurement.json` and `femnist_admission.json` for the channel columns, the `targeted_dose/`, `dose_replication/` and `dose_femnist/` summaries for the ASR column; it also asserts that the identity rung displaces nothing (0.00e+00 for all six rows). The sixth row’s aggregate column comes from `displacement_decomposition.json` and its ASR from `score_only/summary.json`, both read-only; the emitter refuses to merge that arm into the ASR sources it shares cell keys with, so it cannot shadow the first row’s published number. The influence column is **aggregator-specific by necessity** (a selector has no weights) and its instantiation is: adversarial share of the Weiszfeld/reputation weight mass for the averager, adversarial share of selected coordinates for the order statistic, and the indicator that an adversarial update is selected at all for the two selectors. The fifth row replicates the first row’s arm on a second dataset and architecture (FEMNIST with `simple_cnn`) with **its channels measured prospectively, before its ASR existed**; the result is reported below.

A second dataset and a second architecture, pre-registered. The fifth row’s arm is the first row’s arm re-run with EMNIST-byclass (62 classes, $1 \times 28 \times 28$) and `simple_cnn` in place of CIFAR-10 and `cifar_cnn`, everything else held: Mode S, Krum as d_2 , the committed model-scaling attack, the same four frozen rungs $\kappa \in \{0, 0.5, 1, 2\}$. The decision rule lives in `experiments/pre_registration_dose_femnist.md`, git-committed before any FEMNIST ASR existed, and the runner refuses to start unless its `PREREG_COMMIT` matches that hash.

The premise was measured first, with no ASR anywhere. Krum’s selection changes on 0.643–0.714 of rounds across the three disturbed rungs while the support of the adversarial mass it admits is unchanged in all 56 measured rounds (again a floor: no adversary is admitted at any rung), so the arm has a real disturbance to test. Had the decision change come out near zero the arm would have been reported illegible rather than run: that is the `cos_krum` failure mode, and this check is what distinguishes the two.

Suppression shows no evidence of a practically meaningful change. $\Delta = -0.017$ (mean ASR $0.027 \rightarrow 0.010$) against the frozen ± 0.15 equivalence margin, beside CIFAR-10’s -0.026 on the same cell, with mean clean accuracy *rising* $0.767 \rightarrow 0.835$ so the accuracy floor binds on no rung. Under the frozen rule this is

Table A11: **The four channels, all four aggregators, under Mode S.** Endpoint change $\kappa=0 \rightarrow \kappa=2$. **The first column, Δ agg., is the relative displacement of the aggregate the defense emits,** $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$: what enters training, and **not** the value of a scoring statistic; the measured channels are aggregate displacement, decision, admission and influence. The dissociation between the *decision* channel and the *admission* channel is not a property of one aggregator: it holds for a selector, a cosine-invariant selector, a coordinate-wise order statistic and a weighted averager. The table does **not** order the arms ($\Delta\Lambda_a$ and ΔASR disagree in rank, which is §E.2’s refutation stated in one row-pair) and it is *not* an out-of-sample test: all rows are Mode-S ladders reported elsewhere in this paper. **The sixth row is the score-only control:** the same arm as the first with the magnitude channel closed, so its Δ agg. is the re-selection component alone and its decision, admission and influence columns are the first row’s own numbers, score-only leaving the selection untouched.

Aggregator	Kind	Δ agg.	Δ dec.	Δ adm.	$\Delta\Lambda_a$	Δ ASR
Krum	selector (pairwise distance)	0.892	0.733	0.000	0.000	-0.026
Reputation	weighted averager (cross-round state)	0.659	1.000	0.000	0.020	+0.178
Cosine-Krum	selector (cosine distance)	1.297	0.000	0.000	0.000	-0.425
Coord. median	coordinate-wise order statistic	0.670	0.482	0.000	0.033	+0.098
Krum (FEMNIST)	selector (pairwise distance)	0.842	0.643	0.000	0.000	-0.017
Krum, score-only	selector, magnitude channel closed	0.986	0.733	0.000	0.000	-0.023

a **replication on a second dataset and a second architecture**: the pre-registration fixes that wording and explicitly forbids “generalizes”, which one further dataset does not license.

Three disclosures, each of them from the freeze rather than from hindsight. (i) $n=3$ seeds per rung, not 5, fixed in advance to match the FEMNIST payoff matrix’s own trial budget. (ii) The secondary Jonckheere–Terpstra test was specified to detect a monotone *rise* (the outcome that would have refuted the flat prediction with attenuation already closed by construction) and the observed trend is a *fall* ($J=16.0$, $z=-1.563$, $p_{\uparrow}=0.941$). We report it and we do **not** score it as support for flatness; the pre-registration states in advance that at $n=3$ a null from this test is not evidence of flatness. (iii) The two branches had unequal headroom. With the identity rung at ASR 0.027, refuting us required mean $\text{ASR}(\kappa=2) > 0.177$ against a ceiling of 1 and was fully reachable, whereas the indeterminate branch ($\Delta < -0.15$) was arithmetically unreachable. **Equivalence here is to the identity rung, not to a low absolute ASR.**

Harness sanity, which is an identity check and not corroboration. The $\kappa=0$ rung is identity◦Krum, i.e. Krum alone, and it lands *bit-identically* on the FEMNIST Krum figure from our earlier payoff-matrix sweep (0.0269784552 at accuracy 0.7670795973). Both pipelines are deterministic given a seed and both run three trials from base seed 42, so the agreement shows they execute the same protocol rather than a similar one: the strongest form the pre-registered check can take, and for the same reason **not** independent corroboration of that figure: the same computation run twice cannot corroborate itself.

What the instrument does not hold fixed, named in full, and what it pins exactly. Mode S pins the *adversarial* coefficients; it does not pin the benign ones and was never claimed to. So the dose also moves (a) Krum’s pairwise distances, hence the statistic itself; (b) *which* benign client Krum selects, in 0.533/0.800/0.733 of rounds at $\rho=2.72/7.39/54.6$; (c) the **magnitude and geometry of the update actually aggregated**, since the selected client enters scaled by its own coefficient; and (d) the model trajectory, which diverges across rungs in consequence. Channel (c) is not a conjecture but a **number we already measure and print**: the Δ agg. column of Table A11 is $\|\text{agg}(T(U)) - \text{agg}(U)\|/\|\text{agg}(U)\|$, the relative displacement of the aggregate the defense emits, and it reads **0.892** on this cell. The supplement § S8 splits that 0.892, as vectors, into 0.986 of *re-selection* (a different benign client is chosen, and in 0/11 of the rounds where that happens it is an adversary) and 0.973 of *rescaling* (the chosen client arrives scaled by its own coefficient); for `cos_krum` the displacement is *entirely* rescaling, since 0/15 selections change at any rung. The claim this arm licenses is therefore sharper than a claim about the statistic alone: **a 0.892 displacement of the emitted aggregate, together with decision changes in up to 80% of rounds, leaves suppression inside the pre-specified equivalence margin, provided the adversarial coefficient share does not move. What is pinned is pinned exactly:** $c_{\text{adv}}=1.0$ at every rung with maximum deviation 0.00e+00 and the share constant to 3.5e−07 on CIFAR-10, 3.9e−07 on FEMNIST: float32 norm read-back noise, not slack in the instrument (`measure_admission.py`). Naming (a)–(d) is what keeps the claim to the one channel the instrument closes by construction.

Closing channel (c) as well: the pre-registered score-only control. Channel (c) (the magnitude and geometry of what is actually aggregated) is the one confound Mode S leaves open, so we closed it too rather than argue about it. **Score-only Krum** scores on the transformed stack exactly as before and then aggregates the *untransformed* selected update: the dose can change *which* client contributes and nothing whatever about *what* that client contributes. Statistic and decision are disturbed identically to the uncontrolled arm (the selection is

computed on the same transformed stack, so Δ dec. is the same 0.733) while Δ agg. is no longer 0.892 but the re-selection component alone, 0.986 relative to $\|u_{s_0}\|$ (the supplement § S8), the rescaling component being closed by construction. This is the reviewer-suggested control in its own terms: hold the selected update’s magnitude and direction fixed while perturbing the selection statistic independently.

Frozen before the arm existed. The decision rule is experiments/pre_registration_score_only.md, git-committed at 35788d9 before any score-only ASR existed, and the runner refuses to start unless its own PREREG_COMMIT matches that hash. *Primary:* $\Delta = \text{mean ASR}(\kappa=2) - \text{mean ASR}(\kappa=0)$ against the suite’s existing ± 0.15 equivalence margin, on krum/committed_scaling/CIFAR-10/cifar_cnn, the same four rungs and seeds 42–46. $\Delta > +0.15$ would have said that **the published flat result was partly an artifact of the magnitude channel** and that the paper’s central claim narrows to “with benign magnitudes free to move”: that branch is why the arm was worth three hours of compute. The $\kappa=0$ rung is imported rather than re-run, because at the identity the transform returns the stack unwrapped and score-only Krum is Krum alone; that is an exact claim, not an approximation, and it was verified bit-identically at one seed before the arm started.

The negative survives with the magnitude channel closed. $\Delta = -0.023$ (mean ASR 0.062 $\rightarrow$ 0.039) against -0.026 for the uncontrolled arm on the identical cell: **closing the magnitude channel moves the effect by 0.003.** The refuting threshold 0.212 is nowhere approached, and mean clean accuracy clears the 0.35 floor at every rung (0.565/0.629/0.624/0.630), so no rung is a collapsed model. **The central claim is therefore instrumented against two independent confounds** (attenuation, closed by Mode S, and magnitude, closed here) and the sharpened statement is that a 0.986 displacement of the emitted aggregate together with decision changes in 0.733 of rounds leaves suppression inside the pre-specified equivalence margin.

Three disclosures, from the freeze rather than from hindsight. (i) The secondary Jonckheere–Terpstra test was specified to detect a monotone *rise* (the outcome that would have refuted flatness with attenuation already closed) and the observed trend is a *fall* ($J=58.5$, $z=-1.110$, $p_{\uparrow}=0.867$, permutation $p_{\uparrow}=0.874$). We report it and do **not** score it as support for flatness; the pre-registration states in advance that at $n=5$ a null from this test is not evidence of flatness. (ii) The two branches had unequal headroom. With the identity rung at 0.062, refuting us required mean $\text{ASR}(\kappa=2) > 0.212$ against a ceiling of 1 and was fully reachable, whereas the indeterminate branch ($\Delta < -0.15$) was *arithmetically unreachable*: ASR cannot fall below 0, so the largest possible fall on this cell is 0.062. (iii) **Selectors only.** A weighted averager and a coordinate-wise order statistic emit no single selected client, so “the selected update” is undefined for them and the control does not cover them, which means the coord_median arm carrying the $+0.098$ rise is **not** magnitude-controlled, and neither is reputation’s $+0.178$. Score-only Krum is also an instrument and not a defense: it aggregates an update its own scoring stage never saw.

The corollary the pre-registration staked on cos_krum, and exactly what it settles. Section 5 of the same document asserted, before the run, that score-only cos_krum must be *bit-identical* across rungs: its statistic is exactly invariant under positive rescaling, so the selection cannot move, and with the emitted update untransformed the whole 50-round trajectory must repeat. It does (accuracy 0.528400 and ASR 0.003111 at all four rungs, $|\Delta| = 0.00\text{e}+00$ throughout) and the $\kappa=0$ value reproduces the published cell at that seed exactly, so the two pipelines agree at 0.000000 rather than approximately. **What the check does not do is confirm the arm’s published fall.** It runs one seed, as frozen, and that seed is unrepresentative of the arm: under the uncontrolled dose its own ladder *rises* $+0.224$ where the eight-seed mean *falls* -0.425 . Closing the magnitude channel removes all 0.224 of that seed’s movement, which corroborates the *channel* (for cos_krum the emitted displacement is entirely rescaling, so no other route is open) but attributing the arm’s mean fall to magnitude rests on that decomposition rather than on this check, and settling it empirically would need the arm’s other seven seeds. Expanding a pre-registered seed count after seeing the data is precisely what pre-registration exists to prevent, so we report the invariance as confirmed and the arm-level attribution as **corroborated mechanically, not established at the arm mean.**

What this settles, and what it does not. The intervention delivers the controlled isolation of C2 the two cross-defense suites could not: within a single defense, with attenuation excluded by construction *and*, in the score-only arm, the magnitude of the aggregated update held exactly fixed as well, **substantial disturbance of the statistic d_2 reads need not produce a corresponding change in suppression once the adversarial coefficient share is held fixed** (on the adjudicating Krum arm under Mode S and, by pre-registered replication, both on a downstream defense in a different invariance class and on a second dataset and architecture) while *admission* of adversarial input is the channel the evidence points to. The statistic-level reading the framework was first written in is refuted; the admission-level reading is confirmed as a *direction* in these interventions rather than established in general. It does *not* rescue the framework’s positive direction wholesale: the pooled two-channel model over all 26 cells is **not** confirmed ($R^2=0.239$), and the replication refutes its cross-arm ordering outright, so admission is a better single predictor than the statistic without being a quantitative one. It addresses neither recall nor discovery (FG $\rightarrow$ RFA remains a false negative), and nothing in §E.1 is rescored: its pooled refutation, its two refuted arms and its indeterminate arm stand exactly as reported.

F THE COMPOSITION THE SCREEN CANNOT CERTIFY: FOOLSGOLD→RFA

This section is the empirical witness for Proposition 7. FoolsGold→RFA has the *lowest observed committed ASR* of every composition evaluated here (0.093), and **our own criterion classifies it as a false negative**: neither constituent suppresses the attack alone (FoolsGold 0.732, RFA 0.845), so its suppression is *emergent* in the sense of Proposition 7 and C1 cannot hold. **We report it in detail because it is the sharpest illustration of the screen’s scope, not because it is a flagship**: any criterion that certifies inherited suppression must miss compositions of this kind, which is what bounds recall at the reported 40%. The characterization below is therefore descriptive (what this composition does, measured at $n=30$) and is not evidence for the criterion. **The $n=30$ measurement corrected our own earlier figure downward in reliability, and we report the correction rather than the settled number alone**: the per-seed distribution is right-skewed (median 0.060, mean 0.093, sample skew +2.07, 8/30 seeds at or above 0.10, maximum 0.380), and seeds 42–46 fall in its left tail, so the $n=5$ value we first led with understated the pair by a factor of 1.8. Any 5-seed figure for this composition should be read against that spread. We avoid calling it security-maximal: it also sustains the *highest* ASR under our adaptive adversary (0.167, versus 0.093 committed), and it carries a 50% accuracy cost, so it is the right choice only where utility is nearly irrelevant. We validate at $n=30$ seeds with a full adaptive attack suite ($N=10$, $K=5$, $f=0.2$, 50 rounds), in Table A12:

Table A12: FG→RFA under base and adaptive attacks. Base: $n=30$, reported as median [IQR] with P95 tail; adaptive: $n=5$, mean±std. Clean accuracy: test-set accuracy on non-triggered inputs. No-attack baseline for RFA alone: 0.737.

Attack	Config	ASR	P95 / max	Acc	Δ_{mix}
<i>Base composition ($n=30$; median [IQR])</i>					
	scaling	0.060 [0.048, 0.096]	0.253 / 0.380	0.497	−0.778
	pixel	0.052 [0.035, 0.087]	0.127 / 0.200	0.492	−0.683
	oracle	0.056 [0.044, 0.090]	0.189 / 0.236	0.494	−0.839
<i>Projection adaptive ($n=5$)</i>					
	$\epsilon=2.0$ (best)	0.079 ± 0.042	− / 0.160	0.495	
<i>Criterion-aware joint-constraint ($n=5$)</i>					
	$\epsilon=1.0$, decorr	0.167 ± 0.069	− / 0.268	0.562	
	$\epsilon=2.0$, decorr	0.130 ± 0.030	− / 0.185	0.506	
	$\epsilon=2.0$, no decorr	0.050 ± 0.008	− / 0.059	0.494	
<i>Neurotoxin ($n=5$)</i>					
	top-20%, scale = 2	0.048 ± 0.009	− / 0.057	0.495	
<i>Consensus-shift ($n=5$)</i>					
	rate = 0.05	0.045 ± 0.012	− / 0.059	0.495	

Results. Median ASR across all base attacks ($n=30$): 0.056 [IQR 0.044–0.090]; P95 worst-case: 0.253 (committed_scaling). Every adaptive strategy performs *worse* than the base attacks: the composition achieves < 10% median ASR under the full suite including the criterion-aware adversary. The accuracy tradeoff (0.50 vs. 0.74 for RFA alone) and Pareto alternatives are discussed below (Table A13).

Composition-aware adversary (end-to-end). We build an adversary that *knows* the full FG→RFA pipeline is deployed (Kerckhoffs) and optimizes against the complete composition, solving both constraints simultaneously: pairwise cosine similarity below FoolsGold’s down-weighting threshold, and ℓ_2 distance to the geometric median below RFA’s rejection radius. Any gradient-based optimizer faces the same tension (coordinating a backdoor raises FG-similarity, while staying near the median caps the perturbation and weakens the backdoor) and our analytical construction reaches the same boundary, the Weiszfeld iteration being contractive and the gradient at convergence determined by the final-iterate reweighting the joint constraint directly constrains.¹ The ablation is informative: without decorrelation ASR stays at baseline (0.050), RFA-evasion alone failing because FoolsGold is the binding constraint; with it the adversary gains partial signal (0.130–0.167), which Theorem 8 explains as bounded, each adversary’s block-restricted delta being too small to move the median outside the benign cluster.

Contrast with rep→tm. FG→RFA (median ASR 0.056) provides near-absolute suppression; rep→tm (oracle ASR 0.316) provides strategic dominance against all tested adversary policies but does not suppress committed.pixel (0.679). The two serve different roles.

¹A fully gradient-based optimizer differentiating through 50 Weiszfeld iterations would face identical binding constraints at convergence; the analytical formulation is equivalent without requiring a differentiable relaxation.

Accuracy–security tradeoff. FG→RFA achieves the lowest ASR of any tested composition but imposes a substantial clean-accuracy cost (0.50 vs. 0.74 for RFA alone). This cost is intrinsic to FoolsGold’s aggressive per-client downweighting, not to composition itself. Table A13 places FG→RFA on the accuracy–security Pareto frontier alongside alternative compositions:

Table A13: Accuracy–security Pareto frontier. Max ASR = worst-case mean across committed attacks. All at $N=10$, $K=5$, $f=0.2$, cifar_cnn.

Defense / Composition	Clean Acc	Max ASR	Regime
Temporal mix (NC/fedavg)	0.79	0.96	No security
RFA alone	0.74	0.91	Bimodal; unreliable
Rep alone	0.78	0.84	Scaling-only
Rep→CM	0.77	0.35	Moderate
FG→CM	0.66	0.11	Strong
FG→RFA	0.50	0.06	Min ASR (evaluated)

FG→RFA is the correct choice under a *security-first* objective (e.g., safety-critical FL where any backdoor is unacceptable). For operators with accuracy constraints, FG→CM (acc 0.68, max ASR 0.107) provides a Pareto-superior tradeoff: statistically indistinguishable suppression (Welch $p = 0.52$) at 37% more accuracy. Rep→CM (acc 0.76, max ASR 0.423) offers near-baseline accuracy with moderate security. The criterion certifies FG→CM and Rep→CM; FG→RFA reaches the lowest ASR of the three but fails the tightened per-attack C1 and is therefore a false negative of our own criterion rather than a certified recommendation. The choice among them is an operator-specific utility decision, not a criterion failure.

G RELATED WORK

The contrast in one sentence. Prior work asks whether a defense withstands an attack. We ask whether an *evaluation criterion* can causally attribute the security of a composition to preservation of the downstream mechanism.

Byzantine-robust aggregation, studied in isolation. Krum (Blanchard et al., 2017), Trimmed Mean (Yin et al., 2018), RFA (Pillutla et al., 2022), FoolsGold (Fung et al., 2020), FLTrust (Cao et al., 2021) and FLAME (Nguyen et al., 2022) are each proposed and evaluated as a *single* stage. We propose no new defense; we reason about *ordered pairs* of existing ones, which is a different object: the upstream stage’s output is the downstream stage’s input, so the question is no longer whether a mechanism works but whether it *survives* a transform applied before it. Ensemble adversarial training (Tramèr et al., 2018) *assumes* that upstream processing preserves downstream signals; we make that assumption explicit, checkable, and (as it turns out) **the wrong abstraction for security**.

Adaptive attacks are why an inherited-robustness guarantee is not a security guarantee. Optimized model poisoning defeats Krum, trimmed mean and FoolsGold once the attack is chosen *against* the defense (Shewalkar & Houmansadr, 2021; Bhagoji et al., 2019), and the same holds for distributed and durable backdoors (Xie et al., 2020; Wang et al., 2020; Zhang et al., 2022). Our C1 quantifies over a *committed* attack by construction, so this literature bounds what any inherited-suppression criterion (ours included) can promise. Our single whitebox composition-aware stress test (§F) is reported in order to *measure* that gap (0.051 → 0.167 seed-matched, 3.3×), not to defend the criterion against it; a screening condition stated against an adaptive adversary is what the framework most lacks.

Evaluation methodology, which is where this paper sits. The adjacent precedent is adversarial robustness, where reported numbers proved unreliable until evaluations were required to be adaptive (Croce et al., 2021). Our point is methodological in the same way but about **identification** rather than attack strength. Wherever a screen gates candidate pairs on standalone effectiveness and then reads the composition’s outcome as evidence about the *mechanism*, Proposition 6 shows that inference cannot work at *any* sample size: the counterfactual is excluded by the gate, not undersampled. **We do not claim this is how the FL literature generally proceeds**, and we can name one external instance rather than a practice: Fenaux et al. (2025) pair a defense effective against large-deviation updates with one effective against small-deviation updates, and validate the pairing by the combined defense’s attack success rate, which selects components by standalone effectiveness and then reads the composition’s outcome. Composition itself is named as an open problem in the field’s survey of them (Kairouz et al., 2021). The claim we do make is structural rather than sociological, and its clearest instance is *our own* screen: Lemma 1 shows $C0 \wedge C1$ *force* the downstream defense to suppress the attack alone, so the gate is constitutive of an inherited-suppression screen and not an incidental filter on it. Statements of a composition-preservation intuition in that literature are informal; what is new here is the *level hierarchy* at which the intuition fails (Definition 2) and the class of designs that can test it (§C.1, §E.2). The causal-inference template we invoke (conditioning on a common descendant of treatment and outcome) is standard outside this literature (Elwert & Winship, 2014; Pearl, 2009); what is not standard is noticing that an inherited-suppression

screen instantiates it, because such a gate reads as a reasonableness filter (only study defenses that work) rather than as the step that removes identification. That is the reading we held ourselves (§6.1).

Attacks and defenses. Backdoor attacks (Chen et al., 2017; Bagdasaryan et al., 2020; Wang et al., 2020; Xie et al., 2020) and robust defenses (Pillutla et al., 2022; Sun et al., 2019; Blanchard et al., 2017) form the landscape. Beyond our norm-based menu, certified robustness (Xie et al., 2021) and filtering defenses (Nguyen et al., 2022; Fung et al., 2020) build stronger single mechanisms; we compose fixed defenses deterministically rather than strengthening a single mechanism, so these are complements to our object rather than baselines for it. Randomized transformations (Xie et al., 2018; Liu et al., 2018) in centralized settings are the conceptual neighbor of a *randomized-menu* question, which this paper does not ask: every claim here concerns deterministic joint application (§C). **An earlier version of this paper compared against a randomized menu and reported that deterministic composition dominates it; that comparison is not part of this submission and no claim here rests on it.**

Backdoor persistence and durability. The persistence mechanism we formalize has empirical precursors. Bagdasaryan et al. (2020) design model-scaling explicitly to embed triggers that survive gradient updates; Zhang et al. (2022) (Neurotoxin) engineer triggers to occupy “less-used” gradient directions, achieving multi-round persistence even under strong norm clipping. These works establish the phenomenon, and this paper *uses* it rather than modelling it: persistence is why a committed attack stays committed across rounds, which is what makes a single committed attack against a fixed defense pair the right setting for an identification question. **We make no formal claim about persistence.** An earlier version of this paper advanced a $(\gamma, \alpha, p_{\text{eff}})$ model of when persistence eliminates the value of defense switching, together with a heuristic boundary as $\gamma \rightarrow 1$; that material is not part of this submission, and we withdraw the claim rather than leave it asserted without the evidence that supported it. What remains is γ as a measured descriptive quantity (Appendix K) and the DBA persistence replication of §L.3, which returns an *intermediate* verdict and establishes no boundary.

Defense composition and ensembling. Composing defenses is well-studied in adversarial ML for centralized models: adversarial training combined with input purification (Croce et al., 2021), detection-then-rejection pipelines, and ensemble adversarial training (Tramèr et al., 2018). These works implicitly require that upstream processing does not destroy downstream signals; our signal-preservation criterion formalizes this requirement explicitly with measurable margins. In FL the closest *structural* precedent is bucketing (Karimireddy et al., 2022), which restores existing robust aggregators under heterogeneous data by inserting an upstream reshuffling step ahead of them: this is the object we study, an upstream transform feeding a fixed downstream aggregator. Its target is convergence under heterogeneity rather than suppression of a committed attack, so it does not meet Proposition 6; what it shows is that the contrast our protocol demands is the natural one for this object, since establishing that an upstream stage helps at all requires comparing the same aggregator with and without it. In Byzantine-tolerant distributed systems, protocol composition (e.g., layered BFT) has a long history; the key distinction is that Byzantine agreement composes for fault tolerance among replicas, while our setting composes for poison resilience across client updates where the adversarial signal is a continuous gradient, not a binary fault. **A predictive criterion for FL composition does exist in recent work, and we withdraw the priority claim we previously made here.** Fenaux et al. (2025) categorize defenses by the deviation of adversarial updates from the mean of the updates and predict that pairing a large-deviation defense with a small-deviation one closes the gap either leaves alone. Our object is prior to theirs and different in kind: not which pairings succeed, but whether a comparison across defenses can identify *why* one did. Proposition 6 therefore applies to their criterion and to ours alike, and it constrains what validating evidence for such a criterion can establish rather than whether the criterion is correct.

H LIMITATIONS, IN FULL

(L1) Limited scope and architecture-conditional boundaries. Three attack families (model_scaling, pixel, DBA), two architectures (CifarCNN, ResNet18, plus `simple_cnn` on the replicated arm), and one dataset (CIFAR-10) for every composition suite but one. The collapse boundary is architecture-conditional (ResNet18 suppression at NC10/rep90: 0.865 vs. CifarCNN 0.097, a $10\times$ shift; per-seed values in §M.2), so results at other N , K or datasets may differ. Cross-distribution transfer is validated across architecture, scale ($N \in \{10, 100\}$) and adversarial fraction ($f \in [0.05, 0.6]$; §D.4(D)), and *dataset* transfer is now tested for the flagship Mode-S arm alone, EMNIST-byclass with `simple_cnn`, where the negative replicates at $\Delta = -0.017$ (§E.2), while remaining untested for the 42-pair screen and for every other composition suite: the signal-preservation conditions are dataset-agnostic in mechanism (cosine similarity, geometric distance), yet the downstream aggregator’s decision boundary may shift with output dimensionality. *What transfers:* the qualitative PASS/FAIL prediction and the mechanism. *What does not:* the exact suppression ASR. A practitioner at $N=10^4$ should use the criterion to select candidates and then run pilot evaluations at their target scale: the criterion identifies *what to try*, architecture-specific calibration determines *how well it works*.

(L2) The composability criterion is an evaluation-prioritization rule, not a security screen and not a discovery tool. It has no established predictive validity: it orders candidates for evaluation, and the pooled

model that would let it predict is unconfirmed ($R^2=0.239$) with its cross-arm ordering refuted (§E.2). The criterion gives *sufficient* conditions ($C1 \wedge C2$, with $C3$ as a margin diagnostic): it reliably identifies mechanism-preserving compositions in the common case where one defense is known-effective, but it cannot certify synergistic compositions where no single defense suppresses the attack alone. The conservative bias is by design: false negatives are operationally cheap to recover from (try the composition anyway) while false positives would deploy an insecure stack. Accuracy is $16/18 = 89\%$ on the development set, $22/24 = 92\%$ on the pre-registered held-out set (90.5% combined 2-way) and $9/10 = 90\%$ on pre-registered 3-way triples (supplement § S4). **None of those tallies should be read as discrimination:** only 5 of the 42 two-way pairs are low-ASR, so a constant HIGH call scores $37/42 = 88.1\%$ and the combined 90.5% beats it by one pair. Precision (2/2) and recall (40%) are the quantities that carry information, and they are why we present the criterion as a prioritizer. All clean $C1 \wedge C2 \wedge C3$ PASS predictions are correct; the four misses are either the predicted intermediate behavior of rank-based aggregators (Proposition 9: dev-set $NC \rightarrow cm$, $rfa \rightarrow cm$; held-out fedavg $\rightarrow cm$) or one speculative LOW call the criterion does not license (held-out $fg \rightarrow NC$, a $C1$ failure). Formalization is partial: Theorem 8 proves $C2 \wedge C3$ for the weighting $\rightarrow$ rank/geometric class with explicit margins, the remaining families are empirically validated, and $C3$'s threshold condition depends on d_2 's discrimination boundary, which in practice requires calibration runs.

(L3) Per-round aggregator switching only. We test per-round defense selection. Randomized *internals* (DP noise, subsampling smoothing à la CRFL, random participation masks) modify the embedding mechanism directly and may retain value even under persistence. Three adaptive attacks are tested against composition: projection-based (best ASR 0.597 at $\epsilon=1$), Neurotoxin-style (Zhang et al., 2022) top- k gradient projection, and a two-phase consensus-shifting attack against rep+tm that biases updates toward the target at rate r for 25 rounds before injecting full backdoor updates (App. L). At shift_rate $\in \{0.01, 0.05\}$, $n=10$ each: mean ASR 0.684 ± 0.138 and 0.658 ± 0.144 , comparable to committed_pixel (0.679) and below the temporal mix (0.915). At $f=0.2$ the honest 80% anchor the coordinate-wise median, limiting Phase-1 drift to $\sim r/(1-f)$ per round, so Phase 2 degenerates to standard committed_pixel: a mechanism that weakens monotonically with f , and at $f \leq 0.05$ over $T \geq 200$ rounds cumulative drift may eventually overcome it. The robustness claim is therefore specific to the tested regime ($f=0.2$, $T=50$).

(L4) No falsifiable prospective positive evidence, and C0 was added retrospectively. This is the sharpest limitation we can state about our own design, and we state it rather than let a reader find it. Of the prospective suite's 8 predicted-LOW pairs (§D.1), 6 are $\cdot \rightarrow FLTrust$ compositions that are near-*identities*: FLTrust gates on a scale-invariant cosine and renormalizes every update to the server-gradient norm, discarding exactly the magnitude information an upstream positive rescaling modifies (verified bit-identical per seed for pass-through and non-clipping d_1). Proposition 10(a) permits no other outcome, so these confirmations cannot falsify it; the remaining 2 have FLTrust as d_1 , where suppression is inherited and $C0$ puts them out of scope. Two further caveats: $C0$ was formulated *after* seeing this suite, so its 7 exclusions are retrospective rather than pre-registered; and the three matched contrasts that most directly probe Proposition 10 (a 19–20 $\times$ ASR gap between $\cdot \rightarrow FLTrust$ and $\cdot \rightarrow Krum$ at fixed d_1) confound $C1$ with $C2$, because in our menu the invariant-signal d_2 is also the only d_2 effective alone. We had named the decisive experiment as a downstream defense with a rescaling-invariant admission signal that is *weak in isolation*; Lemma 1 shows no such pair can be certified in scope, so that experiment does not exist: the confounding is structural, not a gap in our menu. §D.2 runs the achievable substitute, holding the suppression *principle* fixed while varying only the invariance class, and it did not deliver a positive test either: standalone effectiveness could not be held fixed, because for this defense family the magnitude information the mirrors discard is the same information the originals use to suppress a magnitude attack (8/8 frozen labels, $TP = 0$). So after two pre-registered attempts the criterion's positive direction still rests on in-distribution and out-of-sample PASS evidence, and the strongest $C2$ -specific evidence in the paper is the *post-hoc* within-defense contrast of §D.2, which holds $C1$ exactly fixed but was not pre-registered and is statistically resolved on only one of its two cells at $n=3$. §E is the pre-registered design that answers this, and it delivers exactly one falsifiable prospective positive result (arm 2, reputation under model-scaling, $0.017 \rightarrow 0.779$ across the dose at $p = 1.7 \times 10^{-6}$ with clean accuracy nearly unchanged) while refuting two of its own four predictions and its pooled prediction. So the honest state of the positive direction after the ladder is: one confirmed arm, on the *destruction* half of $C2$'s mechanism, with the preservation half's designated control indeterminate (§E.1). The targeted intervention of §E.2 advances this without closing it: closing the attenuation channel by construction, it isolates $C2$ within a single defense, substantial disturbance of the statistic does not produce a corresponding change in suppression *on the adjudicating Krum arm under Mode S*, with *admission* of adversarial input the channel the evidence points to there (H-statistic refuted, H-admission confirmed on that arm, pooled model not confirmed), and it *refutes* the preservation-half reading the ladder left open, since the exactly-invariant cos_krum neither suppresses nor holds ASR fixed. A third pre-registered attempt (§E.2) was designed to be the falsifiable prospective *positive* test: a replication arm chosen because the statistic reading and the admission reading order its rise *oppositely*, with the ordering frozen in git before the arm was run. It replicated the negative on a defense in a different invariance class and **refuted the positive ordering**: the arm with the largest measured admission change rose *less* (+0.098) than an arm whose admission change was smaller. So what remains absent is still a validated *positive* predictor: after that replication the pooled two-channel model is not confirmed over 26 cells ($R^2 = 0.239$) and its cross-arm

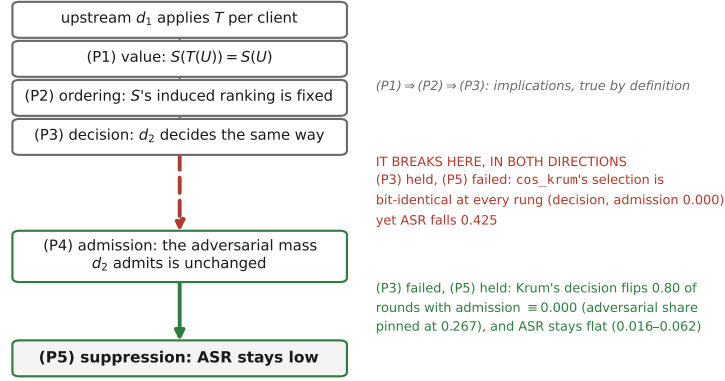
ordering is refuted outright, so the criterion’s positive direction is unestablished as a screen, and admission is a better *direction* than the statistic without being a quantitative one.

(L5) Theorem 8(1)’s condition is not operationally checkable as stated. Regime A’s ordering-survival condition is $\rho < S$ with S the attack’s coordinate extremeness against the benign scale. Building the dose–response ladder (§E) required us to *measure* S rather than stipulate it, and it is not a stable per-attack constant: on model-scaling it ranges 0.00–12.59 across the 12 measured rounds that sampled an adversary (median 0.57), on the pixel backdoor 0.61–1.63 (median 0.92). The $S=10$ of §C.2 is an illustrative stipulation, not a measurement. Two consequences we state rather than work around. First, a practitioner cannot check $\rho < S$ from a single round: the quantity swings by orders of magnitude round to round, and on both committed attacks its median is below 1, so Regime A’s condition fails even at the identity rung under this estimator. Second, we therefore built *no* prediction in §E on S , resting the ladder on directly measured disturbance of d_2 ’s statistic instead. The margin form of the theorem remains valid as a statement about ρ and S ; what is not established is that S admits a stable estimator at the round granularity a deployment would have to check it at, and a usable Regime A needs an S estimator with characterized variance, which we do not provide.

(L6) Measured disturbance is not a transferable predictor, and our dose instrument varies the wrong quantity for three of its four arms. The pre-registered ladder (§E.1) refutes the pooled claim it was built to test ($\rho_s = +0.351$, $p = 0.091$) and refutes two of four per-arm shapes, with ASR *falling* as the dose rises in three arms. Two distinct faults, both ours. First, the abscissa: at matched measured disturbance the real upstream transforms sit up to +0.79 ASR above the synthetic curve, and Krum’s within-margin arm does not reproduce the very contrast (§D.2) we attributed to disturbance of its statistic, so the rate at which a transform flips a defense’s decision is not a sufficient statistic for what the transform does: *which* clients are reweighted matters, and ρ does not measure that. We now measure the quantity that does, Δ_c (the supplement § S9): the two anchors that miss are the two most extremely targeted (-1.21 , -0.72 against the ladder’s -0.34) at ρ that is 3.7 and $13\times$ smaller. That names the axis of the mismatch without supplying a replacement predictor: Δ_c ’s magnitude does not order suppression either. Second, the instrument: mean-1 normalization fixes the aggregate’s scale but not any client’s relative weight, so with the dose assigned independently of adversary status a large ρ mostly dilutes whichever client carries the poison (measured: the ladder’s Δ_c drifts monotonically negative with the dose, $0.000 \rightarrow -0.140 \rightarrow -0.240 \rightarrow -0.336$), and three arms end up measuring Theorem 8’s attenuation regime rather than C2 disturbance. The adversary-targeted dose that separates the two, recorded as not run before the freeze (caveat (iii), the supplement § S6), is run in §E.2, and in retrospect Mode S is the Δ_c -**pinned** instrument (Δ_c exactly 0 at every rung, ρ up to 54.6) and Mode A the Δ_c -**swept** one (-1.14 to $+2.47$): with every adversary’s weight pinned, Krum’s statistic is disturbed in up to 80% of rounds while the admission of adversarial input does not move and ASR stays inside the pre-specified equivalence margin, so *which* clients are reweighted, not how often a decision flips, is indeed the operative variable, and the ladder confound is *explained* (reputation’s rise shrinks from $+0.762$ to $+0.178$ once the adversarial share is held fixed). That isolates C2’s causal content within a single defense; it does not rescue the pooled positive direction ($R^2 = 0.239$ over 26 cells, cross-arm ordering refuted by the replication).

I EVIDENCE STATUS: WHAT IS PROVED, WHAT IS SUPPORTED, WHAT IS NOT

Proved: the invariance classes of Proposition 10; sufficient conditions for mechanism preservation on the weighting→aggregator class (Theorem 8, Lemma 2, Corollary 2); the identification boundary for outcome-gated screening (Proposition 6, with Lemma 1 its FL instance); the recall bound on inherited-suppression screens (Proposition 7); and, formalizing the interventions, statistic-level non-identifiability (Proposition 11) and the independent controllability of ρ and Δ_c (Proposition 12). **The two boundary results are general** (they are statements about the shape of a screen, not about FL), so what carries beyond this paper is the design they license, not the screen. **Empirically supported:** that substantial disturbance of the downstream statistic need not produce a corresponding change in suppression when the adversarial coefficient share is held fixed, with adversarial admission the channel the evidence favours, *on the adjudicating Krum arm under Mode S* and, by pre-registered replication, both on a downstream defense in a *different* invariance class and on a *second dataset and architecture* (§E.2); and that the screen behaves as a conservative prioritizer across 42 pairs. **Not established:** the positive predictive direction of the criterion: the pooled two-channel model is unconfirmed over 26 cells ($R^2=0.239$) and the replication refutes its cross-arm admission ordering, so admission is a direction and not a quantitative predictor; any ability to discover emergent synergy; and generalization beyond star-topology FL, or beyond CIFAR-10 for anything but the one arm replicated on EMNIST-byclass.



Statistic preservation is not security preservation: only the (P4) $\rightarrow$ (P5) link carries security content in our experiments.

Figure A5: **The preservation chain, and where it breaks.** (P1) $\Rightarrow$ (P2) $\Rightarrow$ (P3) holds by definition; **the (P3)–(P4) link breaks in both directions**, and both witnesses come from the oracle intervention that pins every adversary’s coefficient and so closes Lemma 2’s attenuation channel *by construction* (§E.2). The pooled statistic-level prediction we had frozen is separately **rejected** by a pre-registered ladder ($\rho_s = +0.351$, one-sided $p = 0.091$, 16 cells; §E), and a pre-registered replication in a different invariance class reproduces the negative but **refutes the cross-arm admission ordering** ($\Delta = +0.098$; the pooled two-channel model stays unconfirmed, $R^2 = 0.239$ over 26 cells), so **admission is a direction, not a validated predictor**. Every number in the figure and in this caption is recomputed from the scoring scripts’ own JSON and printed by `story_chain_figure.py`.

J THE PRESERVATION CHAIN IN ONE FIGURE

K NOTATION

symbol	meaning
c_i	upstream coefficient applied to client i ’s update, $T(u_i) = c_i u_i$, $c_i > 0$
A	the set of adversarial participants in a round
ρ	statistic-level dispersion, $\max_i c_i / \min_i c_i$: <i>how far apart</i> the coefficients are
Δ_c	adversary-correlated dispersion, $\mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$: <i>who</i> receives them
r	adversary-to-benign weight ratio, $w_a / \max_b w_b$; the Mode-A dial, $r = e^\nu$
S	the attack’s coordinate extremeness (Theorem 8(1))
m	RFA separation margin, $\Delta'_{\text{sep}} / 2(R'_B + \delta')$
Λ_a	aggregate adversarial influence: adversarial share of the Weiszfeld mass
κ	the Mode-S / ladder dispersion dial, $\rho = e^{2\kappa}$
γ	attack <i>persistence</i> , $\text{ASR}(t+1) / \text{ASR}(t)$ after a clean round (≈ 0.66); never a weight

L COMPOSITION AND ADAPTIVE-ATTACK DETAILS

L.1 COMPOSITION BASELINE DETAILS

Simultaneous composition (NormClip $\tau=5$ then reputation every round), $N=10$, $K=5$, $f=0.2$, 50 rounds, cifar_cnn, 5 seeds. Per-seed results:

Seed	model_scaling		backdoor_pixel	
	Accuracy	ASR	Accuracy	ASR
42	0.783	0.766	0.774	0.837
43	0.788	0.783	0.774	0.792
44	0.794	0.835	0.784	0.828
45	0.762	0.889	0.771	0.931
46	0.780	0.763	0.766	0.823
Mean $\pm$ std	0.781 ± 0.011	0.807 ± 0.048	0.774 ± 0.006	0.842 ± 0.047

Composition partially suppresses scaling (ASR 0.807 vs. pure NormClip ~ 0.95 , and 0.961 under the NE3 temporal-mixing policy defined in §L.2) but does not approach pure reputation (0.017). Norm clipping preserves the attack direction while reducing magnitude; the clipped update then passes reputation’s consensus-distance filter. Pixel attacks are similarly persistent under composition (ASR 0.842). Both attack families remain effective despite simultaneous application of both defenses.

L.2 DBA AT $f=0.4$: PER-SEED DETAILS

Three run labels are carried over from an earlier framing and used here only to identify runs, not to make any equilibrium claim: *NE3* is the mixed server policy that framing computed, which randomizes over defenses across rounds; *reputation eq* is the pure-reputation policy; and *BR* is the adversary’s best response to the policy it faces. NE3 BR on CIFAR-10 with DBA in place of model scaling at $f=0.4$ (raising adversarial co-participation to $K \cdot f = 2.0$ from the $f=0.2$ value of 1.0). $N=10$, $K=5$, 5 seeds 42–46, 50 rounds. Per-seed realized ASRs:

seed	DBA realized ASR	backdoor.pixel realized ASR
42	0.222	0.972
43	0.103	0.945
44	0.176	0.913
45	0.603	0.993
46	0.077	0.934
mean $\pm$ std	0.236 ± 0.190	0.952 ± 0.028

The DBA per-seed spread is $7.5\times$ between min and max; only seed 45 clearly persists, three (0.08, 0.10, 0.18) are below the threshold (≥ 0.20) we would call active persistence, and one (0.22) is borderline. Pixel persists cleanly at $f=0.4$ (0.952 ± 0.028), so the persistence mechanism survives an adversarial-fraction shift on the original attack. DBA’s seed-variability reflects per-round co-participation alignment (whether ≥ 2 of the 4 adversarial clients land in the same round to coordinate the distributed trigger) rather than the structural argmax-collapse mechanism we describe for scaling. The cleanest evidence for the headline persistence claim therefore remains model scaling; DBA at $f=0.4$ is suggestive but not a robust second-attack demonstration.

L.3 DBA AT $f=0.6$: PER-SEED DETAILS AND BIMODALITY REPRODUCTION

Follow-up to Appendix L.2, designed to test whether the DBA bimodality at $f=0.4$ is purely a per-round co-participation effect or whether it persists as a per-seed admissibility latent at higher f . At $f=0.6$, expected adversarial co-participation per round is $K \cdot f = 3.0$, making ≥ 2 adversarial co-participation per round near-certain. NE3-style BR with committed adversary playing DBA; 5 seeds (42–46), 50 rounds, $N=10$, $K=5$, `cifar_cnn`. Two deployed policies: *NE3 mix* (FedAvg 26% + NormClip 74%) and the *reputation equilibrium* (NormClip 83% + reputation 17%).

seed	DBA ASR (NE3 mix)	DBA ASR (reputation eq)
42	0.360	0.339
43	0.295	0.278
44	0.386	0.415
45	0.734	0.705
46	0.154	0.130
mean $\pm$ std	0.386 ± 0.192	0.373 ± 0.190

Observations. (i) Mean DBA ASR improves over $f=0.4$ (0.236) to ~ 0.38 on both policies, but does not reach the ≥ 0.85 persistence band scaling achieves. (ii) Per-seed bimodality persists at $f=0.6$: seed 45 alone reaches > 0.7 in both policies; the remaining four seeds cluster in $[0.13, 0.42]$. (iii) Crucially, *the same seed is the strong-persistence seed under both policies* (NE3 mix: 0.734; reputation eq: 0.705), and *the same four seeds are the weak-persistence seeds under both*: the per-seed cross-policy correlation rules out per-round co-sampling alignment as the dominant source of bimodality. The bimodality is a per-seed training-initialization latent in DBA itself: whether the distributed trigger embeds at all in the early rounds. This is the DBA analogue of the model-scaling binding-cell Bernoulli, but at a much lower mean (admissibility prevalence $\approx 1/5$ for DBA at $f=0.6$ vs. $\approx 4/5$ for scaling at $f=0.2$).

Pre-specified verdict: INTERMEDIATE. Overall mean 0.380 falls between PARTIAL (> 0.5) and NEGATIVE (< 0.3). Reading: DBA at $f=0.6$ does *not* replicate the persistence collapse with the cleanliness scaling achieves; the trigger-alignment mechanism has its own admissibility constraint we do not control by raising f .

The cleanest headline-persistence evidence remains model scaling under norm-based menus, as acknowledged in L3.

M SCALE, ARCHITECTURE, AND EXTERNAL VALIDITY

M.1 SCALE VALIDATION ($N = 100$) AND DEFENSE-INFERENCE CHECK

Scale check at $N = 100$ (4×5 game, 5 seeds). Residual material from an earlier framing, recorded rather than dropped. Nothing in the screen, the identification result or the dose-response evidence depends on it. Write VoPD (value of policy diversity) for the gain the server’s equilibrium strategy achieves over its best *pure* defense, so VoPD = 0 exactly when the equilibrium is pure. The equilibrium is pure on seeds 43 to 46 (model scaling vs. RFA, VoPD = 0) and mixed on seed 42 (pixel + scaling vs. FedAvg + RFA, VoPD = 0.012); per-seed values in Table A14.

Table A14: Per-seed NE, CIFAR-10 $N = 100$ (4×5 game). Seeds 43–46 identical.

Seed	Type	Adversary	Server	VoPD
42	Mixed	Pixel + Scaling	FedAvg + RFA	0.012
43–46	Pure	Model scaling	RFA	0.000

M.2 RESNET18 ARCHITECTURE-REPLICATION MINI-SWEEP: PER-SEED DETAILS

Per-seed details for the ResNet18 architecture-replication mini-sweep behind the architecture-conditional collapse boundary in §H. `resnet18` with BatchNorm replaced by GroupNorm (8 groups), conv1 adapted to 32×32 (kernel 3, stride 1, maxpool $\rightarrow$ Identity); 11.2M parameters. Identical FL parameters to `cifar.cnn`: $N=10$, $K=5$, $f=0.2$, 50 rounds, $\tau = 5$. Committed-scaling BR, 3 mix points $\times$ 5 seeds = 15 runs at ~ 100 min/run on MPS.

mix (NC/rep)	s42	s43	s44	s45	s46	ResNet18 mean	cifar.cnn (R50)
90/10	0.926	0.929	0.965	0.974	0.962	0.951 $\pm$ 0.020	0.945 $\pm$ 0.021
50/50	0.924	0.971	0.813	0.920	0.903	0.906 $\pm$ 0.052	0.815 $\pm$ 0.104
10/90	0.917	0.733	0.783	0.975	0.915	0.865 $\pm$ 0.091	0.097 $\pm$ 0.090

NC-dominant endpoint replicates within 0.01; mid-mix is $\sim 9\%$ higher on ResNet18; the reputation-dominant endpoint shows an order-of-magnitude gap (no CI overlap). Seed 45’s accuracy collapses to 0.153 at NC10/rep90 (the persistent attack effectively breaks the model) while trigger ASR remains 0.975. **Verdict: MEDIUM** (monotonic ordering preserved, endpoint criterion fails). The `cifar.cnn` collapse boundary at ~ 70 –90% reputation is therefore a property of the smaller-architecture regime. This sweep bounds the architecture-conditional claim of §H and establishes no boundary of its own; we draw no conclusion here about which mixture of the two defenses survives, since mixture policies are outside this paper’s scope (§C).